DRAUGHTSMAN MECHANICAL

NSQF LEVEL - 4

1st Year

TRADE PRACTICAL

SECTOR: CAPITAL GOODS & MANUFACTURING

(As per revised syllabus July 2022 - 1200 Hrs)

कौशलं बलम्

Directorate General of Training

DIRECTORATE GENERAL OF TRAINING
MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP
GOVERNMENT OF INDIA

Nimi

NATIONAL INSTRUCTIONAL
MEDIA INSTITUTE, CHENNAI

Post Box No. 3142, CTI Campus, Guindy, Chennai - 600 032

Sector : Capital Goods and Manufacturing

Duration : 2 - Years

Trades : Draughtsman Mechanical - 1st Year - Trade Practical - NSQF LEVEL 4

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FOREWORD

The Government of India has set an ambitious target of imparting skills to 30 crores people, one out of every four Indians, by 2020 to help them secure jobs as part of the National Skills Development Policy. Industrial Training Institutes (ITIs) play a vital role in this process especially in terms of providing skilled manpower. Keeping this in mind, and for providing the current industry relevant skill training to Trainees, ITI syllabus has been recently updated with the help of Media Development Committee members of various stakeholders viz. Industries, Entrepreneurs, Academicians and representatives from ITIs.

The National Instructional Media Institute (NIMI), Chennai, has now come up with instructional material to suit the revised curriculum for **Draughtsman Mechanical 1st Year Trade Practical NSQF Level-4 in Capital Goods and Manufacturing Sector under Yearly Pattern.** The NSQF Level - 4 Trade Practical will help the trainees to get an international equivalency standard where their skill proficiency and competency will be duly recognized across the globe and this will also increase the scope of recognition of prior learning. NSQF Level - 4 trainees will also get the opportunities to promote life long learning and skill development. I have no doubt that with NSQF Level - 4 the trainers and trainees of ITIs, and all stakeholders will derive maximum benefits from these Instructional Media Packages IMPs and that NIMI's effort will go a long way in improving the quality of Vocational training in the country.

The Executive Director & Staff of NIMI and members of Media Development Committee deserve appreciation for their contribution in bringing out this publication.

Jai Hind

Ms. TRISHALJIT SETHI,
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Ministry of Skill Development & Entrepreneurship,
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New Delhi - 110 001

PREFACE

The National Instructional Media Institute (NIMI) was established in 1986 at Chennai by then Directorate General of Employment and Training (D.GE & T), Ministry of Labour and Employment, (now under Directorate General of Training, Ministry of Skill Development and Entrepreneurship) Government of India, with technical assistance from the Govt. of Federal Republic of Germany. The prime objective of this Institute is to develop and provide instructional materials for various trades as per the prescribed syllabus under the Craftsman and Apprenticeship Training Schemes.

The instructional materials are created keeping in mind, the main objective of Vocational Training under NCVT/NAC in India, which is to help an individual to master skills to do a job. The instructional materials are generated in the form of Instructional Media Packages (IMPs). An IMP consists of Theory book, Practical book, Test and Assignment book, Instructor Guide, Audio Visual Aid (Wall charts and Transparencies) and other support materials.

The trade practical book consists of series of exercises to be completed by the trainees in the workshop. These exercises are designed to ensure that all the skills in the prescribed syllabus are covered. The trade theory book provides related theoretical knowledge required to enable the trainee to do a job. The test and assignments will enable the instructor to give assignments for the evaluation of the performance of a trainee. The wall charts and transparencies are unique, as they not only help the instructor to effectively present a topic but also help him to assess the trainee's understanding. The instructor guide enables the instructor to plan his schedule of instruction, plan the raw material requirements, day to day lessons and demonstrations.

IMPs also deals with the complex skills required to be developed for effective team work. Necessary care has also been taken to include important skill areas of allied trades as prescribed in the syllabus.

The availability of a complete Instructional Media Package in an institute helps both the trainer and management to impart effective training.

The IMPs are the outcome of collective efforts of the staff members of NIMI and the members of the Media Development Committees specially drawn from Public and Private sector industries, various training institutes under the Directorate General of Training (DGT), Government and Private ITIs.

NIMI would like to take this opportunity to convey sincere thanks to the Directors of Employment & Training of various State Governments, Training Departments of Industries both in the Public and Private sectors, Officers of DGT and DGT field institutes, proof readers, individual media developers and coordinators, but for whose active support NIMI would not have been able to bring out this materials.

Chennai - 600 032 **EXECUTIVE DIRECTOR**

ACKNOWLEDGEMENT

National Instructional Media Institute (NIMI) sincerely acknowledges with thanks for the co-operation and contribution extended by the following Media Developers and their sponsoring organisation to bring out this IMP **(Trade Practical)** for the trade of **Draughtsman Mechanical** under the **Capital Goods and Manufacturing** Sector for ITIs.

MEDIA DEVELOPMENT COMMITTEE MEMBERS

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Shri. V. Gopalakrishnan	-	Manager NIMI - Chennai -32.

NIMI records its appreciation of the Data Entry, CAD, DTP Operators for their excellent and devoted services in the process of development of this Instructional Material.

NIMI also acknowledges with thanks, the invaluable efforts rendered by all other staff who have contributed for the development of this Instructional Material.

NIMI is grateful to all others who have directly or indirectly helped in developing this IMP.

INTRODUCTION

TRADE PRACTICAL

The trade practical manual is intented to be used in workshop . It consists of a series of practical exercises to be completed by the trainees during the course of the **Draughtsman Mechanical** trade supplemented and supported by instructions/ informations to assist in performing the exercises. These exercises are designed to ensure that all the skills in compliance with NSQF LEVEL - 4

The manual is divided into Nine modules.

Module 1	Importance of trade training and Safety
Module 2	Basic Engineering drawing - Types of curves - Methods of dimensioning - Types of scales
Module 3	Projection - Freehand sketching of different part of machines
Module 4	Sectional views
Module 5	Development of surfaces and interpreteration of solids in orthographic projection
Module 6	Isometric projection from orthographic views (viceversa) oblique projection from orthographic views
Module 7	Specification of different types of fasteners and locking devices as per SP 46 2003
Module 8	Allied trade
Module 9	Tolreances - Machining symbol - surface finishing symbols - Geometrical tolerances - sectional view
Module10	Computer Operation

The skill training in the shop floor is planned through a series of practical exercises centred around some practical project. However, there are few instances where the individual exercise does not form a part of project.

While developing the practical manual a sincere effort was made to prepare each exercise which will be easy to understand and carry out even by below average trainee. However the development team accept that there is a scope for further improvement. NIMI, looks forward to the suggestions from the experienced training faculty for improving the manual.

TRADE THEORY

The manual of trade theory consists of theoretical information for the course of the **Draughtsman Mechanical** Trade. The contents are sequenced according to the practical exercise contained in the manual on Trade practical. Attempt has been made to relate the theoretical aspects with the skill covered in each exercise to the extent possible. This co-relation is maintained to help the trainees to develop the perceptional capabilities for performing the skills.

The Trade theory has to be taught and learnt along with the corresponding exercise contained in the manual on trade practical. The indicating about the corresponding practical exercise are given in every sheet of this manual.

It will be preferable to teach/learn the trade theory connected to each exercise atleast one class before performing the related skills in the shop floor. The trade theory is to be treated as an integrated part of each exercise.

The material is not the purpose of self learning and should be considered as supplementary to class room instruction.

CONTENTS

LEARNING OUTCOME

On completion of this book you shall be able to

S.No.	Learning Outcome	Ref. Ex.No.
1	Construct different Geometrical figures using drawing Instruments following safety precautions.	1.1.01 - 1.3.22
2	Draw orthographic Projections giving proper dimensioning with title block using appropriate line type and scale.	1.3.23 - 1.4.28
3	Construct free hand sketches of simple machine parts with correct proportions.	1.5.29
4	Construct plain scale, comparative scale, diagonal scale and vernier scale	1.5.30
5	Draw Sectional views of orthographic projections.	1.5.31 - 1.6.33
6	Develop surface and interpenetration of solid in orthographic projection.	1.6.34 - 1.6.38
7	Draw isometric projection from orthographic views (and vice-versa) and draw oblique projection from orthographic views.	1.6.39 - 1.7.47
8	Draw and indicate the specification of different types of fasteners, welds and locking devices as per SP-46:2003	1.7.48 - 1.8.62
9	Acquire basic knowledge on tools and equipments and their application in Allied trades viz. Fitter, Turner, Machinist, Sheet Metal Worker, Welder, Foundry man, Electrician and Maintenance Motor Vehicles.	1.8.63 - 1.8.72
10	Construct different types of gears, couplings and bearings with tolerance dimension and indicating surface finish symbol.	1.8.73 - 1.9.81
11	Perform computer application and create 2D objects on CAD drawing space using commands from ribbon, menu bar, toolbars and by typing in command prompt.	1.9.82 - 1.10.89

SYLLABUS

Duration	Reference Learning Outcome	Professional Skill (Trade Practical) (With inidcative hour)	Professional Knowledge (Trade Theory)
Professional Skill 120Hrs; Professional Knowledge 26Hrs	Construct different Geometrical figures using drawing Instruments following safety precautions. (CSC/NO402)	1 Importance of trade training, List of tools & Machinery used in the trade. (02 hrs) 2 Safety attitude development of the trainee by educating them to use Personal Protective Equipment (PPE). (05 hrs) 3 First Aid Method and basic training. (03 hrs) 4 Safe disposal of waste materials like cotton waste, metal chips/burrs etc. (02 hrs) 5 Hazard identification and avoidance. (02 hrs) 6 Safety signs for Danger, Warning, caution & personal safety message. (02 hrs) 7 Preventive measures for electrical accidents & steps to be taken in such accidents. (05 hrs) 8 Use of Fire extinguishers. (07 hrs)	Importance of safety and general precautions observed in the industry/shop floor. All necessary guidance to be provided to the newcomers to become familiar with the working of Industrial Training Institute system including stores procedures. Soft Skills: its importance and Job area after completion of training. Introduction of First aid. Operation of electrical mains. Introduction of PPEs. Introduction to 5S concept & its application. Response to emergencies e.g. power failure, fire, and system failure.(04 hrs.)
		Perform assignment using drawing instruments: 9 Draw straight lines of a given length. (01hr) 10 Draw perpendicular, inclined (given angle) and parallel lines. Draw triangles with given sides and angles. (03hrs) 11 Construct regular polygons (up to 8 sides) on equal base. (04hrs) 12 Draw inscribed and circumscribed circles of triangle, pentagon and hexagon. (04hrs) 13 Draw a parallelogram with a given length included angle. (02hrs) 14 Draw an angle bi-sector and a line bi-sector. (08hrs) 15 Divide a line into given equal divisions. (06hrs)	Nomenclature, description and use of drawing instruments & various equipments used in drawing office. Their care and maintenance.(04 hrs.) Lay out and designation of a drawing sheet as per Sp -46 : 2003 Recommended scale of engineering drawing as per Sp -46 : 2003 Types of Lines and their application. Folding of prints for filing Cabinets or binding as per SP: 46-2003. (06 hrs.)
		16 Layout a A3 drawing sheet as per Sp -46 : 2003 with margin and name plate. (05hrs) 17 Draw a sample title block providing details as: (i) Title of the drawing (ii) Sheet number (iii)Scale (iv)Symbol, denoting the method of projection (v) Revision with sign (vi) Name of the firm (vii) Initials of staff drawn, checked and approved. (05hrs)	

		18 Draw different types of lines & write their uses in drawing. (05hrs) 19 Label a drawing views showing most of the types of line. (13hrs)	
		20 Write Block letters & numerals in single & double stroke of ratio 7:4 and 5:4 in drawing sheet. (18hrs)	Type of lettering proportion and spacing of letters and words. (06 hrs.)
		21 Construction of ellipse, parabola & hyperbola in different methods. (10120hrs) 22 Construction of involutes, cycloid curves, helix & spiral. 08hrs)	Definition of ellipse, parabola, hyperbola, different methods of their construction. Definition & method of drawing involutes cycloid curves, helix & spiral. (06 hrs.)
Professional Skill 60 Hrs; Professional Knowledge 15Hrs	Draw orthographic Projections giving proper dimensioning with title block using appropriate line type and scale. (CSC/NO402)	23 Construct object drawing with dimensioning in different alignment as per SP-46. (03hrs) 24 Create dimensions in previous assignments. (15hrs)	Terminology - feature, functional feature, functional dimension, datum dimension, principles. Units of dimensioning, System of dimensioning, Method of dimensioning & common features. (04 hrs.)
		25 Draw orthographic projection of points and lines. (08 hrs) 26 Draw projection of plane figures (lamina). (12 hrs)	Methods of obtaining orthographic view. Position of the object, selection of the views, three views of drawing. Planes and their normal projections. (04 hrs.)
		27 Draw orthographic projection of solids- prisms, cylinders, cones, pyramids. (10 hrs) 28 Draw orthographic projection of cut section/ frustums of solids- prism, cylinders, cones, pyramids. (12hrs)	Orthographic projection. First angle and third angle projection. Principal of orthographic projection. Projection of solids like prism, cones, pyramids and their frustums. (05 hrs.)
Professional Skill 15Hrs; Professional Knowledge 06Hrs	Construct free hand sketches of simple machine parts with correct proportions. (CSC/NO402)	29 Free hand sketch (in proper proportion) of tool post of a Lathe, Bench Vice, Cutting Tools, Bolts, Stud & Nut, gland, Pipe Flange, Hand Wheel, Crane hook, Steel bracket. (15hrs)	Methods of free hand sketching for machine parts. (06 hrs.)
Professional Skill 15Hrs; Professional Knowledge 06Hrs	Construct plain scale, comparative scale, diagonal scale and vernier scale(CSC/NO402)	30 Draw plain scales, diagonal scales, comparative scales, venire scale & scale of chords. (15hrs)	Knowledge of different types of scales, scale of cords, their appropriate uses, Principle of R.F, diagonal &vernier. (06 hrs.)
Professional Skill 30Hrs; Professional Knowledge 12Hrs	Draw Sectional views of orthographic projections. (CSC/NO402)	31 Sketch Conventional sings and symbols. (05hrs) 32 Sketch different types of section lines and abbreviations for different materials as per SP-46:2003. (05hrs) 33 Draw Orthographic drawing of solids (viz., cube, prisms, cone and pyramids) finding out the true shape surfaces cut by oblique planes. (20hrs)	Knowledge of solid section. Types of sectional views & their uses. Cutting plane and its representation. Parts not shown in section. Conventional signs, symbols, abbreviations & hatching for different materials. Solution of problems to find out the true shape of surfaces when solids are cut by different cutting planes. (12 hrs.)
Professional Skill 82Hrs; Professional Knowledge 20Hrs	Develop surface and interpenetration of solid in orthographic projection. (CSC/NO402)	34 Construct the development of surface of cylinder, prisms, Cone, pyramids and their frustum. (18hrs) 35 Draw development of an oblique cone with elliptical base. (05hrs) 36 Draw the development of a 3-pieces pipe elbow, a pipe hole through it, a bucket and a funnel. (13hrs)	Definition of development, its need in industry & different method of developing the surfaces. Development of surfaces bounded by plane of revolution intersecting each other. Development of an oblique cone with elliptical base etc. Calculation of de-

			veloped lengths of geometrical solids. (10 hrs.)
		37 Construct orthographic projection of interpenetrating solids (cylinder, cones, prism & pyramid) of axes right angle to each other and axes inclined to each other. (26hrs) 38 Generate the curves of intersection of cylinder penetrating through a sphere, cone and a cylinder. (20hrs)	Definition of Intersection & interpenetration curves. Common method to find out the curve of interpenetration. Solution of problems on interpenetration of prism, cones, & pyramids with their axes intersecting at an angle. Intersection of cylinder. (10 hrs.)
Professional Skill 82 Hrs; Professional Knowledge 20Hrs	Draw isometric projection from orthographic views (and vice-versa) and draw oblique projection from orthographic views. (CSC/NO402)	39 Construct the isometric view of Polygons and circular lamina. (08 hrs) 40 Draw isometric view of solid geometrical figures from orthographic views with dimension. (08 hrs) 41 Draw isometric views of truncated cone and pyramid. (08hrs)	Principle of isometric projection and Isometric drawing. Methods of isometric projection and dimensioning. Isometric scale. Difference between Isometric drawing & Isometric projection. (04 hrs.)
		42 Construct orthographic views from isometric drawing of solid blocks with holes, grooves, notches, dovetail cut, square cut, round cut, stepped, etc. (10hrs) 43 Construct orthographic views of hanger, bracket & support (08 hrs) 44 Draw isometric view of V-block, Angle plate, sliding block. (10 hrs) 45 Draw isometric drawing of a simple Journal Bearing. (08 hrs.)	Principles of making orthographic views from isometric drawing. Selection of views for construction of orthographic drawings for clear description of the object. (10 hrs.)
		46 Draw oblique projection of circular lamina in receding axis at 30° & 45°. (05hrs) 47 Draw oblique projection of levers and hollow blocks. (17 hrs)	Principle and types of oblique projection. Advantage of oblique projection over isometric. Projection. (06 hrs.)
Professional Skill 130 Hrs; Professional Knowledge 30Hrs	Draw and indicate the specification of different types of fasteners, welds and locking devices as per SP-46:2003 (CSC/NO402)	48 Draw Screw threads with SP-46:2003 conventions. (08hrs) 49 Draw different types of bolts, studs, nuts and washers as per SP-46:2003 conventions. (08hrs) 50 Draw different locking arrangement of nuts, machine screws, caps screw set screw as per convention. (08hrs) 51 Draw a half sectional view of a coupler nut. (04hrs) 52 Draw four different types of foundation bolt. (16hrs)	Screw threads, terms nomenclature, types of screw thread, proportion and their uses, threads as per SP-46:2003 conventions. Types of bolts, nuts and studs, and their proportion, uses. Different types of locking devices. Different types of machine screws, cap screws, set screws as per specification. Different types of foundation bolts and their uses. (10 hrs.)
		53 Draw fillet weld and butt weld joint specifying the basic term of the joint. (05hrs) 54 Draw a weld joint representing the position and dimensioning of the weld with conventional symbols on the drawing. (06hrs) 55 Draw section of welded steel structural column & bracket fabricated by plate. (10hrs)	Description of Welded Joints and their representation (Actual and Symbolic) Indication of Welding Symbol on drawing as per SP-46. (04 hrs.) Different types of keys (Heavy duty and Light duty) cotters, splined shaft, pins and circlips. Calculation of sizes and proportions of keys. (06 hrs.)

		56 Draw a half-sectional view of Cotter joint with socket and spigot ends. (12hrs) 57 Draw different types of Keys, splined shaft, circlips and pins as per convention. (08 hrs)	
		58 Draw the different types of pipe fittings. (06 hrs) 59 Draw pipe joints: flanged joint, welded joint, threaded joint, socket and spigot joint.(18hrs)	Pipe Joints: selection of materials as per carrying fluid and conditions. Description of different pipe joints fitted on pipe. Expansion joint, loop and other pipe fittings. (04 hrs.)
		60 Draw rolled steel sections as per IS specification. (05hrs) 61 Draw the different types of rivet heads indicating the dimensions related to diameter of the rivet as per convention. (08hrs) 62 Draw riveted joints of lap and butt with covers in chain and zig-zag orientation. (08hrs)	Types of rivets, their size proportions and uses. Types of riveted joints, terms and proportions of riveted joints. Conventional representation. Relation between rivet size and thickness of plates and calculation for arrangement of rivets position. Causes of failure of riveted joint efficiency of riveted joints. (06 hrs.)
Professional Skill 130Hrs; Professional Knowledge 30Hrs	Acquire basic knowledge on tools and equipments and their application in Allied trades viz. Fitter, Turner, Machinist, Sheet Metal Worker, Welder, Foundry man, Electrician and Maintenance Motor Vehicles. (CSC/NO402)	Allied Trade- Fitting 63 Use of different types of fitters hand tools. (06hrs) 64 Work on MS plate by filing, hack sawing, check dimensions, mark the plate, punch centre mark, cut a v-notch by chisel, drill a hole on the center mark. (16hrs)	Description and application of simple measuring tools. Description of vices, hammers, cold chisel, files, drills, etc.-proper method of using them. Method of using precision measuring instrument. Maintaining sequence of operation in fitting shop and safety precaution. (04 hrs.)
		Allied Trade Turning 65 Cut a round bar in power saw, centering and facing the bar, perform the turning, grooving, stepped and taper operation on the bar. (20hrs)	Safety precaution for lathes. Description of parts of Lathe & its accessories. Method of using precision measuring instrument such as inside & outside micrometers, depth gauges, verniercallipers, dial indicators, slip gauges, sine bars, universal bevel protractor, etc. (04 hrs.)
		Allied Trade Machinist: 66 Use of jigs and fixtures Simple operations on milling machine such as plain-milling and key way cutting. (10 hrs) 67 Mark out on castings and forgings work piece, set up and perform operation of shaping, slotting and planning machines. (10 hrs)	Brief Description of milling, shaping, slotting and planning machines. Quick return mechanism of these machines. Maintaining sequence of operation in machine shop and safety precaution. (06 hrs.)
		68 Allied Trade: Sheet Metal Use of hand tools such as planishing,hammers stakes, mallet, bricks prick punch etc. Mark and cut a sheet to make a container. (20hrs)	Brief description of common equipment required for sheet metal work. Different types of joints used in sheet metal work. (04 hrs.)
		Allied Trade: Welding 69 Use of hand tools used in gas and in electric arc welding	Maintaining sequence of operation in machine shop and safety precaution. Brief description of the hand tools used gas & arc welding. Different types of

		Weld an object according to drawing. (12 hrs) 70 Foudryman/Moulder Different types of mould, cores and core dressing, use of moulding tools. (12 hrs) Allied Trade: Electrician 71 Prepare a simple wiring for residential room. Identify the electrical equipment and measuring instruments.(12hrs) Allied Trade: MMV- IC Engine 72 Identify different parts of IC Engines (Both spark ignition & compression ignition-2 stroke & 4 stroke engines). (12hrs)	welded joints and necessary preparation required for these. Safety precautions, Hand tools used for molding. The description, use and care of hand tools.(06 hrs.) Safety precaution maintained in electrician shop. A.C & D.C Motors Generators of common types and their uses and brief description of common equipment necessary for sheet metal work. Electrical units and quantities. Laws of electricity. Simple examples of calculation of current voltage, resistance in series and parallel connection (D.C.Circuit). Brief description of internal combustion engines, such as cylinder block piston, carburettor spark plug, camshaft, crank shaft, injector fuel pump etc. (06 hrs.)
Professional Skill 120Hrs; Professional Knowledge 26Hrs	Construct different types of gears, couplings and bearings with tolerance dimension and indicating surface finish symbol. (CSC/NO402)	73 Draw the diagram illustrating basic size deviations and tolerances. (03hrs) 74 Draw symbols for machining and surface finishes (grades and micron values) (03hrs) 75 Draw the system of indication of geometrical tolerances of form and position as per standard: Straightness, flatness, circularity, cylindricity, parallelism, perpendicularity, angularity, concentricity, coaxiality, symmetry, radial run-out, axial run-out. (10hrs) 76 Construct a machine part indicating geometrical tolerance. (08hrs)	Limits, fit, tolerance. Toleranced dimensioning, geometrical tolerance. Indications of symbols for machining and surface finishes on drawing(grades and micron values) Production of interchangeable parts, geometrical tolerance. Familiarization with IS: 919, IS:2709.(06 hrs.)
		Construct the sectional view of: 77 Muff coupling, (06hrs) 78 Flanged coupling, (10hrs) 79 Friction grip coupling. (10hrs) 80 Pin type flexible coupling, (10hrs) 81 Universal coupling. (10hrs) (conventional method)	Couplings, necessity of coupling, classification of couplings. Uses and proportion of different types of couplings. Materials used for couplings. (10 hrs.)
		Draw detailed and assembly drawing of: 82 Simple bearing (03hrs) 83 Foot step bearing. (03hrs) 84 Plummer block. (08hrs) 85 Self-aligning bearing (swivel bearing). (08hrs) 86 Construct tooth profile of a spur gear above 30 teeth. (10hrs) 87 Draw two spur gears in mesh (08hrs) 88 Draw two bevel gears in mesh (10hrs)	Knowledge of bearing to reduce friction, types of bearing, frictional and antifrictional bearings. Material used for frictional bearings. Properties of frictional bearing (sliding bearing) materials. Parts of anti-frictional bearings (ball, roller, thrust ball, needle & taper roller). Materials and proportion of parts. Difference between frictional and anti-frictional bearings. Advantages of anti-frictional bearings. (05hrs.) Gears and gear drives- uses, types, nomenclature and tooth profiles. (05 hrs.)

Professional Skill 56Hrs; Professional Knowledge 15Hrs	Perform computer application and create 2D objects on CAD drawing space using commands from ribbon, menu bar, tool bars and by typing in command prompt.	89 Perform computer application (05hrs) i) create new folder, ii) add subfolders, iii) create application files, iv) change appearance of windows, v) search for files, vi) sort files, vii) copy files, viii) create shortcut folder, ix) create shortcut icon in desktop and taskbar x) move files to and from removable disk/ flash drive. xi) install a printer from driver software in operatingsystem. Create, save and print a document, worksheet and pdf (portable document format) files. (10hrs)	Introduction to computer, Windows operating system, file management system. Computer hardware and software specification. Knowledge of installation of application software. (04 hrs.)

Importance of training - List of tools and machinery used in trade

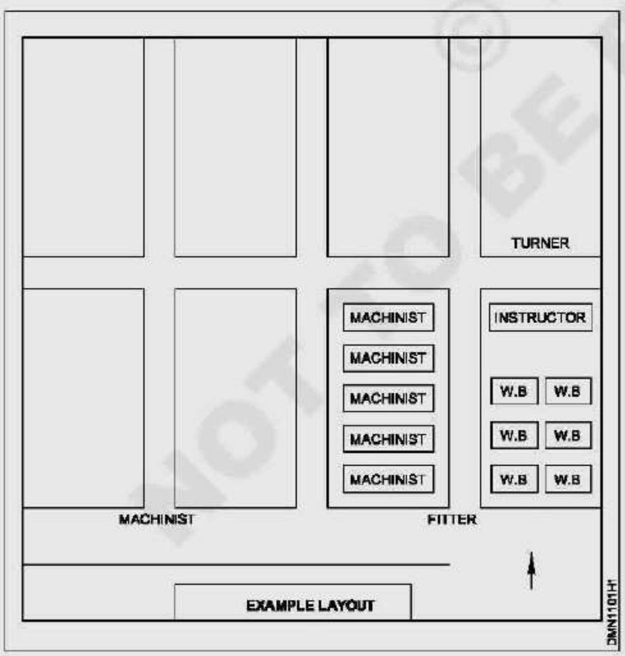
Objective: At the end of this exercise you shall be able to
- **over view about the trade.**

- Draughtsman mechanical trade is basically differ from other trade
- Practicals are conducted in the classroom instead of conducting in shop floor.
- Trainees has been taught basic knowledge of geometrical construction, projections, development of solids and sectional views.
- A trainee in this trade has to under go, Allied training in various trades.
- The trades are fitting - turning - machinist - sheet metal - welding - foundry - Electrician - MMV - IC.
- Being a draughts man mechanical he plays an important role in the capital goods and manufacturing.
- He prepares working drawings and even in CAD with required limits-fits and tolerances.
- Hence the trade of D'man mechanical is a vital trade in I.T.I
- It has more job opportunity and even for self-employment
- The following are the tools and equipment used by a draughtsman mechanical in the course of training.

Visit to various section of the ITI

Objectives: At the end of this exercise you shall be able to
- **identify the ITI staffs their designation and their names**
- **list the trades available at your ITI**
- **mention the location of your ITI with respect to railway station/ bus stand and any land mark**
- **record the telephone numbers of ITI office, hospital, police station and fire station**
- **draw the layout of your section.**

Exercise 1: Visit various section of ITI and acquaint with the staff members



Instructor will lead the new recruits to various section of ITI.

- During the visit collect information like the designation of staff member, their name.
- Identify the sections in the ITI and list the trades in which training is given.
- Identify the location of ITI with respect to railway station and bus stand and list of bus route numbers which ply near the ITI.
- Collect the telephone numbers of ITI office, nearest hospital, nearest police station and the nearest fire station.

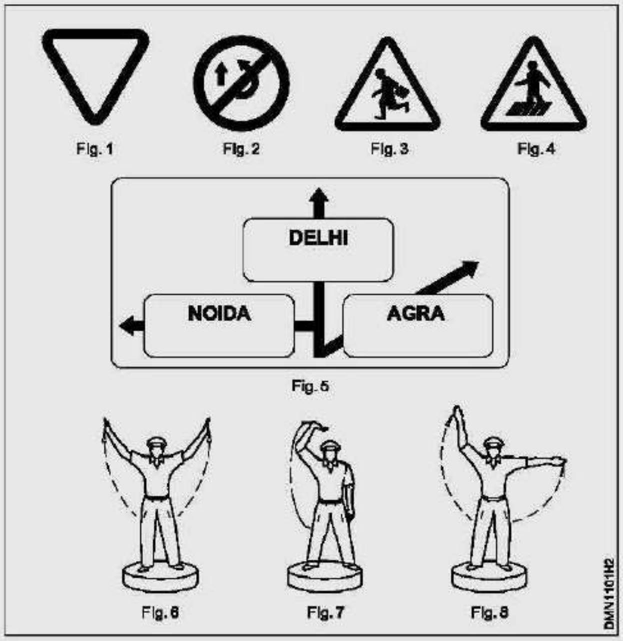
Exercise 2: Draw the layout of your section of ITI

- Draw the plan of your section to a suitable scale in a separate sheet of paper (A4 size).
- Take the length and the breadth measurements of machine foundations, work benches, panels, wiring cubicles, doors, windows, furniture etc.
- Draw the layout of the machines work benches panels and furniture etc. The section plan should be same scale as in step1 as per the actual placement of the machine foundation, panels, furniture work benches etc.

Exercise 3: Road safety signal / Traffic signals

Instructor will explain all the road safety sign and traffic police signals

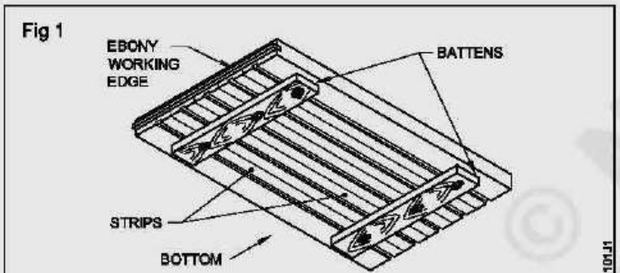
- Read the sign given and mention their kinds and the meaning in the table1.



Get it checked by the instructor.

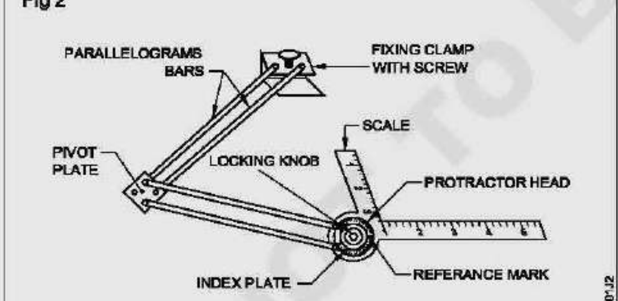
Exercise 4: Drawing equipment and instruments in Table 2

Fig 1



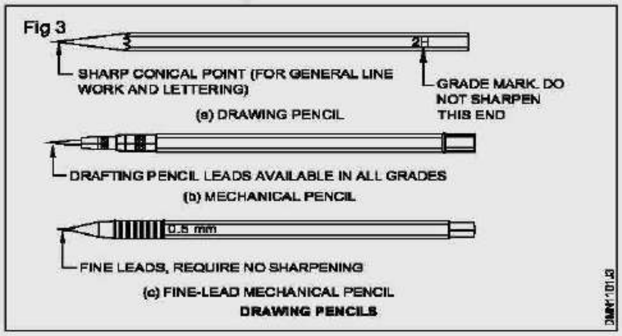
DRAWING BOARD

Fig 2



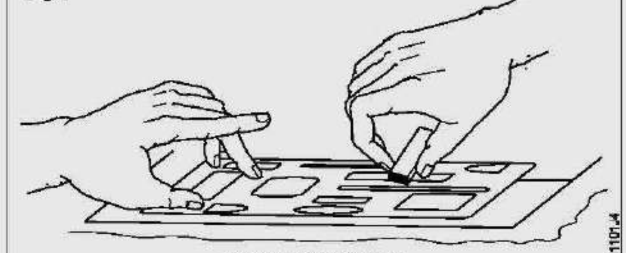
PARTS

Fig 3



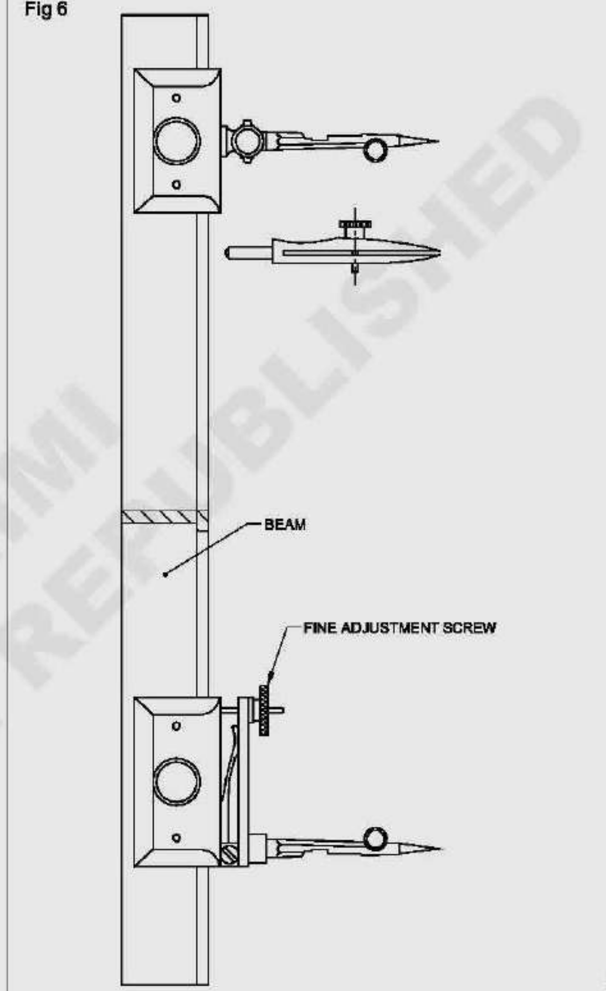
DRAWING PENCILS

Fig 4



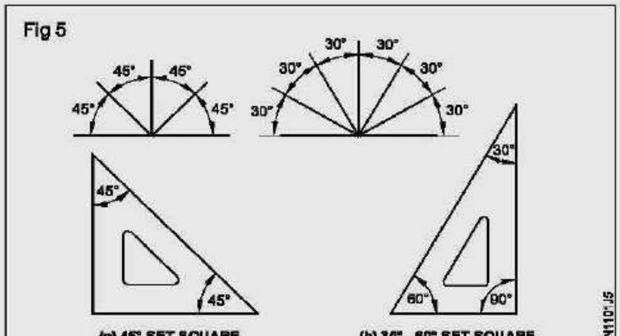
USE OF ERASER SHIELD

Fig 6



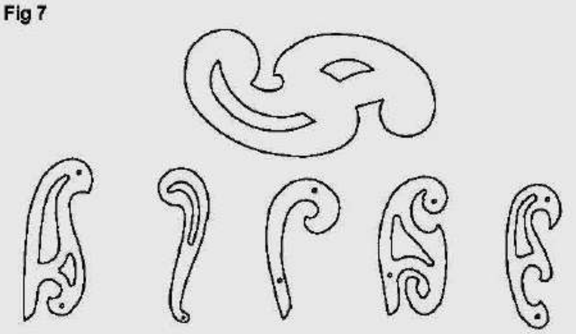
BEAM COMPASS

Fig 5



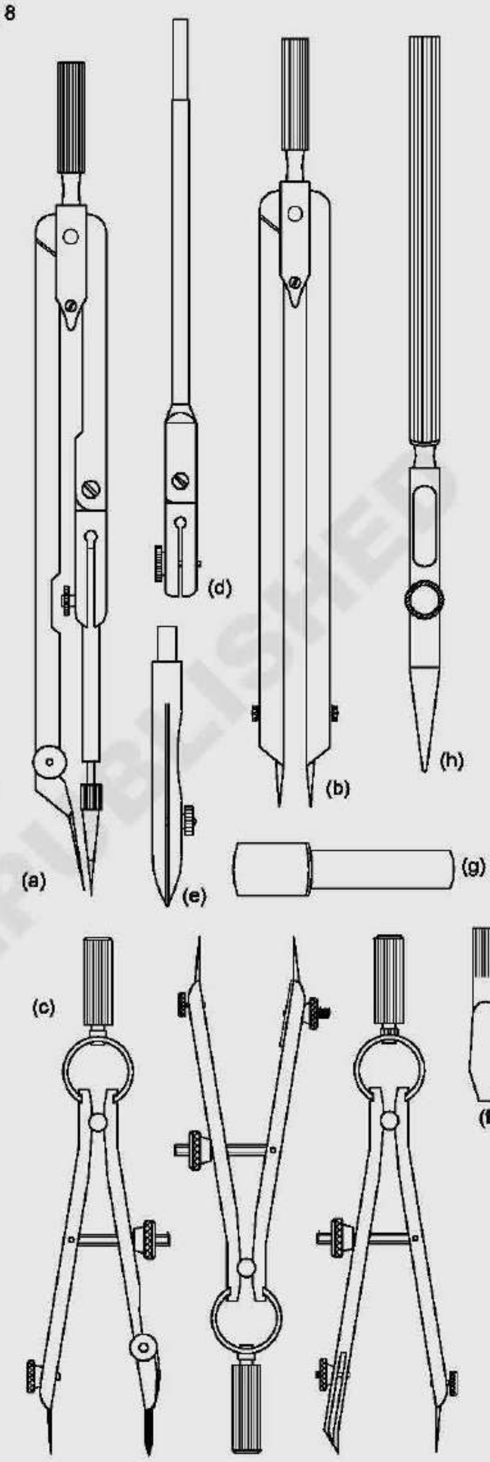
(a) 45° SET SQUARE (b) 30° - 60° SET SQUARE

Fig 7



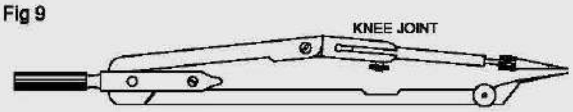
FRENCH CURVES

Fig 8



DRAWING INSTRUMENTS

Fig 9

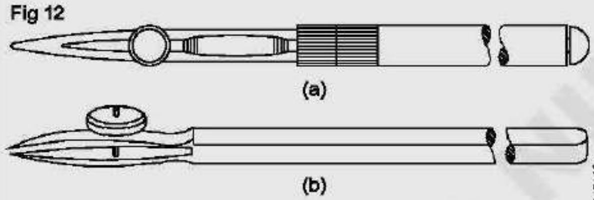


LARGE COMPASS

Fig 10

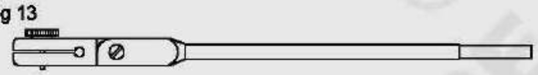
DIVIDER

Fig 12



RULING PENS

Fig 13

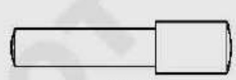


LENGTHENING BAR

Fig 14

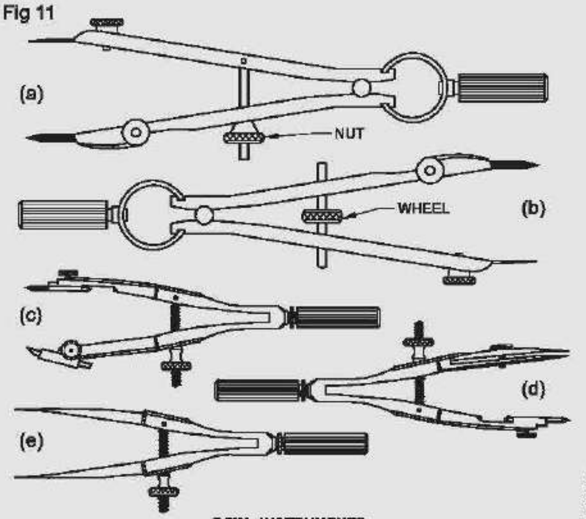
SCREW DRIVER

Fig 15



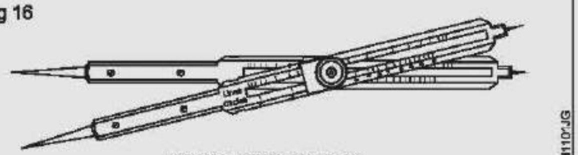
LEAD CASE

Fig 11



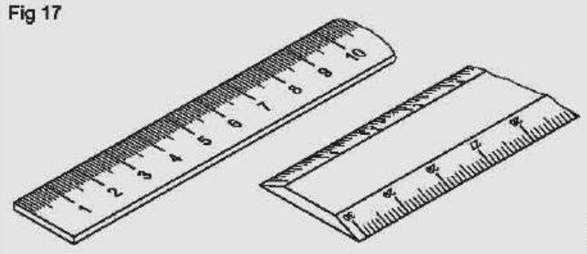
BOW - INSTRUMENTS

Fig 16



PROPORTIONAL DIVIDERS

Fig 17



Note : Instructor should ask the trainees to prepare appropriate table for each topic and fill the details. Guide the trainees to prepare the table.

Safety - Use of personal protection equipment (PPE) and japanese 5s concept

Objectives: At the end of this exercise you shall be able to
- **identify the places/machinery/equipments are to be cleaned**
- **collect the cleaning materials/devices required on cleaning**
- **clean the machines/equipment and devices installed in your section.**

Requirements

Tools / Equipments

- Portable vacuum cleaner/blower - 1 No.

Materials

- Emery sheet 'O' grade - 1 No.
- Dusting cloth - as reqd.
- Dust bin - 3 Nos (labeled)

Exercise 1: Practice on cleanliness

Switch-off all the machinery and equipment before starting the cleaning. Use mask or cover the mouth and nose.

Instructor has to brief the Japanese 5S concept to the trainees before starting the work.

IS.Sort
IIS.Set in order
IIIS.Shine
IVS.Standardise
VS.Sustabin

} **5S concept**

1 Identify the areas/equipment machine have to be cleaned.

2 Keep the movable items at one place and group it.

3 Clean the dust carefully without damaging any part/ connection of the machine / equipment, by using clothes.

4 Use wet dusting cloth to shine the areas to be cleaned/ wired.

5 Remove the rust in any of the parts of equipment or devices by using emery sheet.

Do not remove any lubricants applied to the machine for its function while wiping/cleaning.

Wear the PPE set to avoid accidents

6 Use vacuum cleaner to suck the dust from the places where brush or cloth cannot be reached.

7 Collect the waste materials found in the lab and put it in the dustbin specified for it, as shown in Fig 1.

Dusting and cleaning can be arranged in groups of trainees under the supervision of instructor.

8 Clean the places where water or oil spill over the floor and dusting particles.

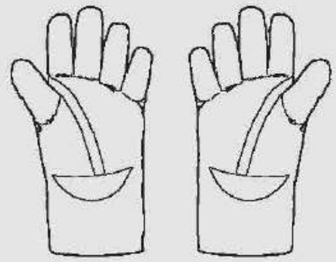
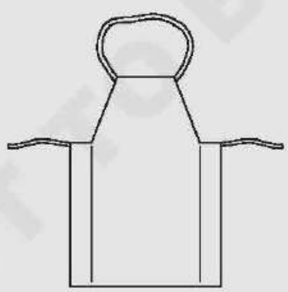
Note down any abnormal things you noticed in particular while doing cleaning and report to the instructor to take action to correct it.

9 Put back all the materials and equipment used for cleaning.

10 Inspect in the presence of instructor and ensure that all the machines are working after cleaning.

11 Discuss with instructor anything you noticed in particular and prepare a report if required to the instructor.

Assign the cleaning work in batch wise daily into the trainees by the instructor in an arranged manner. Dispose the waste as and when required through stores.

Sl. No.	Sketches	Name of PPE	Type of protection	Uses
1	Fig 1  DMN1102H1			
2	Fig 2  DMN1102H2			
3	Fig 3  DMN1102H3			

12 Get it checked by your instructor.

Fig 1

Importance of housekeeping & good shop floor practice

Objectives: At the end of this exercise you shall be able to
- **follow the activities performed for better up keeping of working environment**
- **follow the good shop floor practices.**

Exercise 1: Housekeeping

The following activities to be performed for better up keep of working environment.

1 **Cleaning of shop floor:** Keep clean and free from accumulation of dirt and refuse daily.

2 **Cleaning of machines:** Reduce accidents to keep machines cleaned well.

3 **Prevention of leakage and spillage:** Use splash guards in machines and collecting tray.

4 **Disposal of scrap:** Empty scrap, wastage, SWAB from repective containers regularly.

5 **Tool storage:** Use special racks, holders for respective tools.

6 **Storage spaces:** Identify storage areas for respective items. Do not park material in aisle.

7 **Piling methods:** Do not overload platform, floor and keep material at safe height.

8 **Material handling:** Use forklifts, conveyors and hoist.

Good shop floor practices

- Good shop floor practices are motivating action plans for improvement of the manufacturing process.
- All workers are communicated with daily target on manufacturing activities.
- Informative charts are used to post production, quality standards.
- Manufactured parts are inspected to ensure adherence to quality standards.
- Production processes are planned by engineering to minimize product variation.
- 5S Methods are used to organize the shop floor and production lines.
- Workers are trained on plant safety practices in accordance with OSH standards.
- Workers are trained on "root cause" analysis for determining the causes of non - conformances.
- A written preventive maintenance plan for upkeep of plant machinery & equipment.
- Management meets with plant employees regularly to get input on process improvements.
- Process improvement teams are employed to implement "best practices".

Use of personal protective equipment (Occupational Safety)

Objectives: At the end of this exercise you shall be able to
- **read and interpret the different types of Personal Protective Equipment (PPE) from the chart (or) real PPE**
- **identify and name the PPEs for the corresponding type of protection and write their uses.**

Requirements

Tools / Equipments

- Chart showing different types of PPEs - 1 No.
- Real PPEs (available in section) - as reqd.

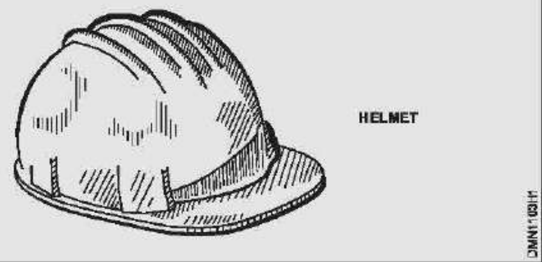
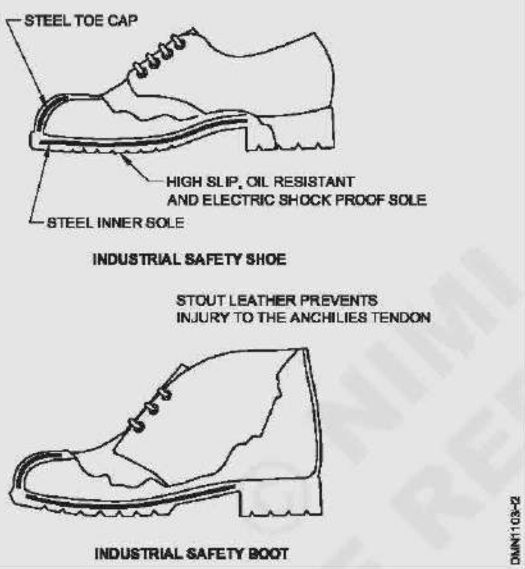
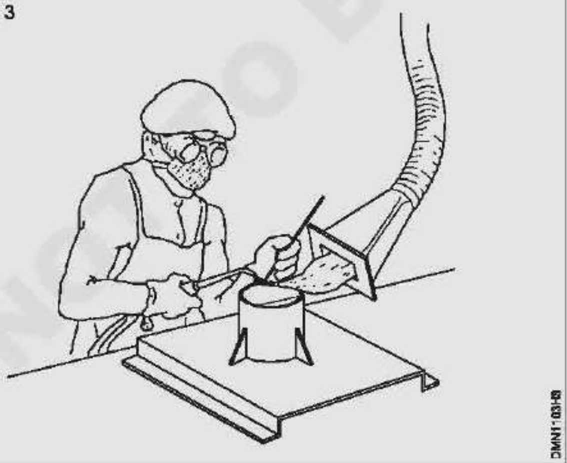
Excercise 1: Identifying and name the PPEs - fill in to table 1

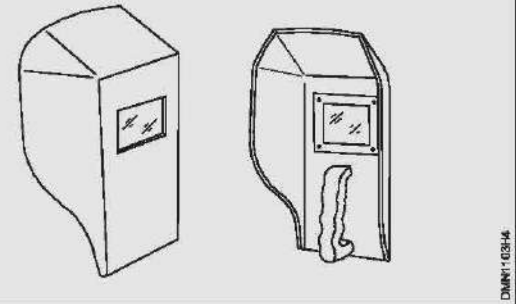
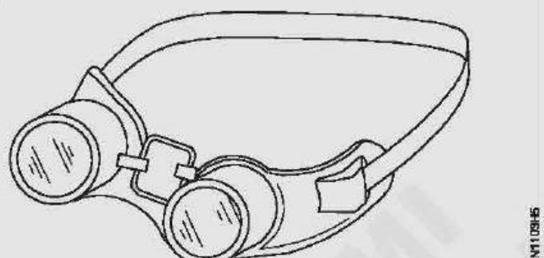
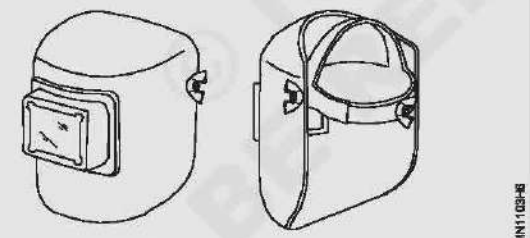
Instructor may arrange the available different types of PPEs in the table (or) provide the chart showing the PPEs. Explain the types of PPEs and their uses for corresponding hazards.

1 Identify the type of PPEs and write their names to the corresponding PPE, by referring from chart (or) read PPEs in Table 1.

2 Write their type of protection and uses in the blank space provided against each PPE in Table 1.

Table 1

Sl. No.	Sketches	Name of PPE	Type of protection	Uses
1	Fig 1 			
2	Fig 2 			
3	Fig 3 			

Sl. No.	Sketches	Name of PPE	Type of protection	Uses
4	Fig 4 			
5	Fig 5 			
6	Fig 6 			

First aid method and basic training

Objective: At the end of this exercise you shall be able to
- **prepare the victim for elementary first aid.**

Requirements

Equipment/Materials

- No. of Persons (Instructor can divide the trainees in suitable No. of groups.) - 20 Nos.
- Control panel arrangement - 1 No.
- Motor - 1 No.
- Rubber mat - 1 No.
- Wooden stick - 1 No.
- 2 persons for demonstration purpose

PROCEDURE

Assumption - For easy manageability, Instructor may arrange the trainees in group and ask each group to perform one method of resuscitation.

TASK 1: **Prepare the victim before giving first aid treatment**

1 Loosen the tight clothing which may interfere with the victim's breathing. (Fig 1)

Fig 1

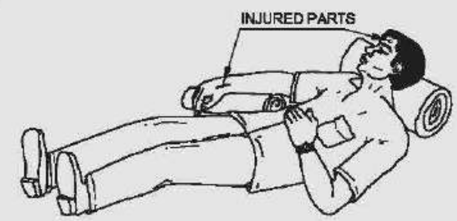
2 Remove any foreign materials or false teeth from his mouth and keep the victim's mouth open. (Fig 2)

Fig 2

3 Bring the victim safely to the level ground, taking necessary safety measures. (Fig 3)

Do not waste too much time in loosening the clothes or trying to open the tightly closed mouth.

Fig 3



4 Avoid violent operations to prevent injury to the internal parts of the victim.

———————

TASK 2: **Prepare the victim to receive artificial respiration**

1 If breathing has stopped, apply immediate artificial respiration.

2 Send word for professional assistance. (If no other person is available, you stay with the victim and render help as best as you can)

3 Look for visible injury in the body and decide on the suitable method of artificial respiration.

4 Have you observed ? (In this case you are told by the instructor)

5 In the case of injury/burns to chest and/or belly follow the mouth to mouth method.

6 In case the mouth is closed tightly, use Schafer's or Holgen–Nelson method.

7 In the case of burn and injury in the back, follow Nelson's method.

8 Arrange the victim in the correct position for giving artificial respiration.

All action should be taken immediately.

Delay even by a few seconds may be dangerous.

Exercise extreme care to prevent injury to internal organs.

9 Place the mock victim in the recovery position.

10 Cover the victim with coat, sacks or improvise your own method. It helps to keep the victim's body warm.

11 Proceed to perform the suitable artificial respiration method.

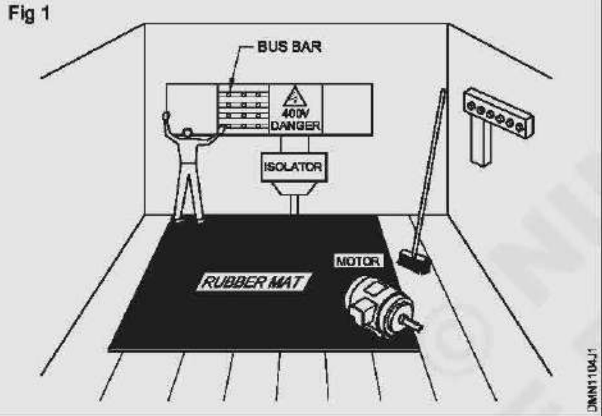
Rescue a person and practice artificial respiration

Objectives: At the end of this exercise you shall be able to
- **disconnect the victim from electric shock**
- **resuscitate the victim by**
  - **nelson's arm - Lift back method**
  - **schafer's method**
  - **mouth to mouth method**
  - **mouth to nose method.**

PROCEDURE

TASK 1 : **Disconnecting a person (mock victim) from a live supply (simulated). (Fig 1)**

Fig 1



1 Observe the person (mock victim) receiving an electric shock. Interpret the situation quickly.

2 Remove the victim safety from the 'live' equipment by disconnecting the supply or using one of the items of insulating material.

Do not run to switch off the supply that is far away.

Do not touch the victim with bare hands until the circuit is made dead or the victim is moved away from the equipment.

Push or pull the victim from the point of contact of the live equipment, without causing serious injury to the victim.

3 Move the victim physically to a nearby place.

4 Check for the victim's natural breathing and consciousness.

5 Take steps to apply respiratory resuscitation if the victim is unconscious and not breathing.

TASK 2 : **Resuscitate the victim by Nelson's arm - Lift back pressure method**

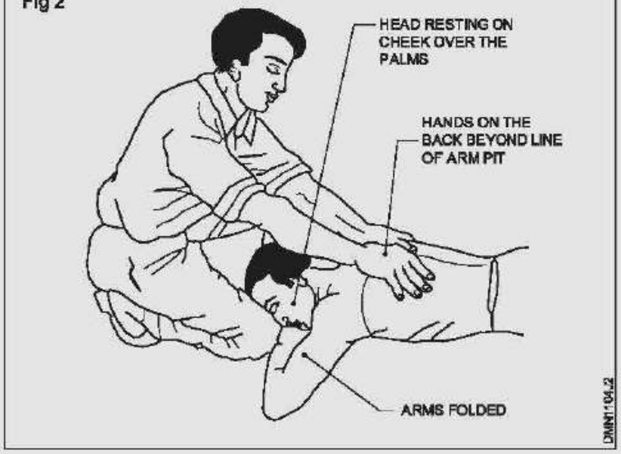
Nelson's arm-lift back pressure method must not be used in case there are injuries to the chest and belly.

1 Place the victim prone (that is face down) with his arms folded with the palms one over the other and the head resting on his cheek over the palms.

2 Kneel on one or both knees near the victim's hand.

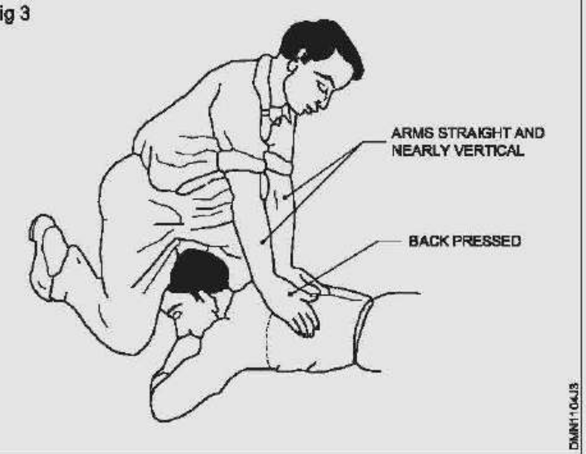
3 Place your hands on the victim's back beyond the line of the armpits, with your fingers spread outwards and downwards, thumbs just touching each other as in Fig 2.

Fig 2



4 Gently rock forward keeping your arms straight until they are nearly vertical, and steadily pressing the victim's back as shown in Fig 3 to force the air out of the victim's lungs.

Fig 3



5 Synchronize the above movement of rocking backwards with your hands sliding downwards along the victim's arms, and grasp his upper arm just above the elbows as shown in Fig 4. Continue to rock backwards.

Fig 4

6 As you rock back, gently raise and pull the victim's arms towards you as shown in Fig 5 until you feel tension in his shoulders. To complete the cycle, lower the victim's arms and move your hands up to the initial position.

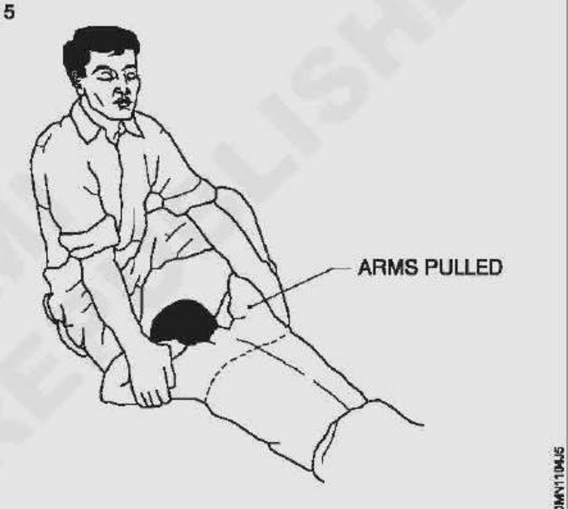
7 Continue artificial respiration till the victim begins to breathe naturally. Please note, in some cases, it may take hours.

8 When the victim revives, keep the victim warm with a blanket, wrapped up with hot water bottles or warm bricks; stimulate circulation by stroking the insides of the arms and legs towards the heart.

9 Keep him in the lying down position and do not let him exert himself.

Do not give him any stimulant until he is fully conscious.

Fig 5



TASK 3 : **Resuscitate the victim by Schafer's method**

Do not use this method in case of injuries to victim on the chest and belly.

1 Lay the victim on his belly, one arm extended direct forward, the other arm bent at the elbow and with the face turned sideward and resting on the hand or forearm as shown in Fig 6.

2 Kneel astride the victim, so that his thighs are between your knees and with your fingers and thumbs positioned as in Fig 6.

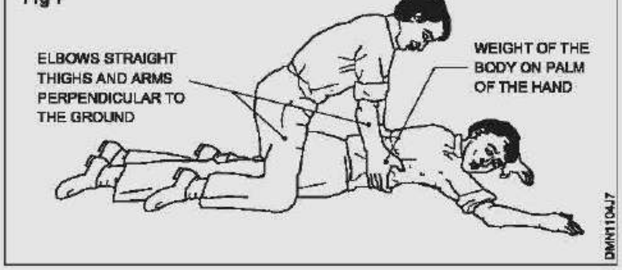
3 With the arms held straight, swing forward slowly so that the weight of your body is gradually brought to bear upon the lower ribs of the victim to force the air out of the victim's lungs as shown in Fig 7.

Fig 6

4 Now swing backward immediately removing all the pressure from the victim's body as shown in Fig 8, thereby, allowing the lungs to fill with air.

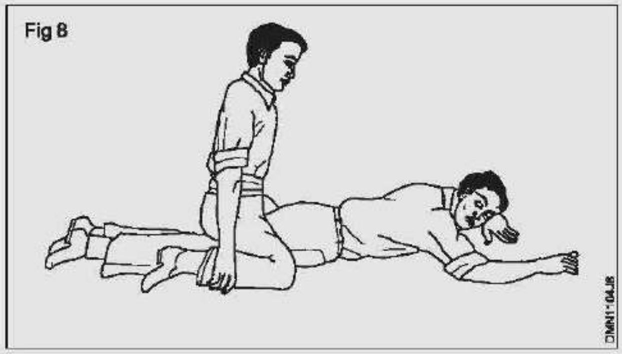
5 After two seconds, swing forward again and repeat the cycle twelve to fifteen times a minute.

Fig 7



6 Continue artificial respiration till the victim begins to breathe naturally.

Fig 8



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TASK 4 : **Resuscitate the victim by mouth-to-mouth method**

1 Lay the victim flat on his back and place a roll of clothing under his shoulders to ensure that his head is thrown well back. (Fig 9)

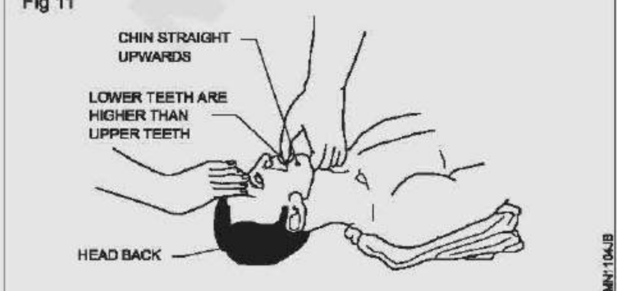
Fig 9

2 Tilt the victim's head back so that the chin points straight upward. (Fig 10)

Fig 10

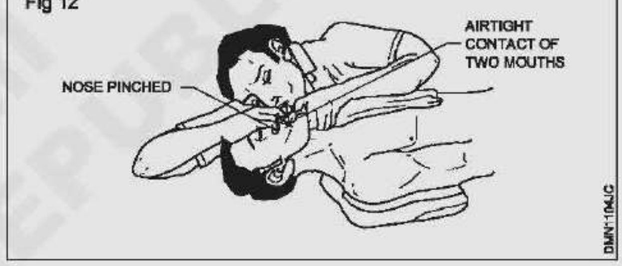
3 Grasp the victim's jaw as shown in Fig 11, and raise it upward until the lower teeth are higher than the upper teeth; or place fingers on both sides of the jaw near the ear lobes and pull upward. Maintain the jaw position throughout the artificial respiration to prevent the tongue from blocking the air passage.

Fig 11



4 Take a deep breath and place your mouth over the victim's mouth as shown in Fig 12 making airtight contact. Pinch the victim's nose shut with the thumb and forefinger. If you dislike direct contact, place a porous cloth between your mouth and the victim's. For an infant, place your mouth over his mouth and nose. (Fig 12)

Fig 12



5 Blow into the victim's mouth (gently in the case of an infant) until his chest rises. Remove your mouth and release the hold on the nose, to let him exhale, turning your head to hear the rushing out of air. The first 8 to 10 breathings should be as rapid as the victim responds, thereafter the rate should be slowed to about 12 times a minute (20 times for an infant).

If air cannot be blown in, check the position of the victim's head and jaw and recheck the mouth for obstructions, then try again more forcefully. If the chest still does not rise, turn the victim's face down and strike his back sharply to dislodge obstructions.

Sometimes air enters the victim's stomach as evidenced by a swelling stomach. Expel the air by gently pressing the stomach during the exhalation period.

— — — — — — —

TASK 5 : **Resuscitate the victim by Mouth-to-Nose method**

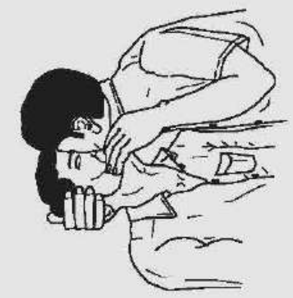
Use this method when the victim's mouth will not open, or has a blockage you cannot clear.

1 Use the fingers of one hand to keep the victim's lips firmly shut, seal your lips around the victim's nostrils and breathe into him. Check to see if the victim's chest is rising and falling. (Fig 13)

2 Repeat this exercise at the rate of 10 - 15 times per minute till the victim responds.

3 Continue this exercise till the arrival of the doctor.

Fig 13



TASK 6 : **Resuscitate a victim who is under cardiac arrest**

In cases where the heart has stopped beating, you must act immediately.

1 Check quickly whether the victim is under cardiac arrest.

Cardiac arrest could be ascertained by the absence of the cardiac pulse in the neck (Fig 14), blue colour around lips and widely dilated pupil of the eyes.

Fig 14

2 Lay the victim on his back on a firm surface.

3 Kneel alongside facing the chest and locate the lower part of the breastbone. (Fig 15)

Fig 15

4 Place the palm of one hand on the center of the lower part of the breastbone, keeping your fingers off the ribs. Cover the palm with your other hand and lock your fingers together as shown in Fig 16.

Fig 16

5 Keeping your arms straight, press sharply down on the lower part of the breastbone; then release the pressure. (Fig 17)

Fig 17

6 Repeat step 5, fifteen times at the rate of at least once per second.

7 Check the cardiac pulse. (Fig 18)

8 Move back to the victim's mouth to give two breaths (mouth-to-mouth resuscitation). (Fig 19)

9 Continue with another 15 compressions of the heart followed by a further two breaths of mouth-to-mouth resuscitation, and so on, check the pulse at frequent intervals.

Fig 18

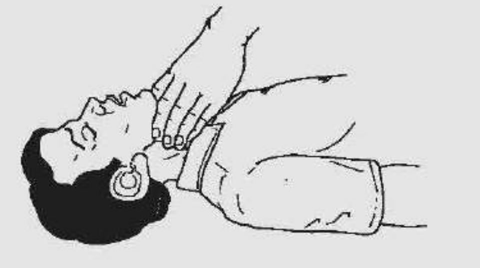


Fig 19

10 As soon as the heartbeat returns, stop the compressions immediately but continue with mouth-to-mouth resuscitation until natural breathing is fully restored.

11 Place the victim in the recovery position as shown in Fig 20. Keep him warm and get medical help quickly.

Other steps

1 Send for a doctor immediately.

2 Keep the victim warm with a blanket, wrapped up with hot water bottles or warm bricks; stimulate circulation by stroking the insides of the arms and legs towards the heart.

Fig 20

Capital Goods and Manufacturing Exercise 1.1.04
Draughtsman Mechanical - Importance of trade training and safety

Disposal of waste materials

Objectives: At the end of this exercise you shall be able to
- **identify the waste material in different category**
- **segregate and arrange the waste materials in it's corresponding bins**
- **dispose non saleable and saleable material separately and maintain record.**

Requirements

Materials

• Shovel	-1 No.	• Trolley with wheels	- 3 Nos
• Plastic/Metal bins	- 4 Nos	• Brush and gloves	- 1 Pair

PROCEDURE

TASK 1: **Disposal of waste materials from the workshop**

1 Collect all the waste materials in workshop.

2 Identify and segregate the different waste like cotton waste. metal chips, all chemical waste and electrical waste etc. (Fig 1) separately and label them.

3 Segregate saleable, non saleable, organic and inorganic materials also.

4 Record the segregated waste material and fill the Table 1.

Fig 1

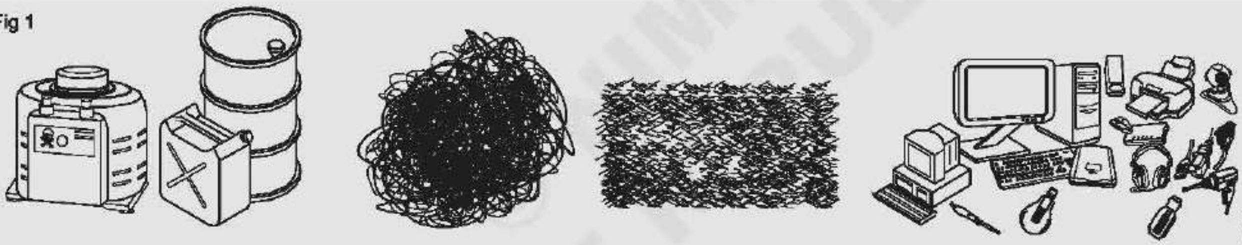


Table 1

Sl.No.	Name of the waste material	Quantity	Saleable or non Saleable
1			
2			
3			
4			
5			
6			

5 Arrange at least 3 trollies with wheel for disposal and stick the lable an each trolly as "Cotton Waste", "Metal chips" and "others" (Fig 2)

6 Put the cotton waste in cotton trolley and similarly put the metal chips waste and others in corresponding trolleys.

7 Keep another 4 bins to collect saleable scarp. non saleable scrap, organic waste and in-organic waste and label them. (Fig 3)

Fig 2

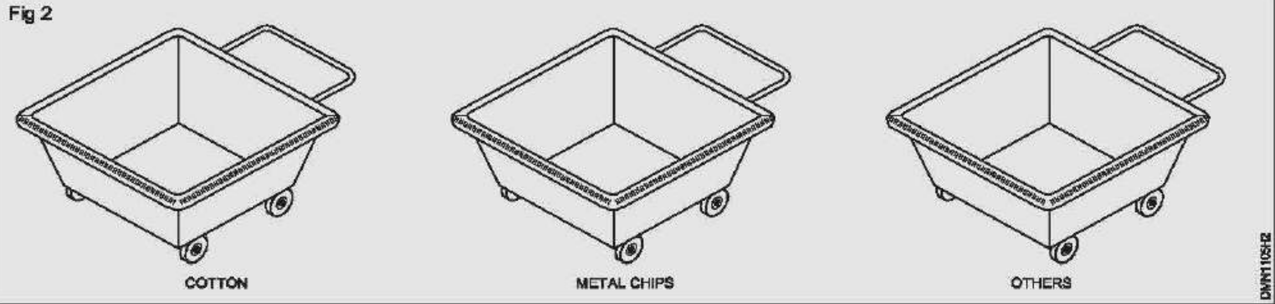


Fig 3

Separate the cotton waste and dispose it

Objective: This shall help you to
- **separate and dispose the cotton waste.**

TASK 1: **Segregating the waste material**

1 Collect the chips by hand shovel with help of brush.

2 Clean the floor if oil is spill.

> **Do not handle the chip by bare hand there may be different metal chips. So separate the chip according to metal.**

3 Separate the cotton waste material and store in the bin provided to store the waste cotton material.

4 Store the each category similarly of metal chip in separate bins.

> **Each bin have respective label.**

5 Collect all the saleable material metal and non - metal separately and keep it's respective bins.

6 Collect all the non - saleable materials like cotton waste, paper waste, wooden pieces etc. and keep it's respective bin.

7 Check the non - saleable material work (organic) and send it for disposal by burning after getting approval.

8 Check the saleable material and segregate like Aluminium, Copper, Iron, Screws, nuts and other items separately and send to stores for disposal by auction (or) as per recommended procedure with approval.

Hazards identification (symbolic) and avidance

Objectives: At the end of this exercise you shall be able to
- **identify the safety symbols from the chart and their basic category**
- **write their meaning and description and the place of use**
- **identify the road safety sign with traffic signal from the chart**
- **read and interpret the different types of occupational hazards from the chart.**

Requirements

Materials

- Basic safety signs chart -1 No.
- Road safety signs and traffic signal chart - 1 No.
- Occupational hazards chart - 1 No.

PROCEDURE

Exercise 1: Identify the safety symbols and interpret their meaning and color with shape

Instructor may provide various safety signs chart for basic categories and road safety with traffic signals. Then explain their categories meaning and color. Ask the trainees to identify the sign and record in table 1.

1 Identify the basic category of each sign from the chart.

2 Write the categories name of the each sign meaning description and the place of use of that safety sign in Table 1.

Table 1

No.	Safety signs	Name of the basic category and sign	Place of use
1	 HOSPITAL		
2	 NO SMOKING		
3	 WEAR HAND PROTECTION		
4	 RISK OF ELECTRIC SHOCK		
5	 DO NOT EXTINGUISH WITH WATER		

No.	Safety signs	Name of the basic category and sign	Place of use
6	 WEAR HEAD PROTECTION		
7	 TOXIC HAZARD		
8	 WEAR EYE PROTECTION		
9	 RISK OF FIRE		
10	 PEDESTRIANS PROHIBITED		
11	 WEAR HEARING PROTECTION		
12	 SMOKING AND NAKED FLAMES PROHIBITED		
13	 DANGER 415V		

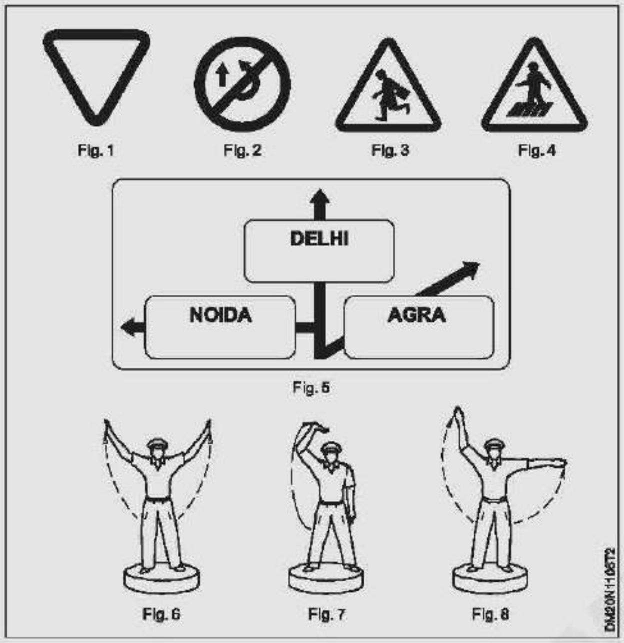
Exercise 2 : **Identify the road safety sign and traffic signals**

Instructor will explain all the road safety sign and traffic police signals.

1 Read the sign given and mention their kinds and the meaning in the table 2.

2 Get it checked by the instructor.

Table 2

No.	Safety signs	Name of the basic category and sign	Place of use
			

Exercise 3: Read and interpret the different types of personal protective devices from the chart

> **Instructor may brief the various types of occupational hazards and their causes.**

1 Identify the occupational hazard to the corresponding situation with a potential harm given in table 3.

2 Fill up and get it checked by your instructor.

Table 3

Sl.No.	Source or potential harm	Type of occupational hazards
1	Noise	
2	Explosive	
3	Virus	
4	Sickness	
5	Smoking	
6	Non control device	
7	No earthing	
8	Poor housekeeping	

Preventive measure for electrical accidents and steps to be taken in such accidents

Objectives: At the end of this exercise you shall be able to
- **practice and follow the preventive safety rules to avoid electrical accident**
- **perform the immediate steps to save the electric shocked victim.**

Requirements

Equipment/Machines
- Fire extinguishers CO_2 - 1 No.

Materials
- Heavy insulated screw driver 200 mm - 1 No.
- Electrical safety chart (or) display - 1 No.
- Gloves - 1 No.
- Rubber mat - 1 No.
- Wooden stool - 1 No.
- Ladder - 1 No.
- Safety belt - 1 No.

PROCEDURE

TASK 1: Practice and follow the preventive safety rules to avoid electrical accident

1 Do not work on live circuits. If unavailable use rubber gloves or rubber mats, etc.

2 Do not touch bare conductors.

3 Stand on a wooden stool or an insulated ladder while repairing live electrical circuits/appliances or replacing fused bulbs.

4 Stand on rubber mats while working, operating switch panels, control gears, etc.

5 Use safety belts always, while working on poles or high rise points.

6 Use wooden or PVC insulated handle screw drivers when working on electrical circuits.

7 Replace (or) remove fuses only after switching off the circuit switches.

8 Open the main switch and make the circuit dead.

9 Do not stretch your hands on any moving part of rotating machine and around moving shafts.

10 Use always earth connection for all electrical appliances along with 3-pin sockets and plugs.

11 Do not connect earthing to the water pipe lines.

12 Do not use water on electrical equipment.

13 Discharge static voltage in HV lines/equipment and capacitors before working on them.

14 Keep the workshop floor clean and tools in good condition.

— — — — — — —

TASK 2: Perform the immediate steps to be taken to solve the shocked victim

1 Proceed with treatment at once without panic emotion.

2 Break the contact either by switching off the power or removing the plug or wrenching the cable free.

3 Remove the victim from contact with the live conductor by using dry non-conducting materials such as wooden bar. (Fig 1 & 2)

Fig 1

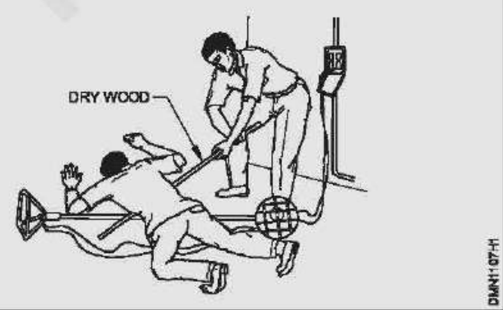
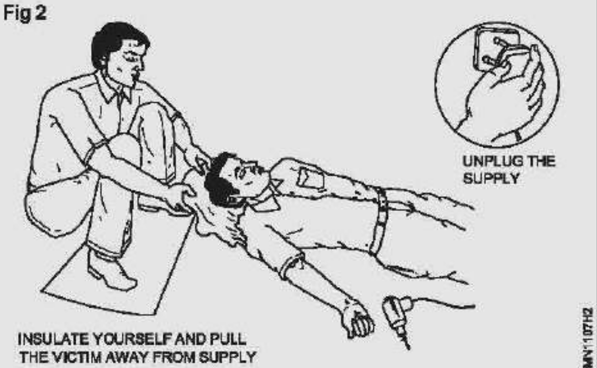


Fig 2



Avoid direct contact with the victim. Wrap your hands in dry material if rubber gloves are not available. If you remain un-insulated, do not touch the victim with your bare hands.

4 Keep the patient warm and at mental rest.

Ensure of good air circulation and comfort. Call for help to shift the patient to safer place. If the victim is alone action to be taken to prevent him from falling.

5 Loosen the clothing about the neck chest and waist and place in recovery position. If the victim is unconscious.

6 Keep the victim warm and comfortable. (Fig 3)

Fig 3

7 Send the person to call doctor, in case of electric burns.

If the victim gets electrical burns due to shock, burns are very painful and dangerous. If a large area of the body is burnt give no treatment. But do the first aid as given below.

8 Cover the burnt area with running pure water.

9 Clean the burnt area by using clean cloth/cotton.

10 Send a person to call the doctor immediately.

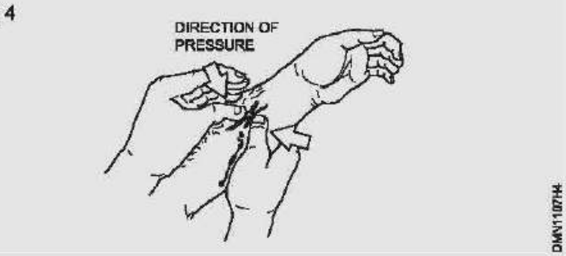
In case of severe bleeding

11 Lay the patient lie down and rest.

12 Raise the injured part above the level of the body. (If possible)

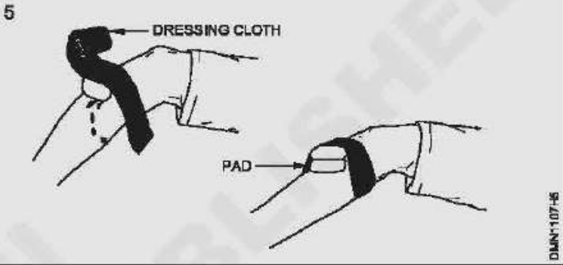
13 Apply pressure on the wound as long as necessary to stop the bleeding. (Fig 4)

Fig 4



14 Apply a clean pad and bandage firmly, if it is large wound. (Fig 5)

Fig 5



If bleeding is severe apply more than one dressing.

14 Proceed to perform the right methods of artificial respiration.

Practice safe methods of fire fighting in case of electrical fire

Objectives: At the end of this exercise you shall be able to
- **demonstrate the ability of fire-fighting for electrical fire**
  - **as a member of the fire-fighting team**
  - **as a leader of the group.**

TASK 1: **General procedure to be adopted in the event of electrical fire**

1 Raise an alarm. Follow the method written below for giving an alarm signals when fire breaks out.

- by raising your voice and shouting Fire! Fire! to call the attention of others
- running towards fire alarm/bell to actuate it
- other means
- switch off the control main switch (if possible)

2 On receipt of the alarm signal:

- stop working
- turn off all machinery and power
- switch off fans/air circulators/exhaust fans. (Better switch off the sub-main)

3 If you are not involved in fighting the fire:

- leave calmly using the emergency exit.
- evacuate the premises
- assemble at a safe place along with the others
- check, if anyone has gone to inform about the fire break to the concerned authority
- close the doors and windows, but do not lock or bolt

As a member of the fire-fighting team

4 If you are involved in fire fighting:

- take instructions for an organised way of fighting the fire.

If taking instructions:

- follow the instructions, and obey, if you can do so safely; do not risk getting trapped.
- do not initiate your own idea.

As a leader of the group

If giving instructions:

- select CO_2 fire extinguisher
- send for sufficient assistance and inform the fire brigade

- locate locally available suitable means to put out the fire
- judge the magnitude of the fire, ensure emergency exit paths are clear of obstructions and then attempt to evacuate (Remove explosive materials, substances that can serve as a ready fuel for fire within the vicinity of the fire break)
- fight out the fire with assistance to put it out, by naming the person responsible for each activity.

5 Report the fire accident and the measures taken to put out the fire, to the authorities concerned.

Reporting all fires however small helps in the investigation of the cause of the fire. It helps to prevent the same kind of accident occurring again.

Use of fire extinguishers

Objectives: At the end of this exercise you shall be able to
- **select the fire extinguisher according to the type of fire**
- **operate the fire extinguisher**
- **extinguish the fire.**

Requirements

Equipment/Machines

- Fire extinguishers CO_2 - 1 No.
- Scissor 100mm - 1 No.
- Cell phone - 1 No.

PROCEDURE

TASK 1: **General Procedure in case of fire**

1 Alert people surrounding by shouting fire, fire, fire when observe fire (Fig 1a & b).

2 Inform fire service or arrange to inform immediately (Fig 1c).

3 Open emergency exist and ask them to go away (Fig 1d).

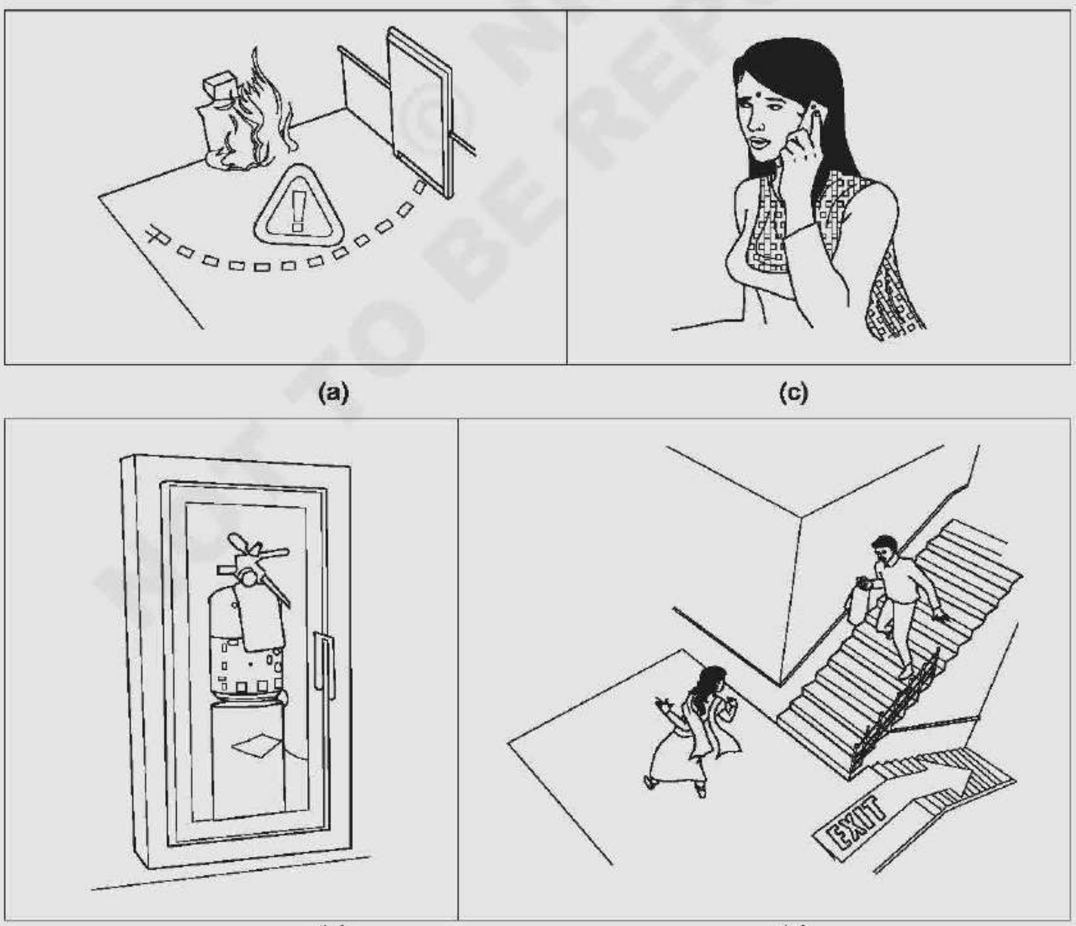
4 Put "Off" electrical power supply.

Do not allow people to go nearer to the fire.

5 Analyze and identify the type of fire. Refer Table1.

6 Assume the fire is D type (Electrical fire).

Fig 1



(a) (c)

(b) (d)

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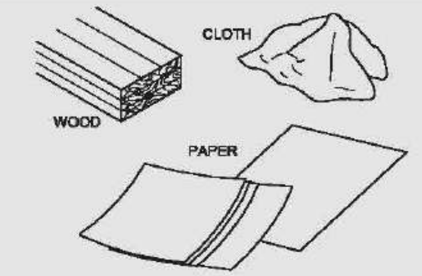
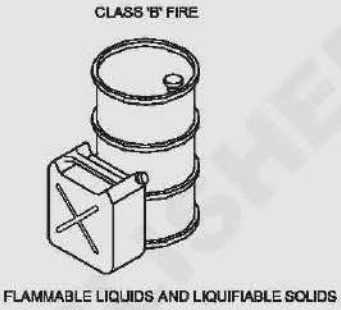
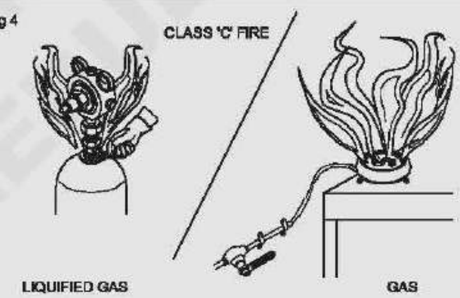
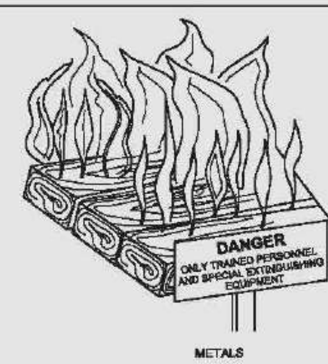
Table 1

Class 'A': Wood, paper, cloth, solid material	Fig 2  DMN1108H2
Class 'B': Oil based fire (grease, gasoline,oil) & liquefiable solids	Fig 3  DMN1108H3
Class 'C' Gas and liquefied gases	Fig 4  DMN1108H4
Class 'D' Metals and electrical equipment	Fig 5  DMN1108H5

7 Select CO_2 (carbon dioxide) fire extinguisher.

8 Locate and pick up CO_2 fire extinguisher. Check for its expiry date.

9 Break the seal. (Fig 6)

10 Pull the safety pin from the handle (Fig 7) (Pin located at the top of the fire extinguisher) (Fig 7)

11 Aim the extinguisher nozzle or hose at the base of the fire (this will remove the source of fuel fire) (Fig 8)

Keep your self low

12 Squeeze the handle lever slowly to discharge the agent (Fig 8)

13 Sweep side to side approximately 15 cm over the fuel fire until the fire is put off. (Fig 9)

Fig 6

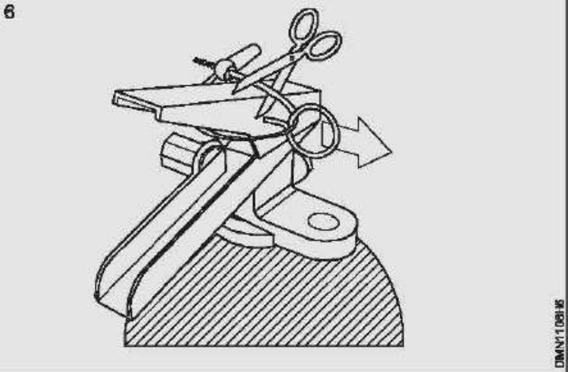


Fig 7

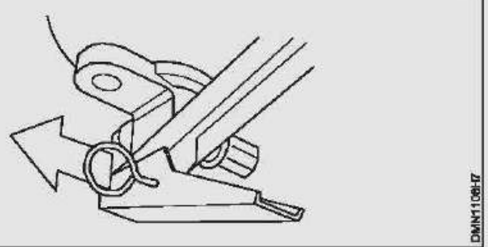


Fig 8

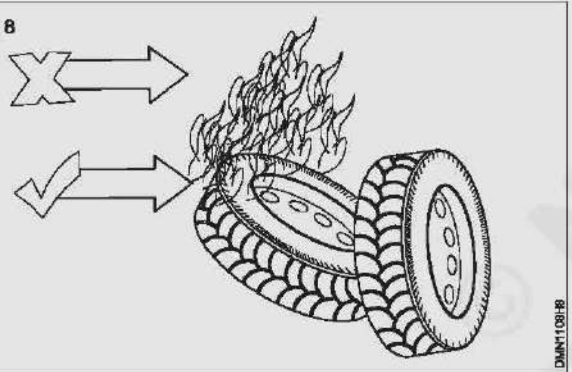
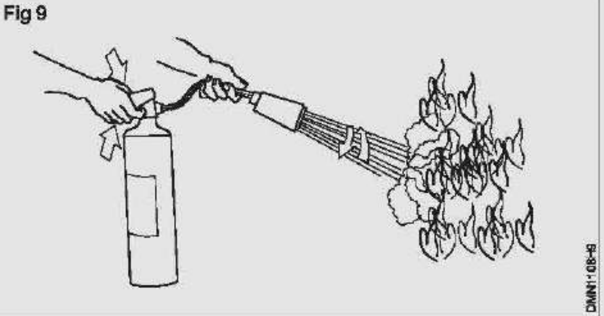


Fig 9



Fire extinguishers are manufactured for use from the distance.

Caution

- While putting off fire, the fire may flare up.
- Do not be panic so long as it put off promptly
- If the fire doesn't respond well after you have used up the fire extinguisher move away your self away from the fire point.
- Do not attempt to put out a fire where it is emitting toxic smoke, leave it to the professionals.
- Remember that your life is more important than property. So don't place yourself or others at risk.

In order to remember the simple operation of fire extinguisher

Remember

P.A.S.S. This will help to use fire extinguisher

P for pull

A for aim

S for squeeze

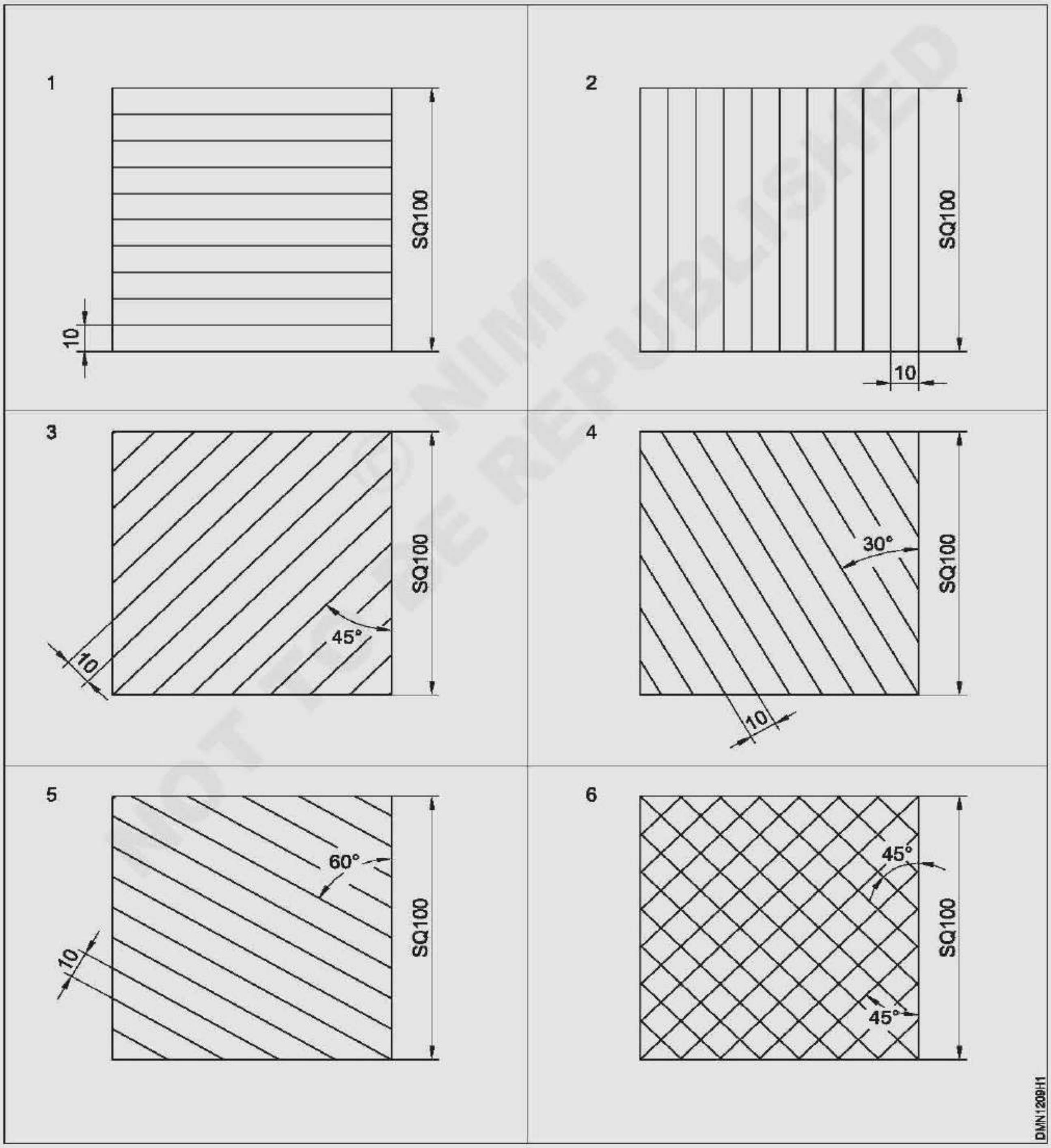
S for sweep

Perform assignment using drawing instruments

Objectives: At the end of this exercise you shall be able to
- **draw figures involving horizontal, vertical and inclined lines**
- **independently using mini drafter, setsquares, scale, divider and protractor.**

PROCEDURE

TASK 1: **Draw the following patterns and components using straight lines**



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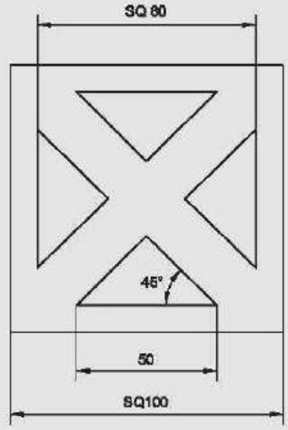

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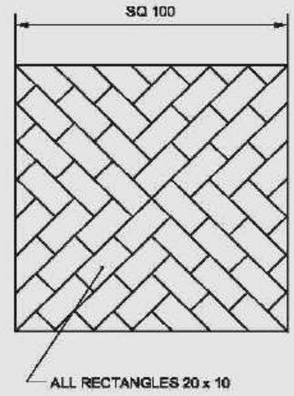

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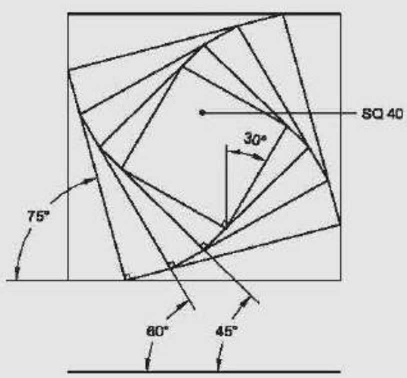

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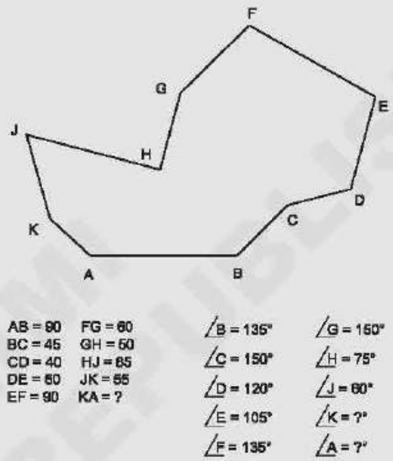

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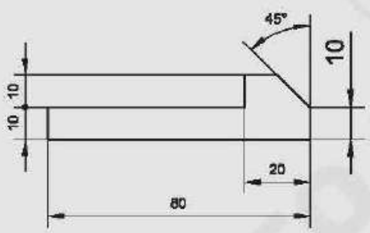

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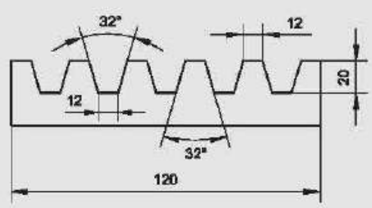

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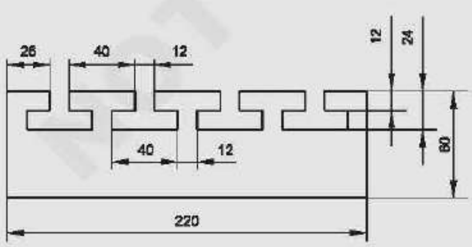

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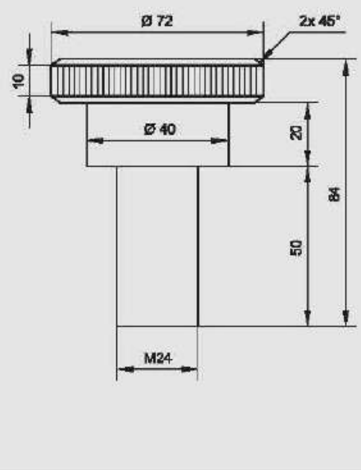

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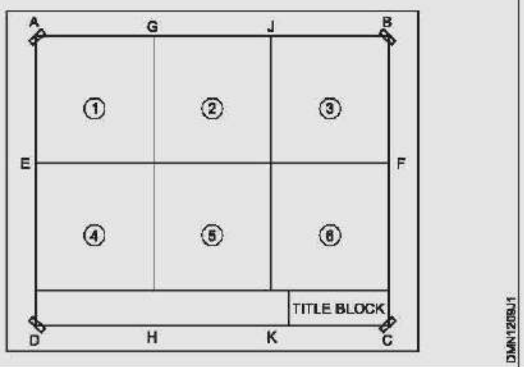
TASK 2: **Layout of drawing sheets**

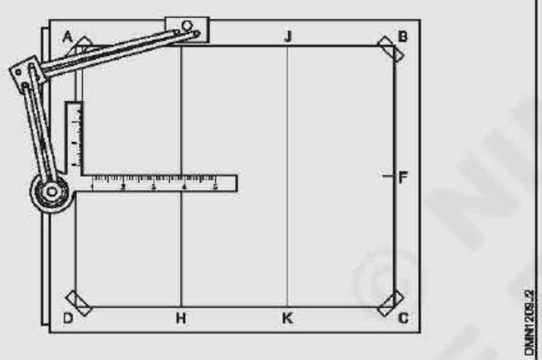
- Layout lines as shown in Fig 1 on an A2 drawing sheet.
- Fix the Minidrafter.

Fig 1



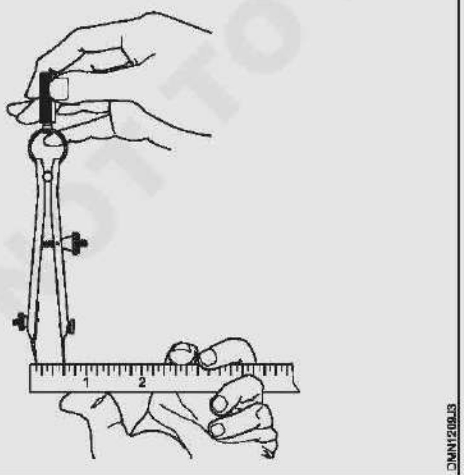
- Draw a horizontal line 100 mm long left to right. (15 mm from AE)
- Draw a vertical line 100 mm long from the left end of the drawing paper as shown in Fig 2.

Fig 2



- Mark of points on the vertical line at 10mm intervals using divider. (Fig 3)

Fig 3



TASK 3: **Horizontal lines**

- Adjust the mini drafter protector to 0°. (Reference mark) & lock using locking knob.
- Hold the pencil approximately at 60° with the paper.
- Draw the horizontal lines from left to right from reference mark giving 10mm gap between them. (Fig 4 &5)

Fig 4

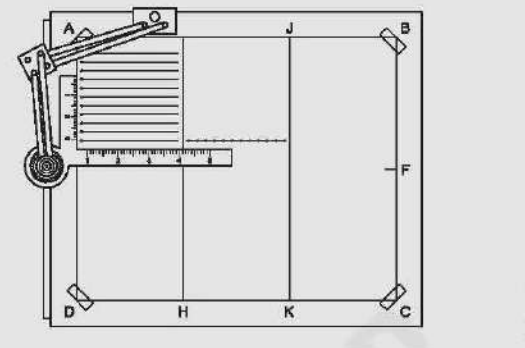
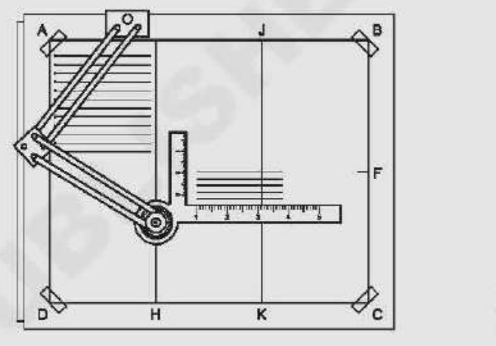


Fig 5



Exercise 4: Vertical lines

- Adjust the Minidrafter protractor head to 0° (Reference mark) & lock using locking knob.
- Hold the pencil approximately at 60° with the paper.
- Draw the horizontal lines from top to bottom from reference mark giving 10mm gap between them. (Figs 6 & 7)

Fig 6

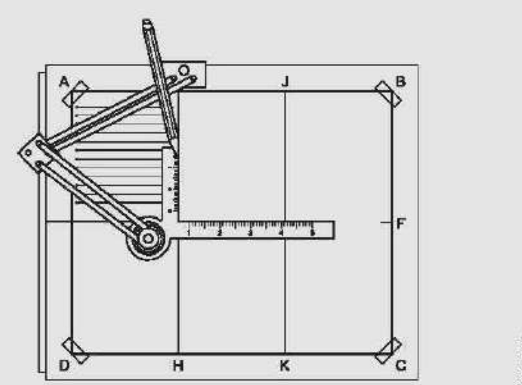
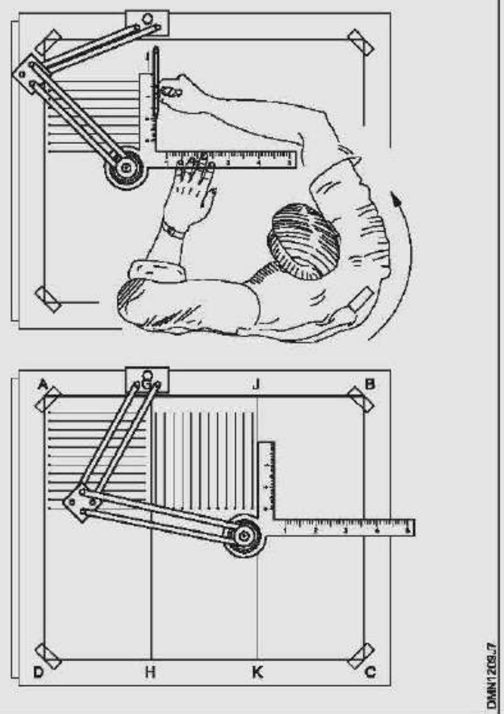


Fig 7



Exercise 5: Inclined lines

For drawing 45° lines.

- Adjust the protector head to 45° from reference mark & lock the using locking knob.
- Draw the 45°. inclined line from reference mark to approximate length. (Figs 8 & 9)

Fig 8

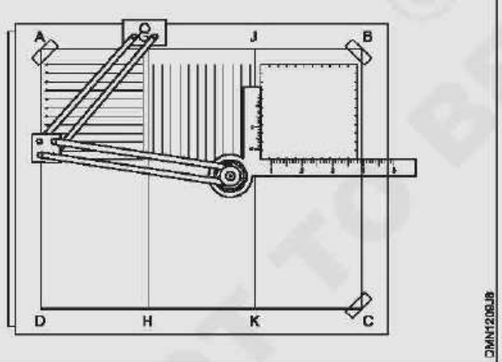
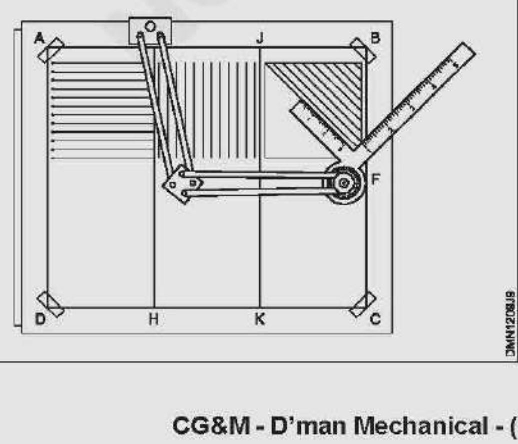
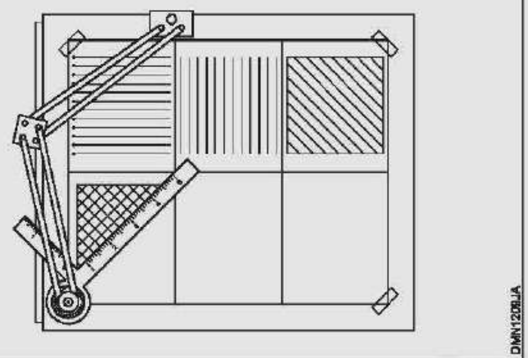


Fig 9



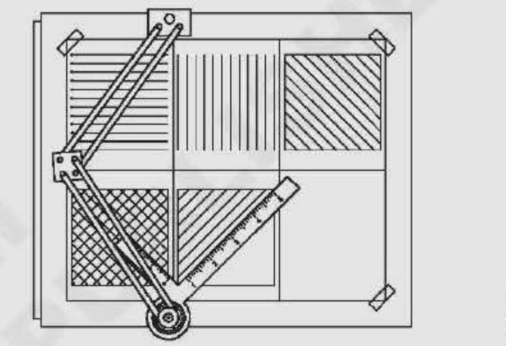
- 30° or/and 60° inclined lines can be drawn using Minidrafter. (Fig 10)

Fig 10



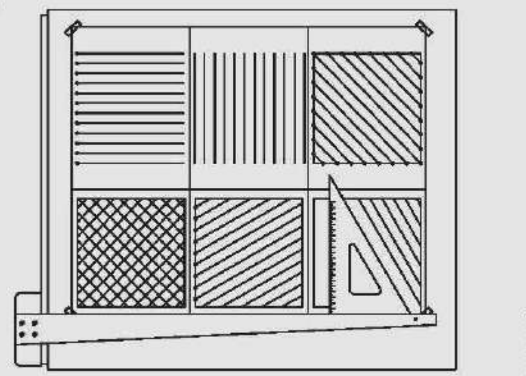
- Draw 30° inclined lines in block 5. (Fig 11)

Fig 11



- In the block 6, draw 60° inclined lines. (Fig 12)

Fig 12



Drawing parallel lines using set squares

Objective: This shall help you to
- **draw parallel lines to a given line through a given point.**

Place any one edge of the setsquare to coincide with the given line.

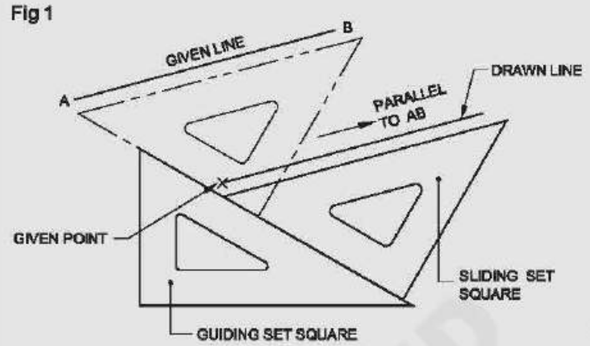
Place the other setsquare (guiding setsquare) with one of its edges butting the first square as shown in Fig 1.

While holding the guiding setsquare firmly, slide the first setsquare (sliding setsquare) till edge touches the given point.

Draw the line along the edge of the sliding setsquare through the given point.

Be sure that the guiding setsquare does not move from its initial position.

Fig 1



TO DRAW A LINE PARALLEL TO A GIVEN LINE

Draw straight lines of given length

Objective: This shall help you to
- **draw a perpendicular to a given line through the given point.**

Method 1 (Fig 1a)

Place one of the perpendicular edges of the setsquare (sliding setsquare) such that in coincides with the given line.

Place the longer edge of the other setsquare (guiding setsquare) against the hypotenuse of the sliding setsquare.

Slide the sliding setsquare till the other edge forming right angle touches the given point.

Through the given point, draw the required perpendicular line along the edge of the sliding setsquare.

Method 2 (Fig 1b)

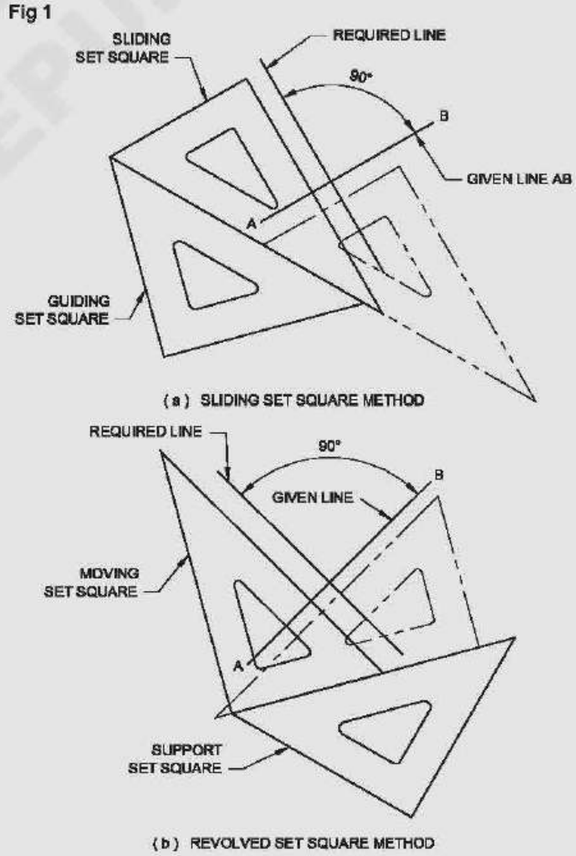
Place the hypotenuse of one setsquare to coincide with the given line.

Place the other setsquare (moving setsquare) with one of its edges butting against one of the perpendicular edges of the moving setsquare as shown in figure.

Holding the supporting setsquare firmly, revolve the moving setsquare and place it on the supporting setsquare such that the hypotenuse of the setsquare passes through the given point.

Draw the required perpendicular line as shown in Fig 1b.

Fig 1



TO DRAW A LINE PERPENDICULAR TO ANOTHER LINE

Drawing perpendiculars lines, inclined lines (given angle) - parallel lines

Objectives: At the end of this exercise you shall be able to
- **draw horizontal and vertical lines of given length with given interval**
- **draw inclined lines at given angle.**

PROCEDURE

Exercise 1: Draw six horizontal parallel lines of 50mm long with 10 mm intervals. (Fig.1)

- Draw a vertical line AB 50 mm long, using setsquare on left side.

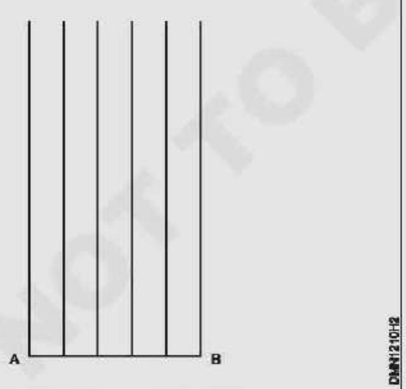
Fig 1

- Mark points on the vertical line AB with 10 mm intervals. Butt a setsquare on the line AB.
- Using another setsquare, draw parallel lines through the points marked.

> **Use sharpened conical point pencil**
>
> **Keep the pencil slightly inclined towards the direction of the movement.**
>
> **While drawing rotate the pencil to keep the constant thickness**
>
> **Maintain uniform pressure on the lead of the pencil**

Exercise 2: Draw six vertical parallel lines of 50mm length with 10mm intervals. (Fig 2)

Fig 2

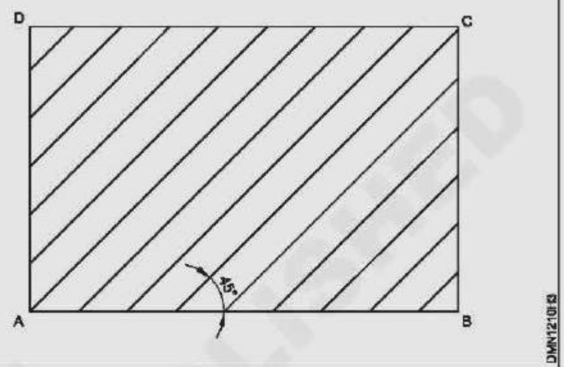


- Draw a horizontal line AB 50 mm long.
- Mark the points with 10 mm intervals.
- Butt a setsquare on the line AB.
- Using another setsquare draw vertical parallel lines from left to right.

> **Draw the vertical lines from bottom to top.**

Exercise 3: Draw 45° inclined lines. (Fig 3)

Fig 3



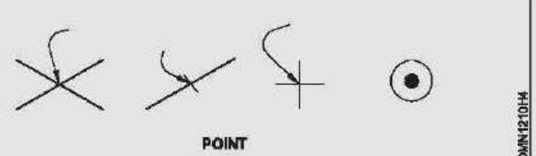
- Draw a horizontal line AB 60 mm long.
- Butt a setsquare on the line AB, draw vertical lines from the points A and B using another setsquare.
- Set off AD and BC equals to 40 mm and complete the box.
- On lines AB and DC mark 10 mm points.
- Butting the 60° setsquare on the line AB, using 45° setsquare draw inclined parallel lines through the marked points.

> **Draw lines from bottom to top.**

Types of lines and angles

A point represents a location in space, having no width or height. It is represented by drawing intersection of lines or a dot. (Fig 4)

Fig 4



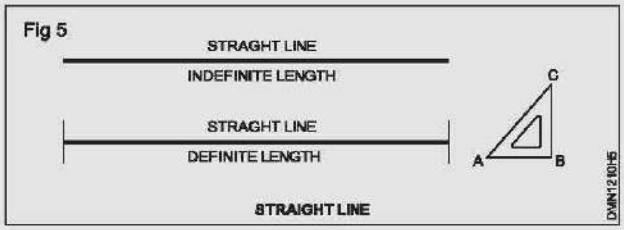
Line is the path of a point when it moves. It has no thickness and are of two types:

– Straight line

– Curved line

Straight line: It is the path of a point when it is moving in a particular direction. It has only length and no width. (Fig.5) Also a straight line is the shortest distance between two points. Straight line, depending on its orientation are classified as Horizontal, Vertical and Inclined or Oblique line.

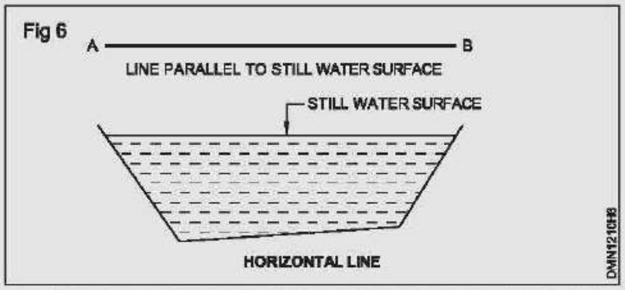
Fig 5



STRAIGHT LINE

Horizontal line (Fig 6: Horizontal lines are those which are parallel to a horizontal plane. Example of horizontal plane is the surface of a still water. (Fig 6)

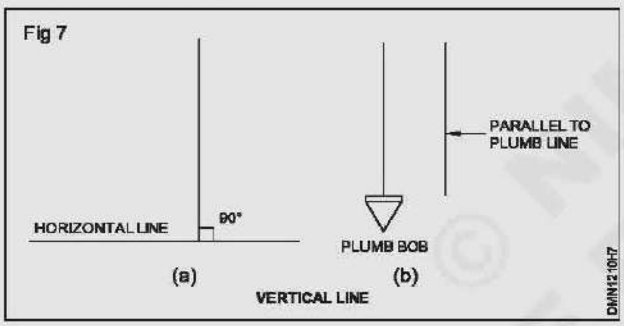
Fig 6



HORIZONTAL LINE

Vertical line (Fig 7a): Lines which are perpendicular to horizontal lines are called vertical lines. It can be treated as a line along the plumb line of the plumb bob or parallel to a plumb line. (Fig 7b)

Fig 7



VERTICAL LINE

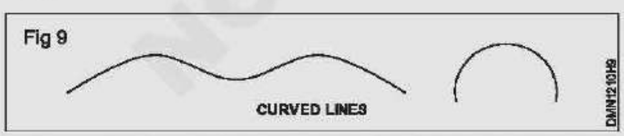
Inclined line or Oblique line: A straight line which is neither horizontal nor vertical is called an inclined line. (Fig 8)

Fig 8

INCLINED OR OBLIQUE LINE

Curved line: It is the path of a point which always changes its direction. Examples of curved lines are shown in Fig 9.

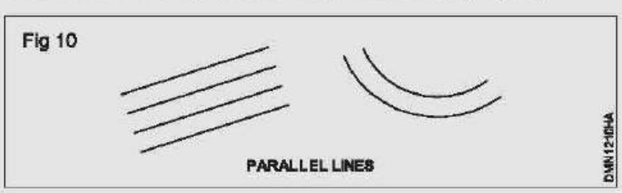
Fig 9



CURVED LINES

Parallel lines: They are the lines with same distance between them. They may be straight lines or curved lines. Parallel lines do not meet when extended. (Fig 10)

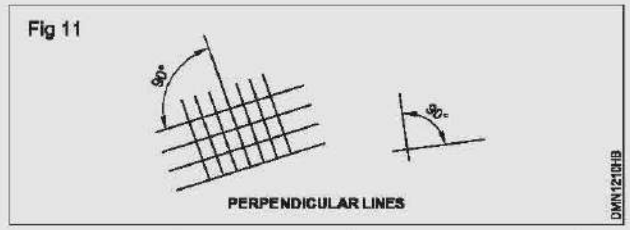
Fig 10



PARALLEL LINES

Perpendicular lines: When two lines meet at 90°, the two lines are said to be perpendicular to each other. One of this lines is called as reference line. (Fig 11)

Fig 11

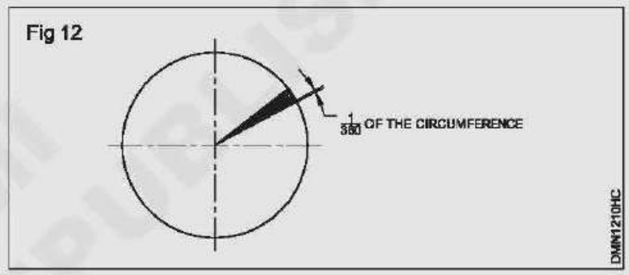


PERPENDICULAR LINES

Angles: Angle is the inclination between two straight lines meeting at a point or meet when extended. AB and BC are two straight lines meeting at B. The inclination between them is called an angle. The angle is expressed in degrees or radians.

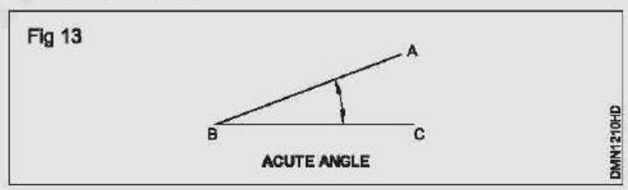
Concept of a degree: When the circumference of a circle is divided into 360 equal parts and radial lines are drawn through these points, the inclination between the two adjacent radial lines is defined as one degree. Thus a circle is said to contain 360°. (Fig 12)

Fig 12



Acute angle: If an angle which is less than 90° is called an acute angle. (Fig 13)

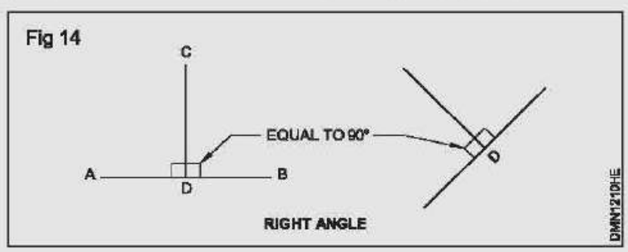
Fig 13



ACUTE ANGLE

Right angle: Angle between a reference line and a perpendicular line is called right angle. (Fig 14)

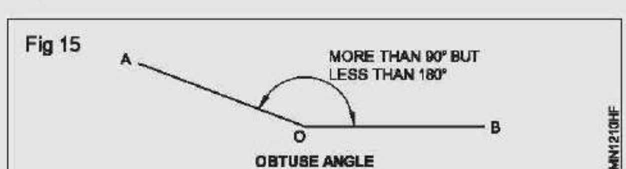
Fig 14



RIGHT ANGLE

Obtuse angle: This refer to an angle between 90° to 180°. (Fig 15)

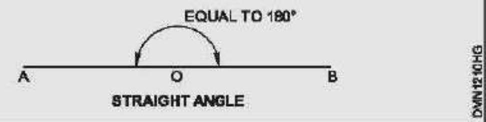
Fig 15



OBTUSE ANGLE

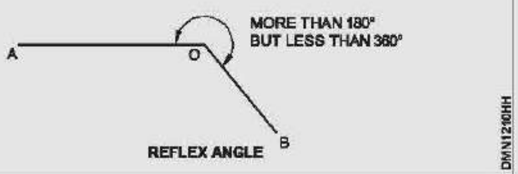
Straight angle: This refers to an angle of 180°. This is also called as the angle of a straight line. (Fig 16)

Fig 16



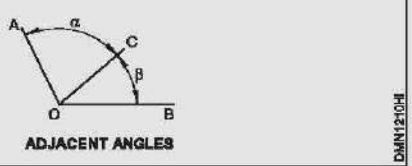
Reflex angle: It is the angle which is more than 180°. (Fig 17)

Fig 17



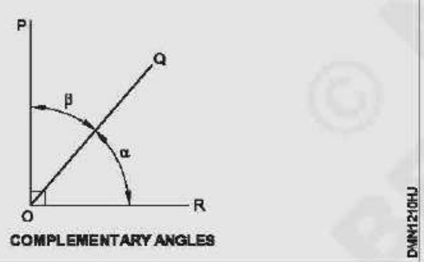
Adjacent angles: These are the angles lying on either side of a line. (Fig 18)

Fig 18



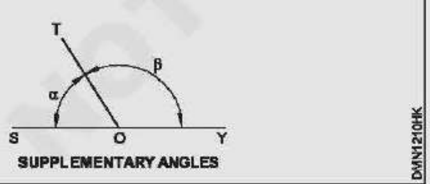
Complementary angles: When the sum of the two angles is equal to 90°, angle POQ + angle QOR = 90° angle POQ and angle QOR are complementary angles to each other. (Fig 19)

Fig 19



Supplementary angle: When the sum of the two adjacent angles is equal to 180°, example angle SOT + angle TOY = 180°, angle SOT and angle TOY are supplementary angles to each other. (Fig 20)

Fig 20



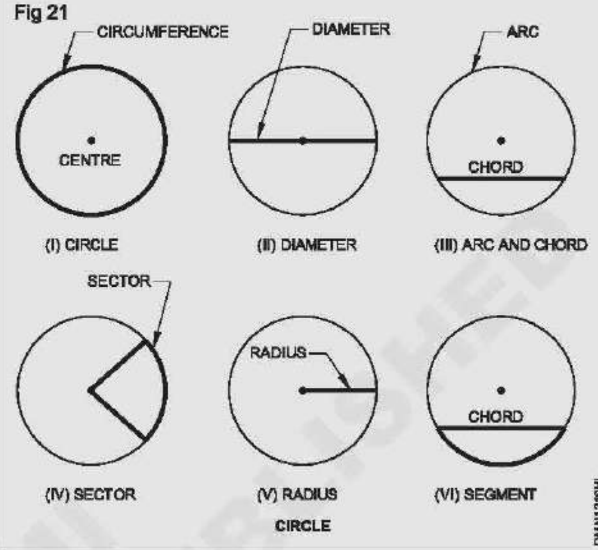
Circle

A moving point, which remains at equal distance from a fixed point called centre forms a circle. (Fig 21)

Some definitions

1. Chord — A line formed by joining two points on the circumference of a circle, is called chord. (Fig 21 iii)

Fig 21



2. Centre — A point in the middle of a circle is called centre. A circle is drawn by placing pin leg of a compass at the centre point. (Fig 21i)
3. Diameter — A line formed by joining two points on the circle and passing through the centre point of the circle is called diameter. (Fig 21ii)
4. Radius — Any line from the centre of the circle and meeting any point on the circumference of 'the circle, is called radius. (Fig 21v)
5. Segment — A line passing through any point on the circumference of a circle is called tangent.(Fig 21vi)
6. Sector - it is a part a circle bounded by two radis (plural of radius) meeting at an angle and an arc.

Drawing triangles of different types and a parallelogram with a given length and angle

Objectives: At the end of this exercise you shall be able to
- **construct scalene triangles**
- **construct right angled triangles**
- **construct isosceles triangles**
- **construct equilateral triangles**
- **construct quadrilateral and parallelograms.**

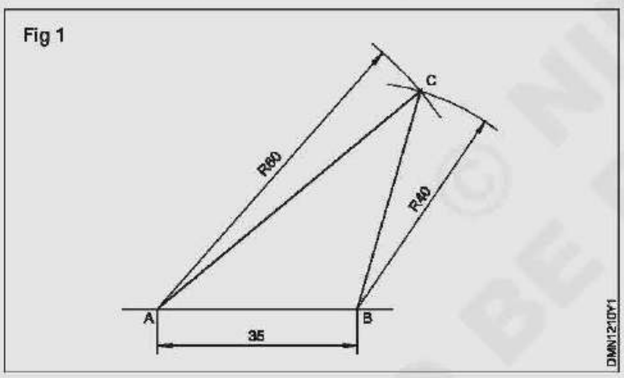
1 Scalene triangle: AB = 35 mm; BC = 60 mm & CA = 40 mm

- Draw base AB = 35
- 'A' as centre draw an arc of radius 60.
- 'B' as centre draw an arc of 40 cutting the previous arc at 'C'.
- Join CA and CB

AB = 35, BC = 40 and AC = 60

ABC is the required triangle. (Fig 1)

Fig 1

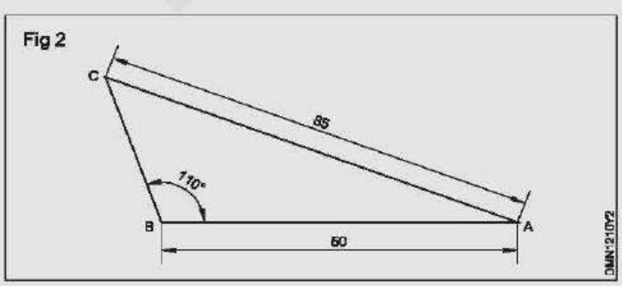


2 Scalene triangle: AB = 50mm; AC = 85 mm & $\angle ABC = 110^\circ$

- Draw AB = 50 mm
- Set an angle 110° at B
- 'A' as centre and with radius AC = 85 mm, draw an arc cutting line at C. Join CB and CA.

ABC is the required triangle. (Fig 2)

Fig 2

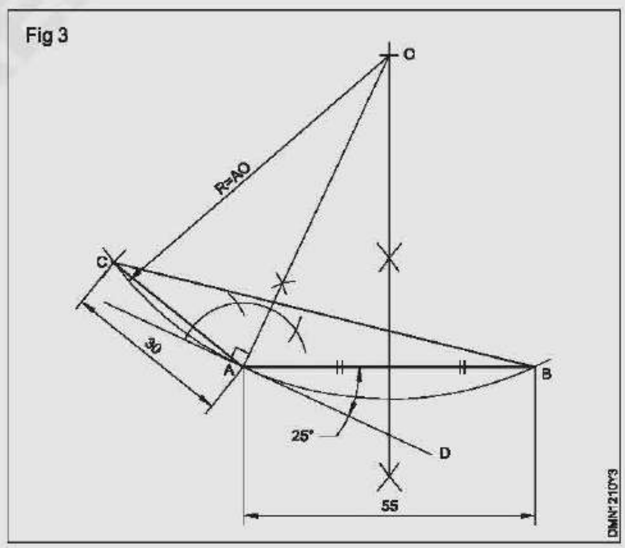


3 Scalene triangle: AB = 55 mm; AC = 30 mm & $\angle BCA = 25^\circ$

- Draw base AB (55 mm) and perpendicular from its mid-point.
- Set/draw the given angle 25° such that angle BAD = 25° (Angle C)
- Erect perpendicular to AD at A.
- Extend both perpendiculars to meet at '0'.
- AO as radius and '0' as centre, draw a circle or an arc.
- Side AC (30 mm) as radius and A as centre, draw an arc cutting the previous arc at C.
- Join CA and CB.

ABC is the required triangle. (Fig 3)

Fig 3

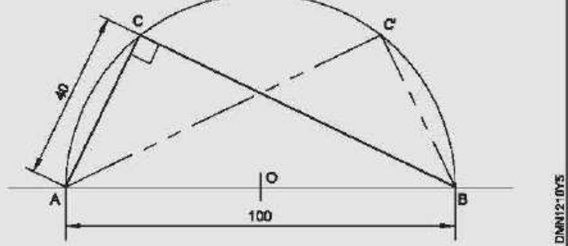


4 Right angled triangle: AB = 100 mm; AC = 40 mm

- Draw a straight line AB of given length 100 mm (hypotenuse).
- Bisect AB and mark centre of AB as '0'.
- AO or OB as radius, draw a semi-circle on AB.
- With centre either A or B draw an arc of radius equal to the side (40 mm) cutting the semi-circle at C.
- Mark the point C and join it to A and B.

ABC is the required right angled triangle. (Fig 4)

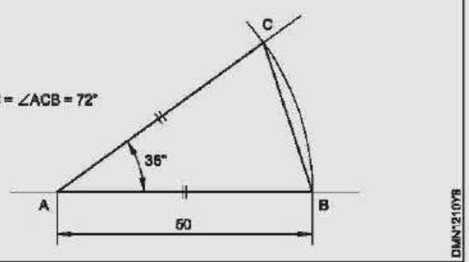
Fig 5



5 Isosceles triangle: AB = AC = 50 mm & $\angle BAC = 36°$

- Draw the side AB equal to 50 mm. A as centre, draw an arc of radius AB.
- Draw a line AC at 36° to BA.
- Join BC to form the triangle ABC. (Fig 6)

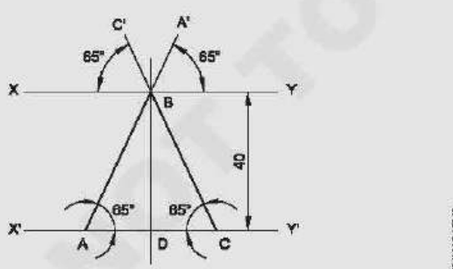
Fig B



6 Isosceles triangle (Fig 6): Altitude = 40 mm & $\angle BCA = \angle BAC = 65°$

- Draw any line X'Y' and erect a perpendicular DB of height 40 mm at a convenient point D.
- Draw a parallel line XY to X'Y' through B.
- Draw A'B at 65° to XY and extend to meet at A on line X'Y'.
- Erect another point C on line X'Y' same way as in the previous step and complete the triangle ABC.

Fig 10

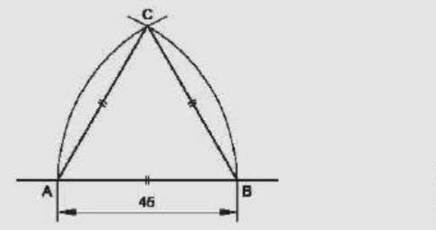


7 Equilateral triangle (Fig 7): AB = BC = CA = 45 mm

- Draw a line and mark AB 45 mm the side of triangle.
- With radius AB and centre A and B, draw arcs cutting at C.
- Join CA and CB.

ABC is a required triangle.

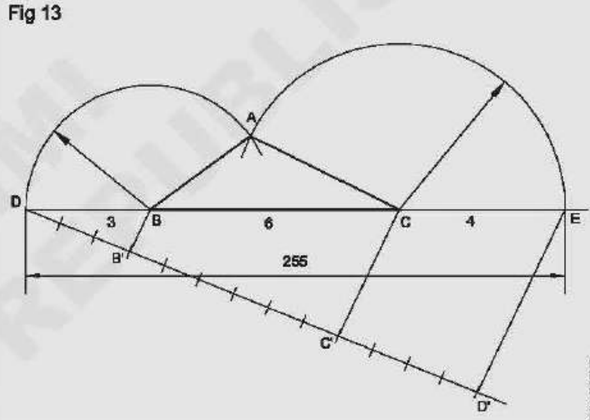
Fig 12



8 Scalene triangle (Fig 8): AB + BC + CA = 255 mm AB:BC:CA = 3:6:4

- Draw a line DE equal to 255 and divide it into 13 equal parts. (sum of the ratio of sides)
- Mark the points B and C at 3[rd] and 9[th] divisions.
- B as centre, BD as radius, draw an arc.
- Similarly C as centre, CE as radius, draw another arc.
- Both arcs meet at A.
- Join AB and AC and complete the triangle.

Fig 13



Problems for further practice

Exercise

1 Construct a right angled triangle of hypotenuse 125 mm and one side 75 mm. Measure the other side.

2 Construct a right angled isosceles triangle of side(s) 80 mm. Find hypotenuse.

3 Construct a right angled triangle of adjacent sides 90 mm and 60 mm. Measure the hypotenuse.

4 Construct a right angled triangle of sides 75 mm and 48 mm.

5 Construct a right angled triangle whose hypotenuse is 100 mm and the sides are equal.

6 Construct a right angled triangle of hypotenuse 75 mm and one angle is 30°.

7 Construct an isosceles triangle of sides 80 mm and the angle between them is 50°.

8 Construct an isosceles triangle of one side 70 mm and the other sides are 90 mm each.

9 Construct an isosceles triangle of base angles 55° and the altitude 90 mm.

10 Construct an isosceles triangle of base angles 75° and altitude is 110 mm.

11 Construct an isosceles triangle of perimeter 150 mm and altitude 65 mm.

12 Construct an isosceles triangle of perimeter 120 mm and altitude 40 mm.

13 Construct a equilateral triangle of side 100 mm using protractor.

14 Construct an equilateral triangle of side 90 mm using compass.

15 Construct an equilateral triangle of perimeter of 240 mm.

16 Construct a triangle of adjacent sides 80 mm and 55 mm and angle between them is 55°.

17 Construct a triangle of sides 90, 75 & 60 mm. Measure the angle opposite to them.

18 Construct a triangle of one side 80 mm and the adjacent angles are 75° and 45°. Measure the other sides.

19 Construct a triangle of side 100 mm and the adjacent angles to it are 125° and 30°. Measure the other sides.

20 construct a triangle of base 75 mm, one side is 40 mm and vertical angle 35°.

21 Construct a triangle of sides 90 mm, 60 mm and the vertical angle 55°.

Construction of quadrilateral

Objective: At the end of this exercise you shall be able to
- **construct square, rectangle, parallelogram, rhombus, trapezium and quadrilateral from the given conditions and data.**

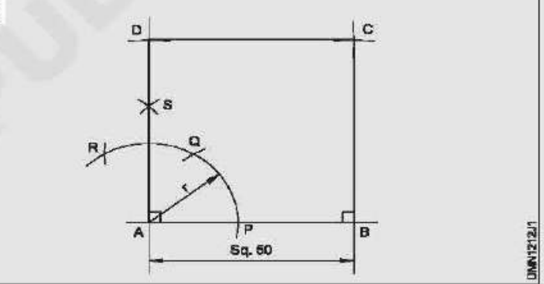
PROCEDURE

Exercise 9 (Fig 9)

A square of side 50 mm by erecting perpendicular

- Draw a line AB 50 mm long.
- 'A' as centre, draw an arc of convenient radius 'r' touching the line AB at 'P' as shown in Fig 9.
- 'P' as centre and radius 'r' draw another arc cutting the earlier drawn arc at 'Q'.
- 'Q' as centre and radius 'r', draw another arc 'R'.
- Bisect QR at S and extend.
- Mark 50 mm on AS extended line. AD = 50 mm.
- From points B and D, draw parallels to AD and AB and complete the square ABCD.

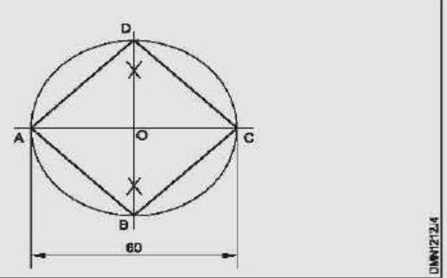
Fig 9



Exercise 10: Square having diagonal 60 mm (Fig 10)

- Draw horizontal and vertical centre lines intersecting at '0'.
- '0' as centre, draw a circle of radius 30 mm passing through centre lines at A,B,C and D.
- Join points A-B, B-C, C-D and D-A. ABCD is the required square, whose diagonal is 60 mm.

Fig 10

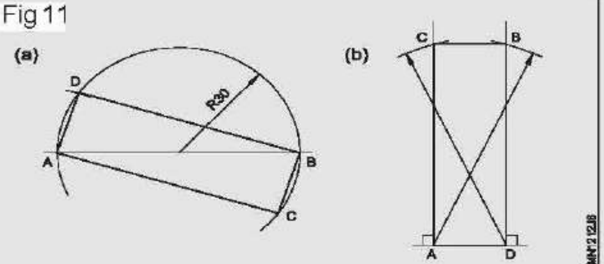


Exercise 11 : (Fig 11)

Rectangle - Diagonal - 60 mm, one side 18 mm

- Draw a line AB 60 mm.
- Draw a circle with AB as its diameter.
- 'A' as centre, draw an arc of R18, cutting the circle at D.
- Join AD and BD.

Fig 11



- Draw AC parallel to DB.
- Join BC and complete the rectangle.

Exercise 12: **Parallelogram** (Fig 12)

Sides = 75 mm and 40 mm

Angle between them: 50°

- Draw line AB 75 mm long.
- Draw line AD equal to 40 mm to 50° angle to AB.
- D as centre, draw an arc of radius equal to AB.
- 'B' as centre, draw an arc of radius equal at AD, upwards such that they meet at a point 'C'.
- ABCD is the required parallelogram.

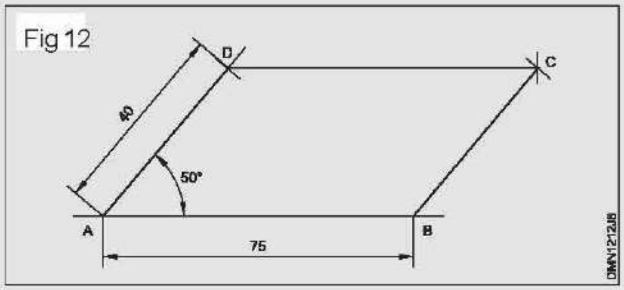


Fig 12

Exercise 13 : Parallelogram (Fig 13)

Parallelogram - Side AB = 60 mm

Diagonal AC = 90 mm ∠ABC= 120°

- Draw a line AB = 60 mm.
- Draw a line from B at angle of 120° to AB.
- 'A' as centre with radius 90 mm, draw an arc cutting 120° line from B at C.
- `C' as centre, radius = AB, draw an arc.
- Similarly `A' as centre and BC as radius, draw another arc, both arcs meet at `D'.
- Join AD and DC.

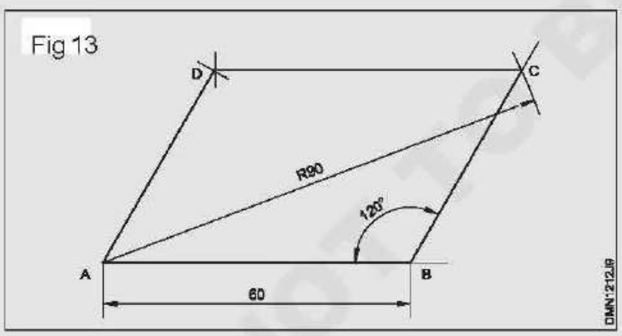


Fig 13

ABCD is the required parallelogram.

Exercise 14 : Rhombus (Fig 14)

Diagonals AB = 80 mm
CD = 50 mm

- Draw a line AB equal to 80 mm
- Draw perpendicular bisector of AB, passing through 0.
- mark OC = OD = 25 mm.
- Join the points AC, CB, BD and DA to complete the rhombus.

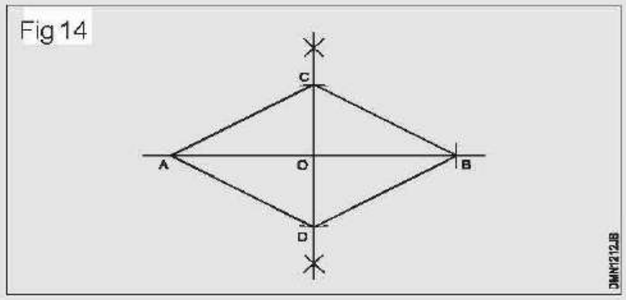


Fig 14

Exercise 15 : Trapezium (Fig 15)

Parallel sides AB = 60 mm; CD = 30 mm

Distance between parallel sides = 40 mm

Side DA = 45 mm.

- Draw the base AB equal to 60 mm.
- With radius 40 mm, draw arcs from A and B.
- Draw a tangential line (Parallel to AB)
- Draw an arc with radius 45 mm and A as centre, cutting the line at two places D and D'.
- From D or D' mark length of 30 mm towards right side, mark it as C or C'.
- Join B and C or C'.
- Join A and D or D'.

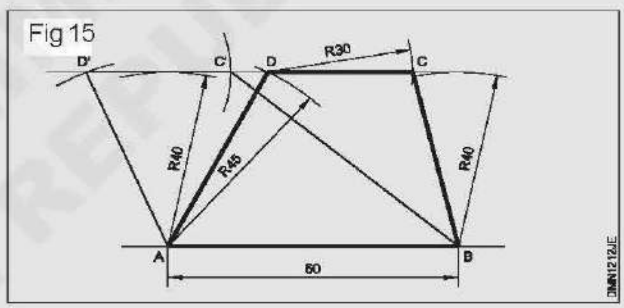


Fig 15

ABCD/ABC'D' is the trapezium.

Exercise 16 : Quadrilateral (Fig 16)

Sides AB = 80 mm; BC = 60 mm; CD = 40 mm; DA = 50 mm and ∠ABC = 75°

- Draw line AB equal to 80 mm.
- Draw line BC equal to 60 mm at an angle of 75°.
- C as centre and radius 40 mm, draw an arc.
- A as centre and radius 50 mm, draw another arc intersecting the previous at D.
- Join CD and AD to form the required quadrilateral.

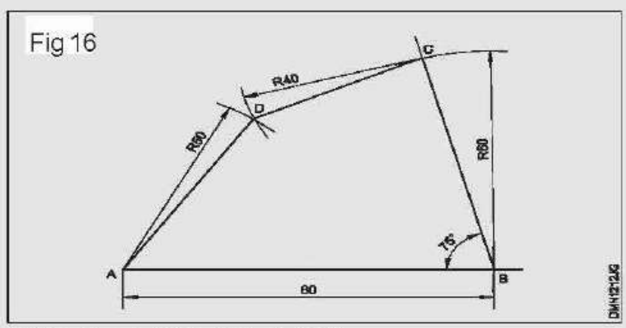


Fig 16

Construct regular polygons (up to eight sides) one equal base

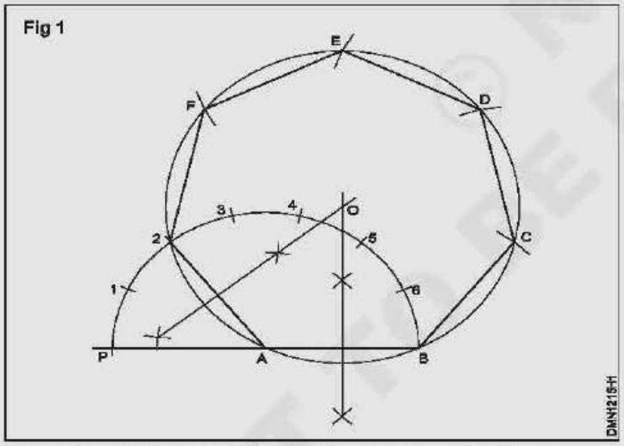
Objective: At the end of this exercise you shall be able to
- **construct a regular polygon from given data by different methods.**

PROCEDURE

Exercise 1: Regular heptagon of side 30 mm Semi-circular method - Type A (Fig 1)

- Draw a line AB equal to 30 mm.
- Extend BA to a convenient length.
- A as centre and radius AB describe a semi-circle.
- Divide the semi-circle into seven equal parts (number of sides) using divider.
- Number the points as 1,2,3,4,5,6 starting from P.
- Draw the perpendicular bisectors from 2A and AB intersecting at 0.
- 0 as centre and OA or OB as radius describe a circle.
- Mark the points C,D,E,F and 2 on the circle such that BC = CD = DE = EF = F2 = AB.
- Join the line BC, CD, DE, EF and F2.
- ABCDEF2 is required heptagon.

Fig 1

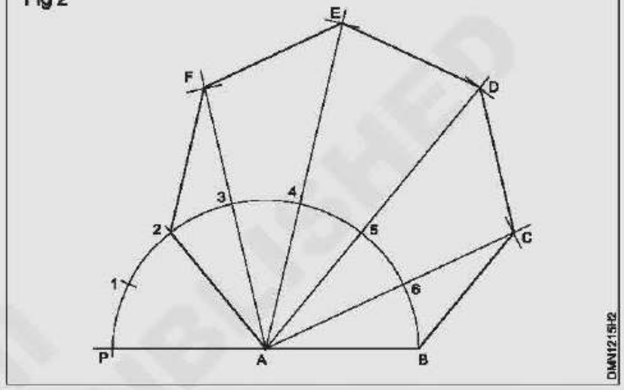


Exercise 2: Semi-circle method - Type B (Fig 2)

Follow the procedure of TYPE A upto dividing the semi-circle into number of equal parts.

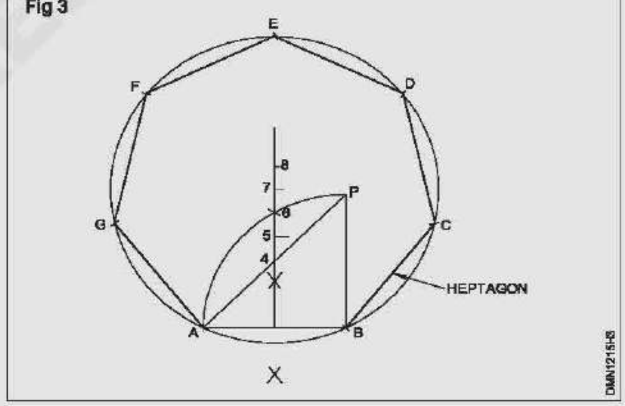
- Join A2
- Join A3, A4, A5 and A6 and extend to a convenient length.
- With centre B and radius AB draw an arc cutting A6 extended line at C.
- C as centre and same radius, draw an arc cutting the line A5 at D.
- Locate the points E & F in the same manner.
- Join BC, CD, DE, EF and F2.
- ABCDEF2 is the required heptagon.

Fig 2



Exercise 3: Perpendicular bisector method - Type A (Fig 3)

Fig 3



- Draw a line AB equal to 30 mm.
- At B, draw a line BP perpendicular AB and equal to AB.
- Join AP
- B as centre BA as radius, draw an arc AP.
- Bisect AB and draw the bisector cutting the line AP and the arc AP at 4 & 6 respectively.
- Mark 5 the mid point of 4-6.
- Set off 6-7, 7-8, 8-9, 9-10 equals to 4-5.
- 7 as centre, 7A as radius, draw a circle on AB.
- On the circumference set off BC, CD, DE, EF, FG equals to AB.
- Join BC, CD, DE, EF, FG and GA.
- ABCDEFG is the required heptagon.

Exercise 4:Perpendicular bisector method - Type B (Fig 4a)

- Draw a semi-circle on line AB, the side of the polygon (in this case it is heptagon)
- Describe an arc with B as centre and AB as radius.

Fig 4a

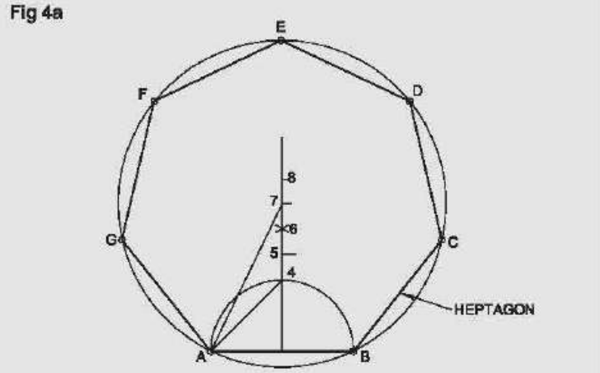
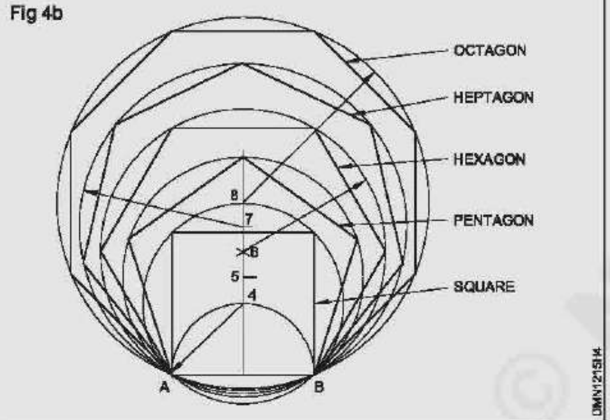


Fig 4b



- Draw a perpendicular bisector of AB.
- Perpendicular line cut the semi-circle at point 4 and the arc at point 6. (AB as radius, B as centre)
- Mark point 5 at the mid-point of 4 and 6.

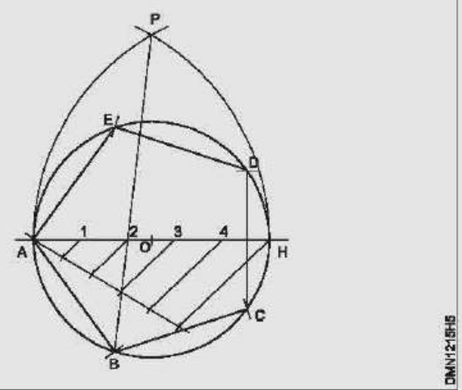
Follow the procedure of and complete the heptogon.

In this method also any regular polygon of different sides can be constructed. (Fig 4b)

Exercise 5: Pentagon inside a circle of diameter 80mm (Fig 5)

- Draw the line AH equals to 80 mm. (Diameter of circle)
- `O' as centre OA as radius describe a circle.
- Divide AH into 5 equal parts (as many equal parts as the sides).
- A and H as centres, AB as radius describe arcs intersecting at P.
- Join P2 and extend it to meet the circle at B.
- Set off BC, CD, DE, EF equals to AB on the circle.
- Join the points
- ABCDEF is the required pentagon.

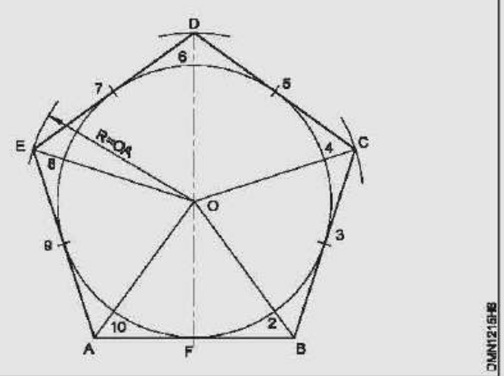
Fig 5



Exercise 6: Pentagon outside a circle of diameter 80mm (Fig 6)

- O as centre and OF as radius describe a circle of dia 80 mm.
- Draw the line DF vertically beyond the top of the circle.
- Divide the circle into 10 equal parts. (Twice as many equal parts as the number of sides)
- Points 1,3,5,7 and 9 are the tangent points of the pentagon.
- Join 02, 04, 06, 08, 010 and extend to a convenient length.
- Draw a tangent to the circle through point 1 (F).
- The tangent cuts the lines 0-2 and 0-10 lines at A & B.
- In same manner draw tangents on points 3,5,7,9 & locate C,D & E.
- Join BC, CD, DE, EA
- ABCDE is the required pentagon.

Fig 6



Exercise 7: Three circle method (Fig 7) **Pentagon of 38mm side**

- Draw the line AB equal to side of polygon 38 mm.
- Draw two circles of radius equal and AB, with centre A and B, cutting at two points F and G.
- Join G and F extend upwards.
- AB as radius, G as centre, draw a circle passing through A and B cutting both the circles at H and J, and also cutting the line FG at K.

Draw Inscribed and circumscribed circles of traingle pentagon and hexagon

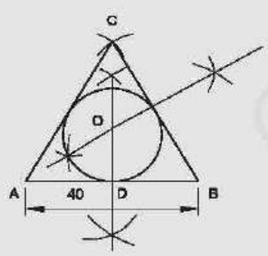
Objectives: At the end of this exercise you shall be able to
- **inscribing a circle in a equilateral triangle**
- **inscribing a circle in a scalene triangle**
- **circumscribing a circle about a equilateral triangle**
- **circumscribing a circle about a scalene triangle**
- **inscribing a circle in a regular pentagon**
- **circumscribing a circle about a regular pentagon**
- **circumscribing a circle about a regular hexagon.**

PROCEDURE

Exercise 1: Inscribe a circle in a equilateral triangle of given side 40mm (Fig.1)

- Construct the equilateral triangle A B C of side 40mm.
- Bisect the side A B & BC and the bisectors intersect at 0,
- Draw perpendicular from the point 0 to any one side (OD)
- With 0 as centre OD as radius inscribe a circle in the triangle and the sides are tangential to the circle.

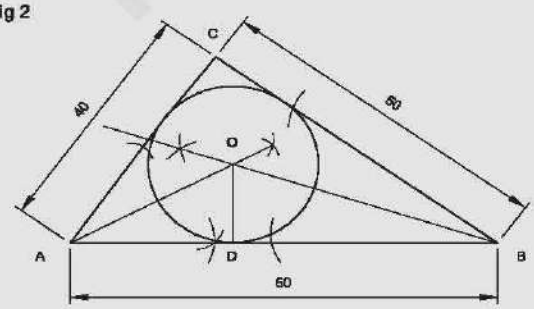
Fig 1



Exercise 2: Inscribing a circle in a scalene triangle of ABC give that AB = 60mm, BC = 50mm, CA = 40mm. (Fig.2)

- Construct the triangle ABC such that AB = 60, BC = 50, CA=40
- Bisect any two angles of the triangle ie ∠CAB & ∠ABC and let the bisectors intersect at O.
- O as centre OD as radius inscribe a circle in the ΔABC

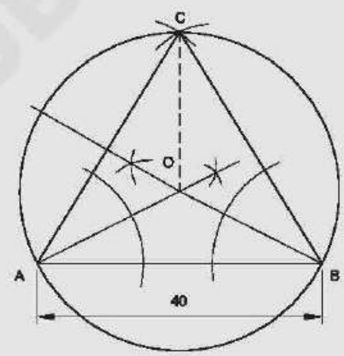
Fig 2



Exercise 3: Circumscribing a circle about a equilateral triangle. ABC of side 40mm (Fig 3).

- Construct the equilateral triangle ABC of side 40mm.
- Bisect any two angles of the triangle ABC ie ∠A & ∠B and the bisector of the angle intersect each other at 'O'
- 'O' as centre OA, or OC or OB as radius describe a circle about the triangle.

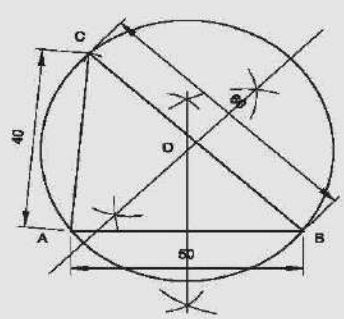
Fig 3



Exercise 4: Circumscribing a circle about a triangle ABC of AB = 50, BC = 60, CA = 40 (Fig 4)

- Construct the scalene triangle ABC of scale AB = 50, BC, 60, CA 40.
- Bisect any two sides of the triangle ABC ie AB & BC.
- Let the bisectors intersect each other at 'O'.
- 'O' as centre OA, OB or OC as radius describe a circle about the triangle ABC.

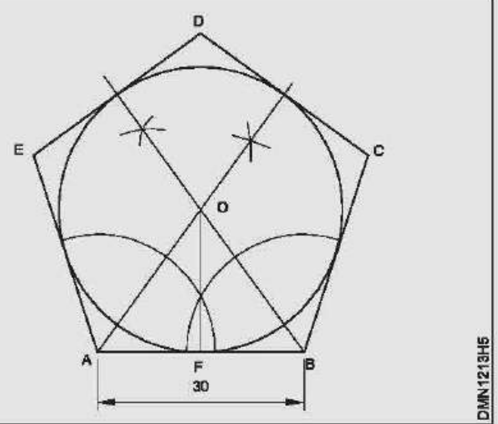
Fig 4



Exercise 5: Inscribe a circle in a given pentagon of side 30mm (Fig 5)

- Construct the pentagon of side equals to 30mm.
- Bisect any two interior angle of the pentagon
- Let the bisector intersect at 'O'.
- From O draw a perpendicular any side of the pentagon (say to AB)
- Meets at F.
- O as centre OF as radius inscribe a circle inside the pentagon.

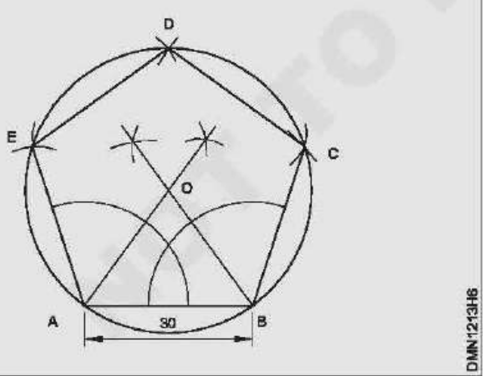
Fig 5



Exercise 6: Circumscribe a circle about a given pentagon of side 30mm (Fig 6)

- Construct a pentagon of ABCDE of side equals to 30mm
- Bisect any two interior angles of the pentagon.
- Let the bisector intersect at 'O'.
- With 'O' as centre and OA, OB, OC, OD or OE as readius describe a circle about the pentagon.

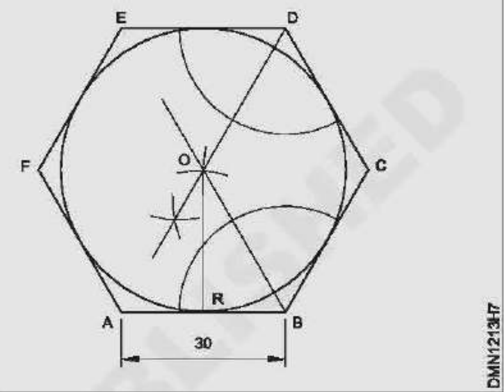
Fig 6



Exercise 7: Inscribe a circle is a given hexagon of side 30mm (Fig 7)

- Construct a hexagon ABCDEF of side 30mm.
- Bisect any interior angle of the hexagon.
- Let the bisector intersect at O.
- From the point 'O' draw a perpendicular to any side of hexagon (ie to AB)
- 'O' as centre and OR as radius inscribe a circle in the given hexagon.

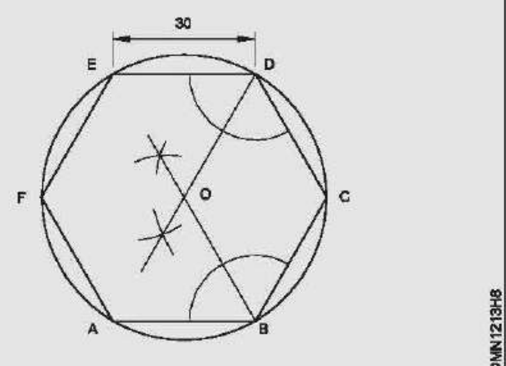
Fig 7



Exercise 8: Circumscribe a circle about a given hexagon of side 30mm (Fig 8)

- Construct the given hexagon ABCDEF of side 30mm.
- Bisect any two interior angles of the hexagon.
- Let the bisector intersect each other at O.
- 'O' as centre OA as radius (or any corner points) as radius circumscribe a circle about the hexagon.

Fig 8



Draw an angle of bisector and a line bisector and divide a line into any number of equal parts

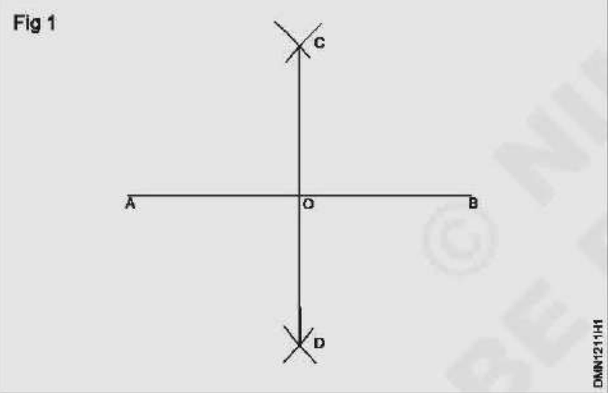
Objectives: At the end of this exercise you shall be able to
- **bisect a given angle**
- **trisect a given right angle**
- **bisect a given line**
- **divide a line into any number of equal parts.**

PROCEDURE

Exercise 1: Bisect a given straight line (Fig 1)

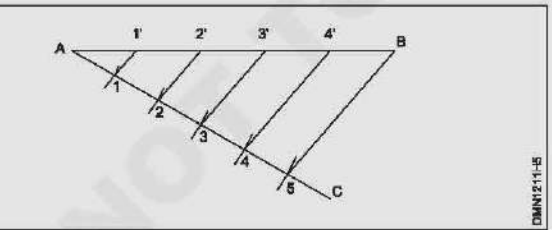
- Draw a line AB of 70 mm long
- With A and B as centres, more than half of AB as radius describe arcs an either side of line AB.
- Let the arcs intersect at C & D
- Join CD, Bisecting the line AB at 'O'
- CD is the bisector of the line AB and AO = OB.

Fig 1



Exercise 2: Divide a line into any number of equal parts (say 2). Fig 2

Fig 5



- Draw a line AB to a convenient length (say 65 mm).
- At `A' draw a line AC to a required length, forming an angle BAC. (Always it is better to form an acute angle)
- Set off 5 equal arcs on the line AC meeting at 1,2,3,4 & 5. (As many equal parts as required)
- Join 5 & B.
- From the points 4,3,2 & 1 draw lines parallel to 5-B meeting the line AB at 4', 3', 2' & 1'.
- Now the line AB is divided into 5 equal parts.

Exercise 2 : Bisect a given angle. (Fig 3)

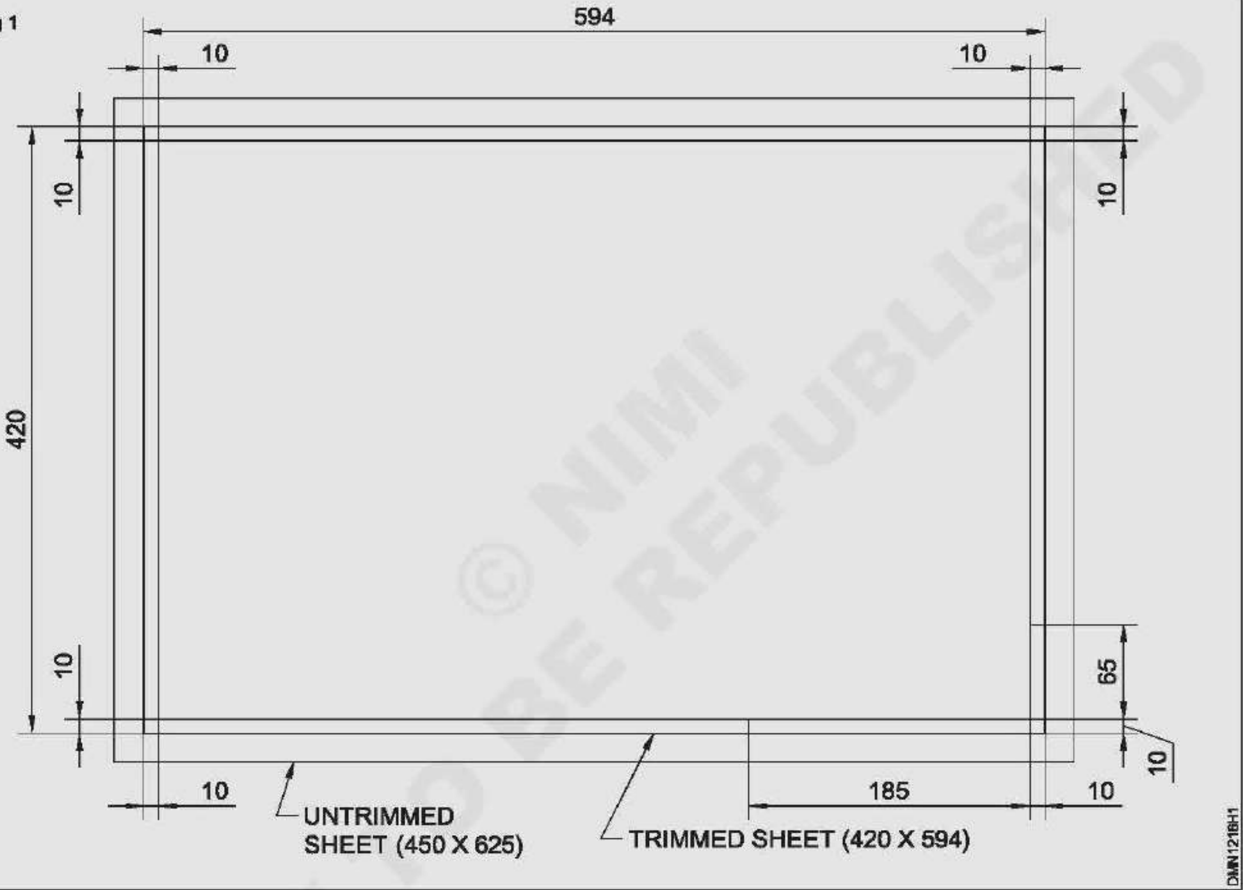
- Construct an angle BAC (say 30°).
- `A' as centre to a convenient radius draw an arc to cut line AC at `E' and AB at `D'.
- Bisect the arc DE at `O'.
- Join AO.
- AO is the bisector of the angle BAC.
- Now $\angle OAB = \angle OAC$.

Layout of A3 drawing sheet as per SP 46-2003 and title block with all informations details and folding of sheets

Objectives: At the end of this exercise you shall be able to
- **set and fix drawing paper on the drawing board**
- **draw margins and title block frame.**

PROCEDURE

Fig 1



Exercise 1: Prepare the layout as shown below on A2 size paper (Fig. 1)

- Place the drawing paper centrally on the drawing board.
- the drawing board and align the top edge of the drawing sheet.
- Hold the drawing sheet by hand in the same position and fix the sheet in this position with drawing cellulose tape. (Fig 1)
- Set off the margin distance using scale..
- Draw four border lines as shown above
- Mark and draw the title block.

Draw the title block providing details

Objectives: At the end of this exercise you shall be able to
- **layout the drawing sheet**
- **draw the title block as per I.S.**

Exercise 1: Draw this title block in position. In the remaining area of paper print the following (Fig 1)

1 All dimensions are in mm
2 Ask if in doubt.
3 Six holes diameter 8 mm equally spaced 60 mm pitch circle diameter.

4 This drawing confirms to IS:9609-1983.

5 Bureau of Indian Standards (BIS) is our national standard.

6 General deviations as per IS:2012;(medium)

7 All thick lines-0.5 mm.

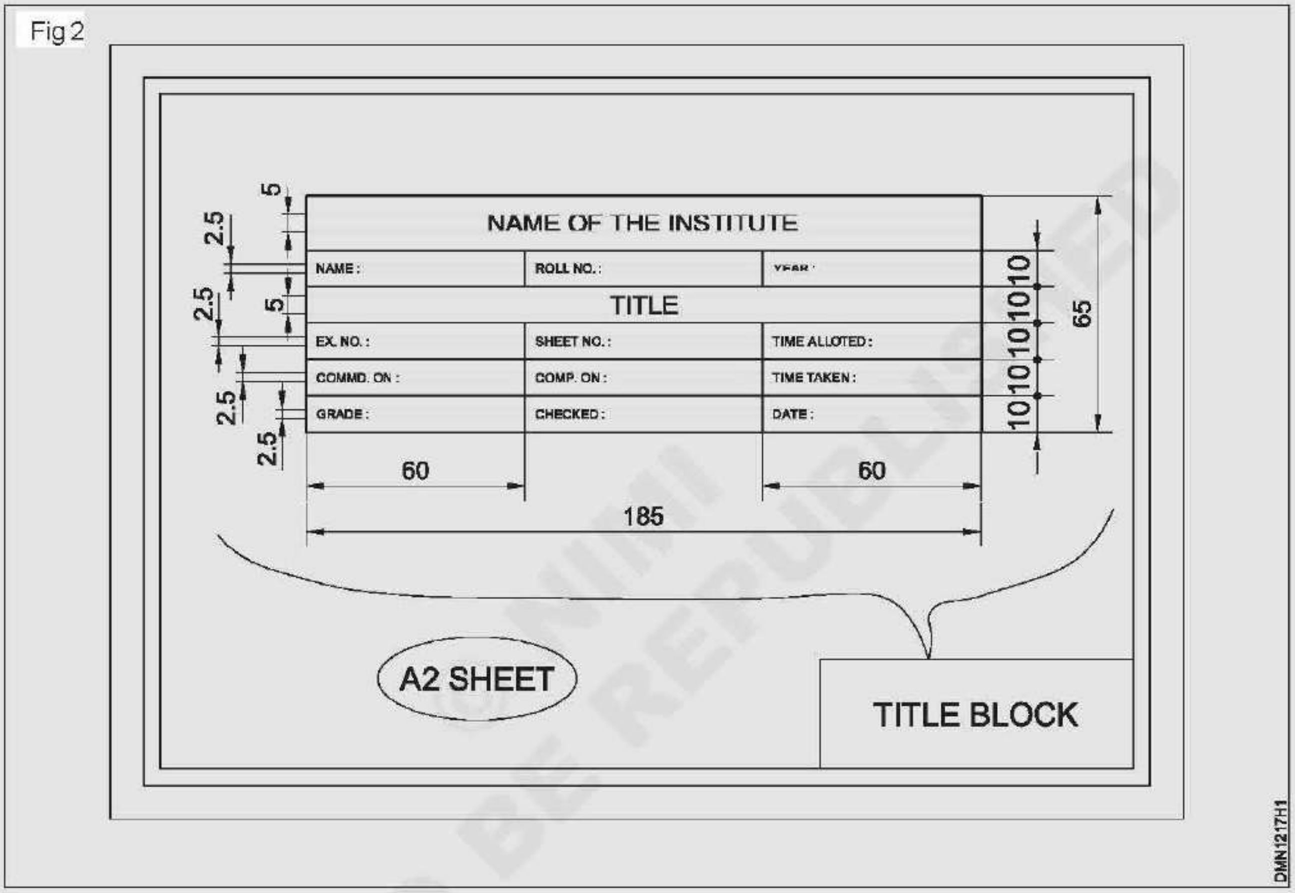
8 Chamfer to bottom of thread.

9 Rough will the surface marked 'X'.

10 Punch roll number and part number.

- Calculate the width of the each letter.
- Draw the guidelines for the required size:
- Mark the width and spacing for each letter.
- Draw vertical guidelines.
- Print the letter free hand, using HB pencil.
- Draw the subsequent squares with the same

Fig 2



Folding of drawing sheets

Objective: At the end of this exercise you shall be able to
- **method of folding at drawing sheets.**

Scope

This section covers two methods of folding of drawing sheets.

The first method is intended for drawing sheets to be filed or bound, while the second method is intended for sheets to be kept individually in filing cabinet

Basic Principles

The basic principles in each of the above methods are to ensure that

a) all large prints of sizes higher than A4 are folded to A4 sizes:

b) The title blocks of all the folded sheets appear in topmost position; and

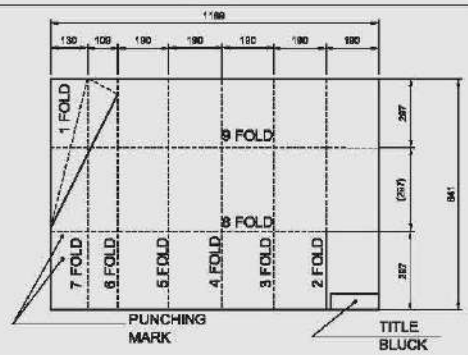
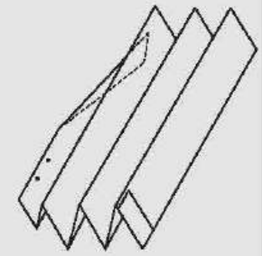
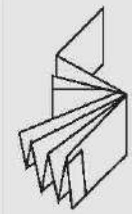
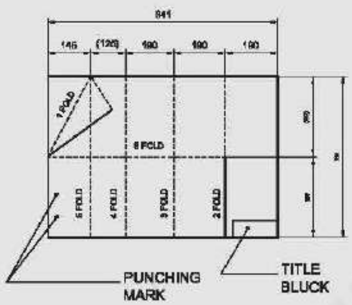
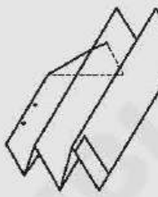
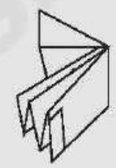
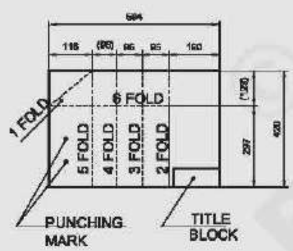
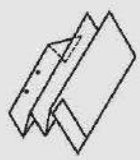
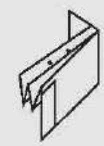
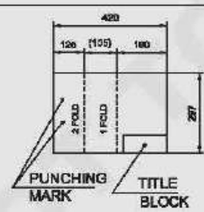
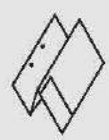
c) The bottom right corner shall be outermost visible section and shall have a width not less than 190mm.

Depending on the method of folding adopted, suitable folding marks are to be introduced in the tracing sheets as guide

Method of Folding of Drawing sheets

The methods recommended for folding are indicated in Exercise (Fig.1) and Exercise (Fig.2).

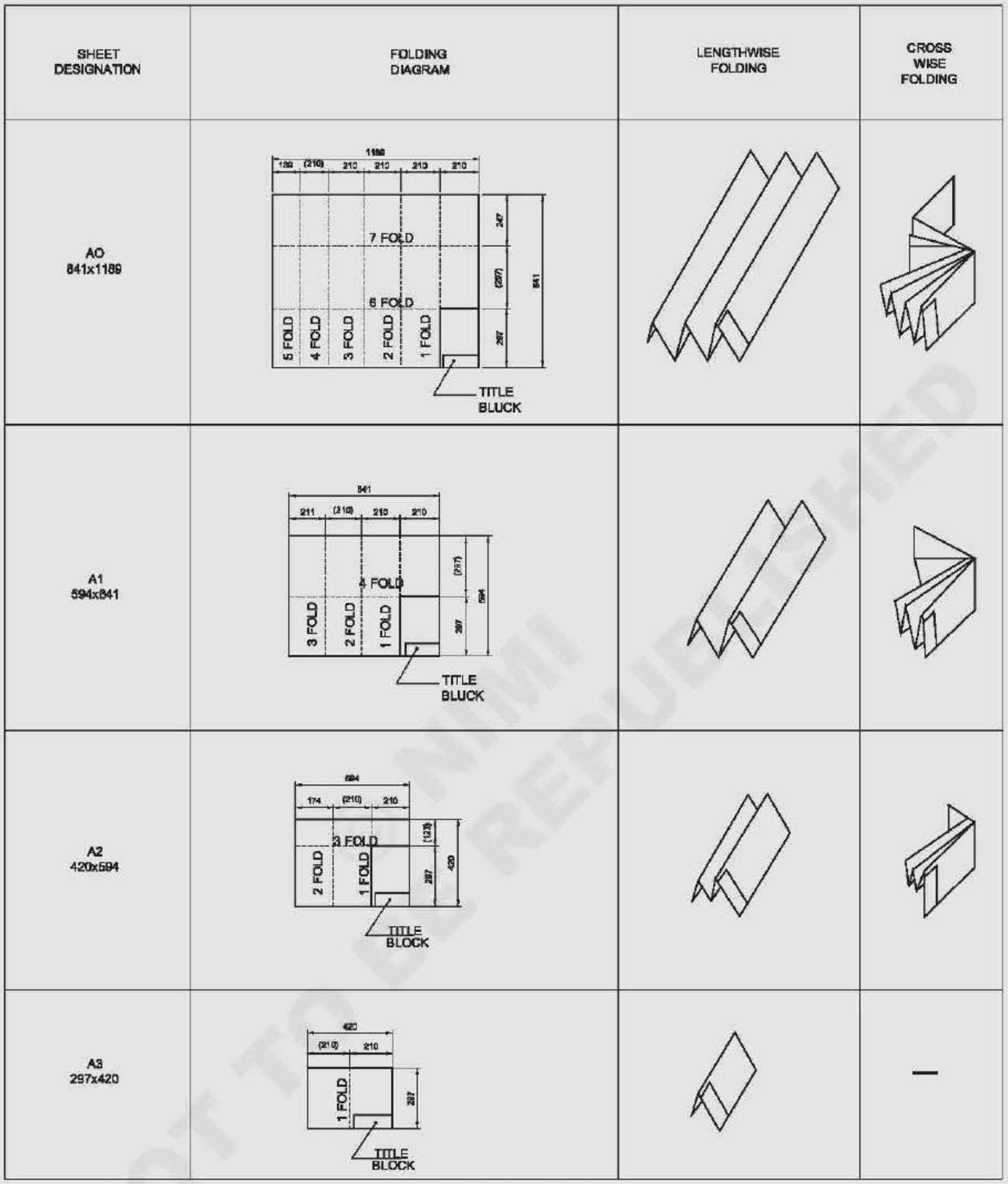
Fig 1

SHEET DESIGNATION	FOLDING DIAGRAM	LENGTHWISE FOLDING	CROSS WISE FOLDING
AO 841x1189			
A1 594x841			
A2 420x594			
A3 297x420			—

ALL DIMENSIONIN MILLIMETRES
FOLDING OF SHEETS FOR FILING OR BINDING

DMN218H1

Fig 2



ALL DIMENSIONIN MILLIMETRES
FOLDING OF SHEETS FOR STORING IN FILING OR BINDING

DM1507J2

Draw different types of lines and write their uses - show most of the lines in drawing view

Objectives: At the end of this exercise you shall be able to
- **draw the different types of lines**
- **draw the views of an object and indicate the application of lines.**

PROCEDURE

Exercise 1: Prepare the Table 1

Drawings are made up of different types of lines. Just as language with alphabets and grammar

Lines of different thickness and features are used for specific use.

Technical drawings are drawn with different types of lines. By proper choice and application of lines product features can be correctly defined in a drawing. Different types of lines recommended for specific applications are given in Table 1.

Table 1

Line	Description	General applications See figure and other relevant figure
A	Continuous thick	A1 Visible outlines A2 Visible edges
B	Continuous thin (straight or curved)	B1 Imaginary lines of intersection B2 Dimension lines B3 projection lines or extension line B4 Leader lines B5 Hatching B6 Outlines of revolved sections in place B7 Short centre lines B8 Thread line B9 Diagonal line
C	Continuous thin free hand	C1 Limits of partial or interrupted views & sections, if the limit is not a chain thin
D	Continous thin (Straight) with zig-zags	D1 Line (See figures)
E	Dashed thick	E1 Hidden outlines E2 Hidden edges
F	Dashed thin	F1 Hidden outlines F2 Hidden edges
G	Chain thin	G1 Centre lines G2 Lines of symmetry G3 Trajectories
H	Chain thin, thick at ends & changes of direction	H1 Cutting planes
J	Chain thick	J1 Indication of lines or surfaces to which a special requirement applies
K	Chain thin double-dashed	K1 Outlines of adjacent parts K2 Alternative and extreme positions of movable parts

K	K3 Centroidal lines K4 Initial outlines prior to forming K5 Parts situated in front of the cutting plane

1 This type of line is suited for production of drawings by machines.

2 Although two alternatives are available, it is recommended that on any one drawing. Only one type of line be used.

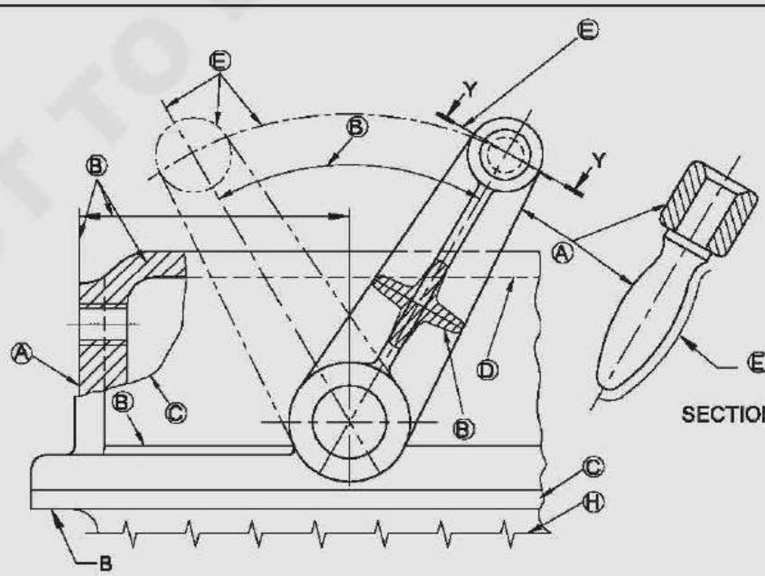
Exercise 2: Reproduce the type of line, illustration application of drawing. (Table 2)

Table 2

Type of Line	Illustration	Application
A Continuous thick	A	Visible outlines
B Continuous thin	B	Dimension lines, leader lines, external lines, construction lines, outlines, adjacent parts, hatching and revolve section.
C Continuous thin-way	C	Irregular boundary and edges
D Short dashes medium	D	Hidden outlines and edges
E Long chain thin	E	Central lines, locus lines, extreme positions of the moveable parts, parts situated in front of the cutting plane and pitch circles.
F Long chain thick at ends and thin elsewhere	F	Cutting plane lines
G Long chain thick	G	To indicates surfaces which are receive additional treatment
H Ruled lines and short Zig zag thin	H	Long break lines

Application of lines

Fig 1



EXAMPLES FOR THE USE OF VARIOUS TYPES OF LINES

Draw block letters of numerals in single stroke and double stroke of ratio 7:4 and 5:4

Objective: At the end of this exercise you shall be able to

- **print letters and numerals of given size in different styles by free hand.**

PROCEDURE

Exercise 1: Reproduce single stroke vertical gothic and double stroke vertical gothic letters and double stroke italic gothic letter

Production

Lettering, in engineering drawing, is an important part of drawing. Lettering enhances the clarity of drawing.

Lettering

The method of writing letters A, B, C, D and numerals 1, 2, 3, 4 etc., titles, notes, scale etc. on drawing is called Lettering.

Types of lettering

1 Gothic Lettering

2 Free Hand Lettering

3 Roman Lettering

1. **Gothic Lettering** — Single stroke letters of equal thickness and width are called Gothic letters. Gothic letters may be single stroke and double stroke letters. These letters can be written on a scale of 7:4 or 5:4. Letters inclined at 15° to vertical axis are italic gothic letters. All these types of letters are shown in figure below.

Fig 1

SINGLE STROKE VERTICAL GOTHIC LETTERING

DOUBLE STROKE VERTICAL GOTHIC LETTERING

INCLINED LINES

DMN'220H1

Exercise 2: Reproduce the roman lettering as given below Fig.2

Fig 2

ROMAN LETTERING

DMN1220H2

Fig 3

DOUBLE STROKE ITALIC GOTHIC LETTERING

DMN1220H3

Exercise 3: Reproduce the given style of letters as given below (Fig.4)

Fig 4

ABCDEFGHIJKLMN

OPQRSTUVWXYZ

1234567890

DOUBLE STROKE ITALIC FREE HAND CAPITAL LETTERING AND NUMERAL

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

1 2 3 4 5 6 7 8 9 0

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

1 2 3 4 5 6 7 8 9 0

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

1 2 3 4 5 6 7 8 9 0

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

1 2 3 4 5 6 7 8 9 0

ABCDEFGHIJKLMNOPQRSTUVWXYZ

1234567890

DIFFERENT SIZES OF FREE HAND SKITCH

a b c d e f g h i j k l m n o p q r s t u v w x y z

1 2 3 4 5 6 7 8 9 0

LOWER CASE FREE HAND

DMN1220H4

Capital Goods & Manufacturing — Exercise 1.2.18
Draughtsman Mechanical - Basic Engineering Drawing - Types of curves - Methods of dimensioning - Types of scales

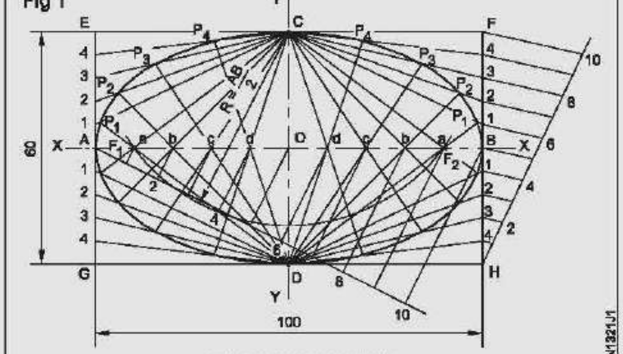
Construction of ellipse - parabola - hyperbola in different methods

Objective: At the end of this exercise you shall be able to
- **construct ellipse using different methods.**

PROCEDURE

Exercise 1: **Rectangle/Oblong method** (Fig 1)

Fig 1



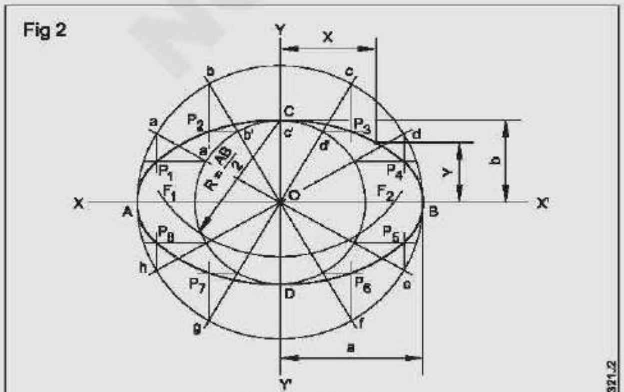
RECTANGLE/OBLONG METHOD

Construct an ellipse of major axis 100 mm and minor axis 60 mm.

- Draw a rectangle EFGH of sides 100 mm and 60 mm.
- Draw the major axis AB and minor axis CD and mark the intersection as `0'.
- Divide AO and OB into 5 equal parts each and name them as shown.
- Divide AE, AG, BF and BH into 5 equal parts and number them as shown.
- Draw lines and form C1, C2, C3, C4, D1, D2, D3, D4.
- Draw lines such as Ca, Cb, Da, Db etc to meet the corresponding lines drawn from C and D at points P_1, P_2 etc.
- Join A,$P_1$$P_2$ etc. with a smooth curve and form the ellipse.

Exercise 2: Concentric circle method (Fig 2)

Fig 2



CONCENTRIC CIRCLE METHOD

Major axis = 100 mm
Minor axis = 60 mm

- Draw the major axis AB (100 mm) and minor axis CD (60 mm), bisecting at right angle at 0.
- '0' as centre OA and OC as radius, draw two concentric circles.
- Draw a number of radial lines through '0' say 4 cutting the two circles.
- Mark the points on the outer circle as a,b,c.
- Similarly mark the intersecting points on inner circle as a',b',c'.
- From points such as a,b,c... draw lines parallel to minor axis.
- From points such as a', b'.... draw lines parallel to the major axis to intersect with the corresponding vertical lines at points P_1, P_2.... etc.
- Join all these points with a smooth curve using "french curve" and form the ellipse.
- To find the 'Foci' - with half the major axis (a) as radius and with 'C' on the minor axis as centre, draw an arc cutting the major axis, at two points, mark them as F_1, F_2, the focus points of the ellipse.

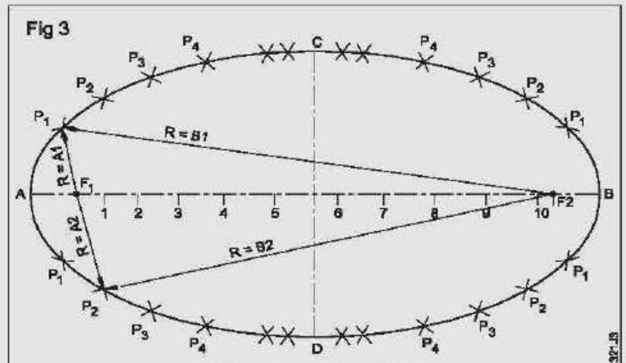
Check

Mark any point P on the curve and measure its distances from X axis and Y axis.

You will observe that $\frac{X^2}{a^2}+\frac{Y^2}{b^2}=1$
where a = 50 mm and b = 30 mm.

Exercise 3: Intersecting arc method (Fig 3)

Fig 3



INTERSECTING ARC METHOD

Constructing parabolic curves by different methods

Objective: At the end of this exercise you shall be able to
- **constructing parabolic curves by different methods.**

Exercise 1: A parabola from the given focus is at 50 mm from the directrix (Fig 1)

Construct a Parabola from a given focus is at 50 mm from the directrix.

Construct a parabola given the base and axis.

Rectangle method

Tangent method

Parallelogram method

Offset method

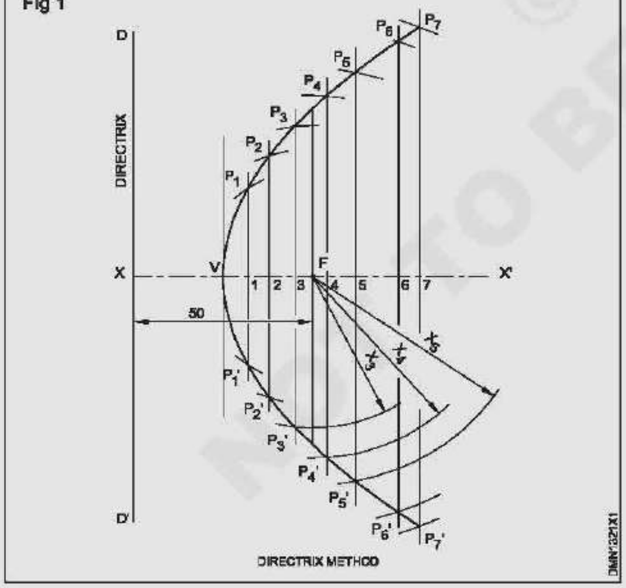
Construct Parabolic curves from the two given points.

– Points forming right angle, obtuse angle and acute angle.

Procedure

- Draw a vertical line D-D^1 the directrix.
- Draw horizontal line XX^1, the axis through any point X on the directrix.
- Mark the focus 'F' on XX^1 = 50 mm from X (on the directrix).
- Mark the midpoint of XF, as V.
- Mark a number of points from V towards right side on the axis as 1,2,3,4.....

Fig 1

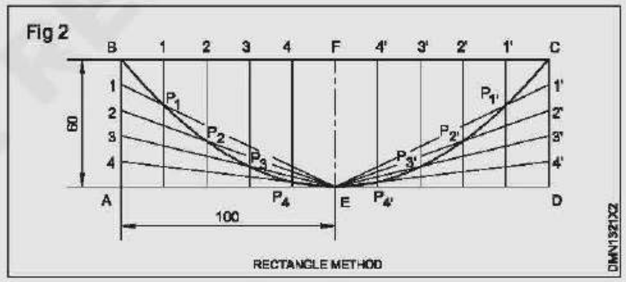


- Draw vertical lines through these points as shown, forming double ordinates.
- Point 'F' as centre, X-1 as radius, draw arcs on the co-ordinates (vertical lines) passing through 1, mark points P^1 & $P_1{}^1$.
- X-2 as radius, F as centre, draw arcs on the 2nd ordinate, mark P^2 & $P_2{}^1$.
- Similarly get point P_3, P_4.... $P_3{}^1$, $P_4{}^1$ etc. on the axis as above.
- Join all the points with a smooth curve, using french curve and form the parabola curve.

Exercise 2: Parabola, given the base and axis of a rectangle; base 200 mm axis 60 mm - Rectangle method (Fig 2)

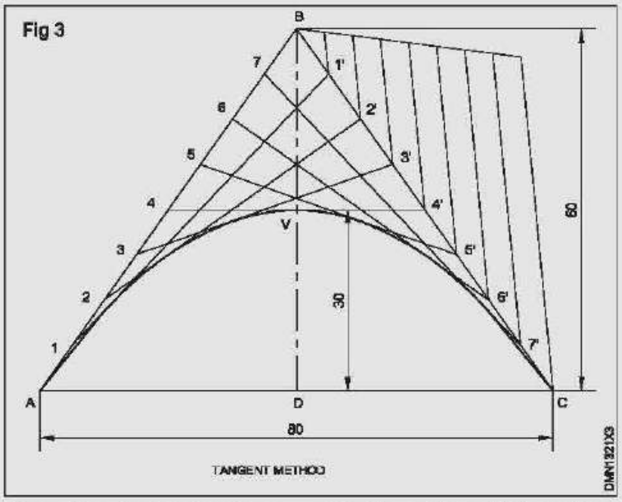
- Draw a rectangle ABCD of sides 200 mm & 60 mm.
- Mark centre points of AD and BC, as E and F, join EF.
- Divide AB & CD and into any number of equal parts say 5. Also divide AE and ED into the same number of equal parts and number them as shown.
- From point E on AD, draw lines to the divisions on AB & CD.
- From the points on AED, draw parallel lines to EF.
- Mark the intersecting points P_1, P_2, P_3, P_4 on either side of axis.
- Form the parabola by joining the points BEC and intersecting with a smooth curve, passing through P_1, P_2.

Fig 2



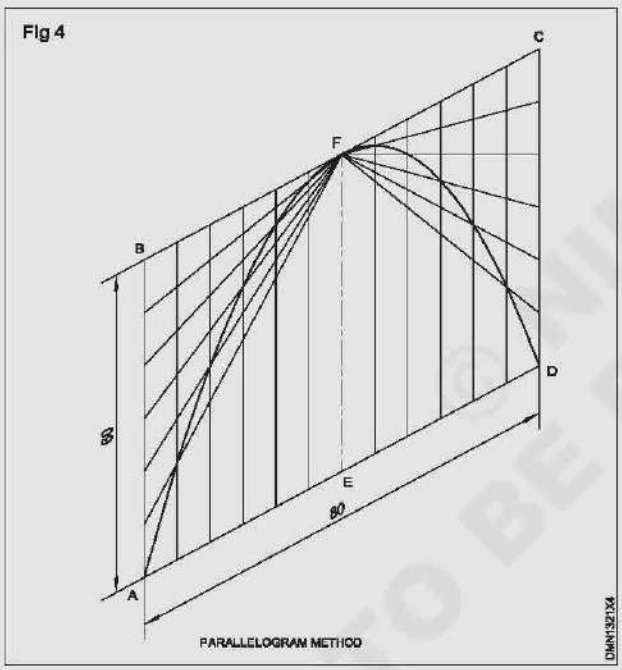
Exercise 3: Parabolic curve with base as 80 mm and axis 30 mm (Tangent method) (Fig 3)

- Draw an isosceles triangle of base 80 mm and altitude 60 mm (double the abscissa).
- Join BD and mark mid point V, the vertex.
- Divide AB and BC into same number of equal parts using divider/other methods.
- Mark the points on AB as 1,2,3 etc in ascending order..
- Similarly mark 1', 2', 3' etc on CB but in descending order.
- Draw lines 1-1', 2-2'..... 7-7'.
- Join the points with A, V and C with a smooth curve. AC is tangential to line 1'1', 2 2' etc and form the required parabola.



Exercise 4: Parabolic curve of sides 80 and 60 making 60°/120° - Parallelogram method (Fig 4)

Procedure is similar to the previous exercise



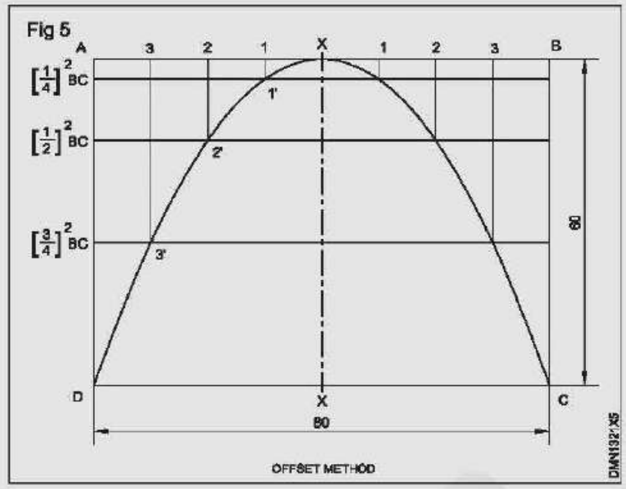
Exercise 5: Draw a parabola given double ordinate 80 mm and abscissa 60 mm 'offset method' (Fig 5)

- Draw the rectangle ABCD and draw XX through the midpoint of AB and CD.
- Divide AX and XB into same number of equal parts say 4 and mark them as 1,2,3 as shown.
- From the points 1,2 & 3, draw parallels (offset) to XX.
- On these offsets mark off distances as below:

  1-1' equal to $(1/4)^2$ of BC = 1/16 x 60 = 3.75 mm

  2-2' equal to $(2/4)^2$ of BC = 1/4 x 60 = 15 mm

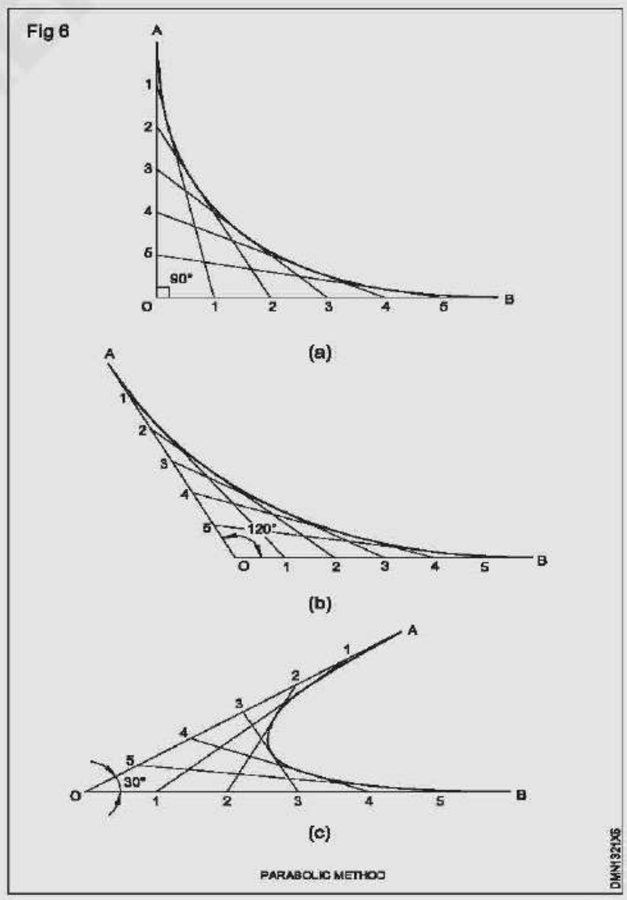
  3-3' equal to $(3/4)^2$ of BC = 9/16 x 60 = 33.75 mm
- Join D-X-C through parts 3', 2', 1' etc with a smooth curve and form the parabola.



Exercise 6: Parabolic curves joining two points A & B as shown (Fig 6)

Let the points A and B are in different positions as shown.

- Assume any point O.
- Join points A & B to O by straight lines.
- Divide AO and BO into same number of equal parts and number them as shown.
- Join the corresponding points i.e 1-1, 2-2.....5-5.
- Draw a smooth curve, tangential to line 1-1, 2-2, 3-3, 4-4, 5-5 etc and form this.
- Check the curve and draw thick parabola curve.



Constructing hyperbolic curves by different methods

Objective: At the end of this exercise you shall be able to
- **construct hyperbola curve using the various conditions given.**

Exercise 1: Given the eccentricity and the distance of focus from the directrix (Fig 1)

Given the eccentricity and the distance of focus from directrix.

Given the double ordinates obscissa and distance

between vertices (transverse axis)

Given the transverse axis and the distance between focii.

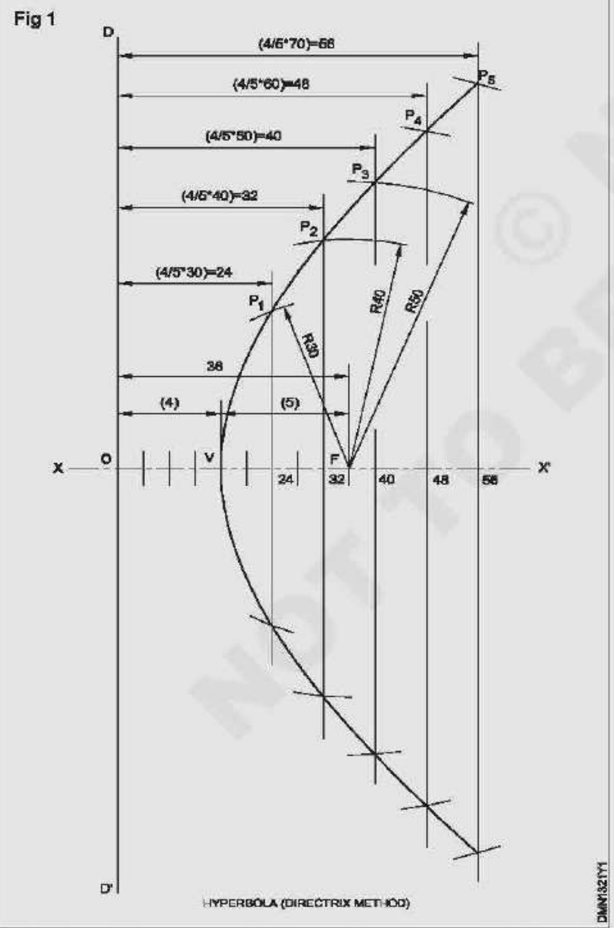
Rectangular hyperbola given point on the curve.

Hyperbola passing through a given point located between

the two asymptotes making any angle other than 90°

- Draw the locus of a point which moves so that its distance from a fixed point (Focus) and a line (directrix) bears a constant ratio of 5/4 (i.e eccentricity). Assume the focus at a distance of 36 mm from the directrix.
- Draw a directrix DD^1 and perpendicular XX^1 to it at 0.
- Mark F on XX^1 at a distance of 36 mm from 0.

Fig 1



- Divide 'OF' into nine equal parts and mark 4th division as 'V'.
- Prepare a table as shown below such that the ratio of the value in each column is 4:5

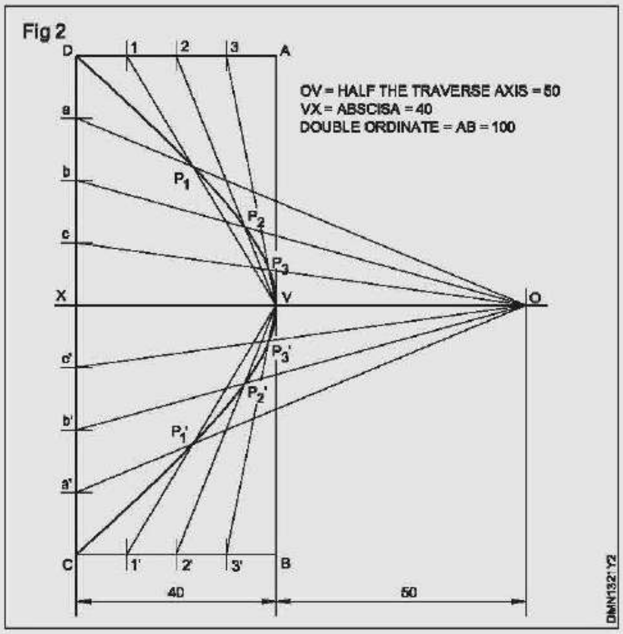
From directrix	24	32	40	48	56
From focus	30	40	50	60	70

- Draw parallels to directrix at distances 24,32 etc.
- F as centre and radius equal to 30, 40 etc. draw arcs to intersect the corresponding lines drawn in the previous step.
- Mark the intersecting points as P_1, P_2...etc.
- Complete the hyperbola by a smooth curve passing through these points.

Exercise 2: Given the double ordinate, abscissa and distance between vertices (traverse axis) (Fig 2)

- Draw a hyperbola with double ordinate 100 mm and abscissa 40 mm (rectangle method) and the transverse axis 100 mm.
- Draw the rectangle ABCD 100 x 40 mm.
- Mark the mid points V, X on AB and CD.
- Join VX and extend it outside to '0' 50 mm 100/2 from V.
- Divide AD and BC into 4 equal parts. Mark them as $1,2,3,1^1,2^1,3^1$.
- Join these points to V by straight lines.
- Divide DX;XC into 4 equal parts, each mark them as a,b,c,c^1,b^1,a^1.
- Join '0' to these points by straight lines.
- Mark the intersecting points as P_1,P_2 etc as shown.
- Join V-D with a smooth curve through P_1,P_2,P_3 etc.

Fig 2

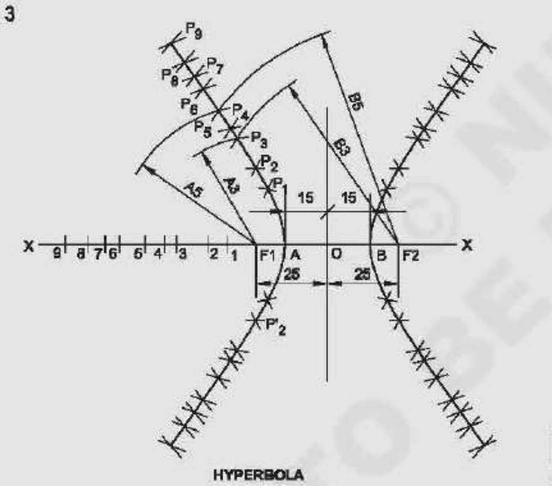


Exercise 3: Given the transverse axis and the distance between focus (Fig 3)

Draw a hyperbolic curve with transverse axis 30 mm and focus 50 mm apart.

- Draw the axis XX' and mark a point '0'.
- Mark OA = OB = 15 mm.
- Mark OF1 = OF2 = 25 mm.
- Mark any number of points on AX as 1,2,3,4....9
- Distance A-1 as radius, with centre F_1 and F_2, draw arcs on either side of XX'.
- Distance B -1 as radius, with centre F1 and F2. Draw arcs to intersect the previous arcs at P_1. (four places)
- A-2 as radius, F1 and F2 as centres, draw arcs on either side of XX'.
- Similarly B-2 as radius, F1 and F2 as centres draw arcs to get the point P2. (four places)
- Repeat as above and mark points P_3, P_4....P9 on four positions on either sides of XX'.
- Join the points in an order forming a pair of hyperbola curves.

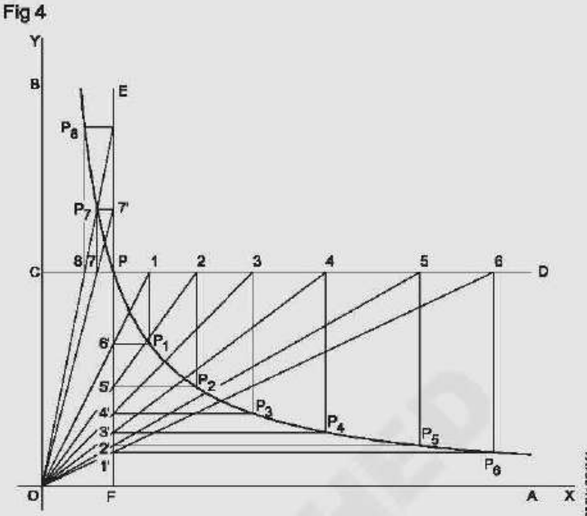
Fig 3



Exercise 4: Rectangular hyperbola given point on the curve (Fig 4)

- Draw a rectangular hyperbola given a symptotes as OX and OY are at right angles and a point P on the curve is 30 mm and 10 mm from OX and OY respectively.
- Draw the symptotes OA and OB at right angles to each other and locate the given point P. (10 mm from OX and 30 mm from OY)
- Draw the lines CD and EF passing through P and parallel to OA and OB respectively.
- Locate number of points 1,2,3 etc. (not necessarily equidistant) along the line CD.

Fig 4



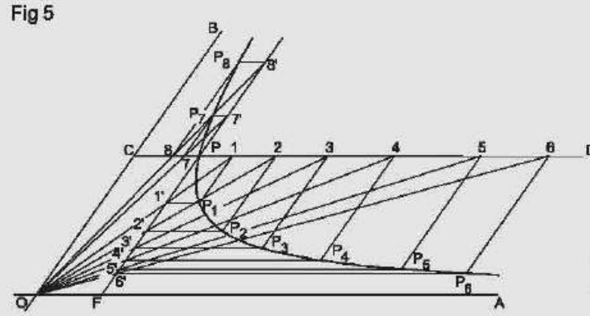
- Join 1,2,3 etc to O and extend if necessary till these lines meet the line EF at points 1', 2', 3' etc.
- Draw lines through 1,2,3 etc. parallel to EF and through 1', 2', 3' etc. parallel to CD to intersect at P_1, P_2, P_3 etc.

  A smooth curve passing through P_1, P_2, P_3 etc. is the required rectangular hyperbola.

Rectangular hyperbola is a graphical representation of Boyle's law, PV = constant. This curve also finds application in the design of water channels.

A hyperbola passing through any given point located between the two asymptotes making any angle other than 90° may also be constructed as shown in Fig 5.

Fig 5



Construction of involutes - cycloid curves - helix and spiral

Objectives: At the end of this exercise you shall be able to
- **construct the following geometrical curves (involutes) under the given condition/data**
  - **involute of circle by radial line and concentric circle method**
  - **involute of plane figures such as square and heptagon.**

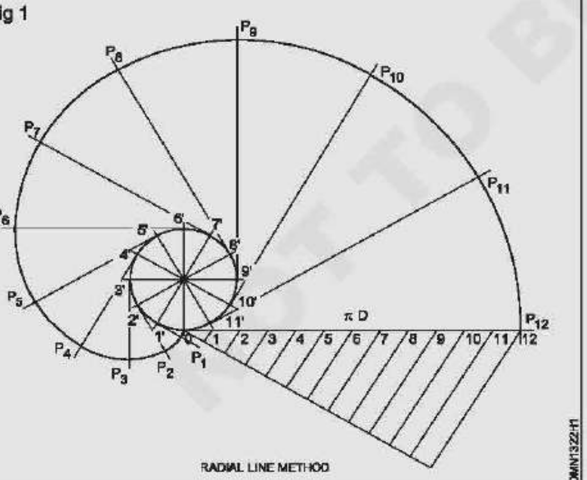
Draw an involute of a circle of diameter 40 mm. Radial line method Concentric method	Draw involutes of (regular plane figures) Square of side 40 mm Heptagon of side 40 mm

PROCEDURE

Exercise 1: Construct an involute of a circle of diameter 40 mm Radial line method (Fig 1)

- Draw a circle of diameter 40 mm.
- Divide the circle into a number of (say 12) equal parts and number than as 1', 2', 3'......12'.
- Draw a tangent through any of the points 1' , 2' etc and set a length equal to π D (graphically) on it, preferable draw the tangent from the point `O'.
- Divide the circumference (D) into equal parts as was done for circles and number them as 1,2,3....12.
- Draw tangents from points 1'2'3' etc and mark their lengths respectively equal to 01, 02, 03....011 etc and get points such as P_1,P_2...P_{12} smoothly and form the involute of circle.
- Follow the first four steps of the Radial line method.

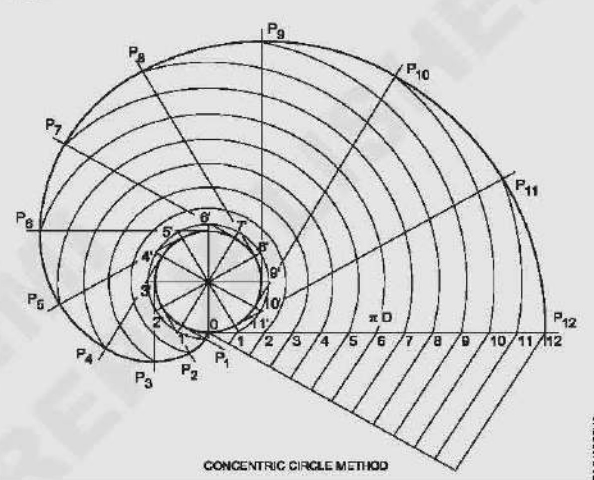
Fig 1



Exercise 2: Concentric circle method (Fig 2)

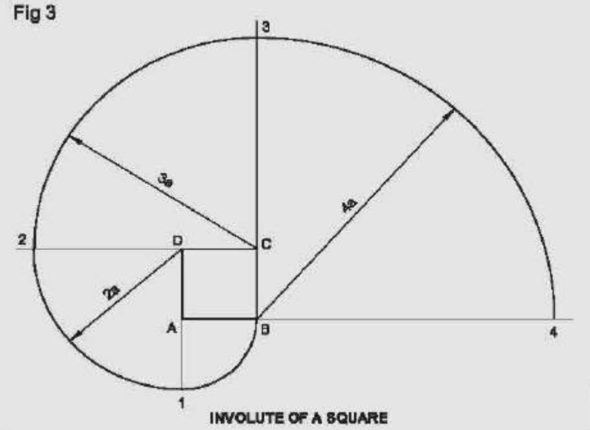
- Draw the concentric arcs, from the centre of the circle through the divisions 1,2,3.... etc to cut the tangential lines drawn from 1',2',3' etc at points P_1,P_2,P_3 etc.
- Join points P_1, P_2, P_3 by a smooth curve and form the involute of the circle.

Fig 2



Exercise 3: Construct an involute of square of side 40mm (Fig 3)

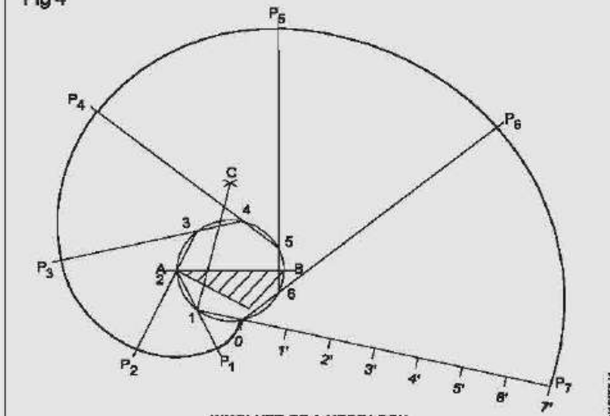
Fig 3



- Draw a square ABCD and extend the sides.
- 'A' as centre and radius 40 mm, draw the quadrant B.
- 'D' as centre and radius D-1, draw a second quadrant 1-2.
- 'C' as centre and radius C-2, draw a third quadrant 2-3.
- 'B' as centre and radius B-3 and draw forth quadrant 3-4.
- Now the curve 1-2-3-4 is the involute of the square.

Exercise 4: Construct an involute of a heptagon (Polygon) in a circle of diameter 60mm (Fig 4)

Fig 4



INVOLUTE OF A HEPTAGON

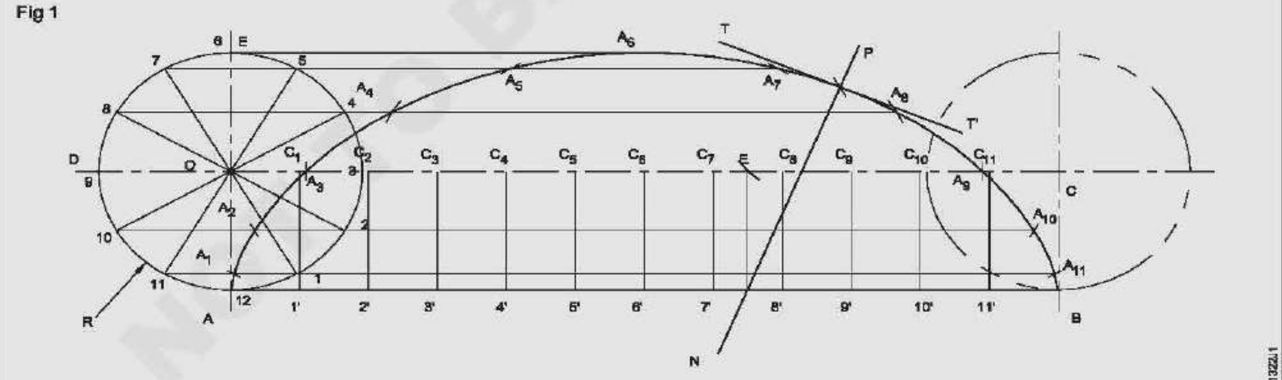
Constructing cycloidal curves, Helix and spiral by different methods

Objective: At the end of this exercise you shall be able to
- **construct given radius/diameter of generating/directing circles.**

Exercise 1: The cycloid of a point on a rolling/ generating circle of dia 60 mm along a straight line (Fig 1)

- Draw the generating circle of diameter 60 mm with centre '0'.
- Locate the initial position of the point A anywhere on the circumference of the circle.
- Draw a line AB tangential and equal to the circumference of the circle.
- Divide the line AB and the circle to the same number of equal parts (say 12) and number them as shown in figure.
- Draw the line OC parallel and equal to AB.
- Erect perpendiculars at $1^1, 2^1$.... etc to meet the line OC at $C_1 C_2$....etc.
- Through the points 1,2,3 etc draw lines parallel to AB.
- C_1 as centre and radius 30 mm, draw an arc intersecting the horizontal line 11-A_{11} at A_1.
- C_2 as centre and same radius (R30 mm) draw an arc intersecting the horizontal line 10-A_{10} at A_2. Similarly locate points A_3, A_4...A_{12} etc with centres C_3, C_4....etc.
- Join A, A_1, A_2....A_{11}, B with a smooth and complete the required cycloid.

Fig 1



CYCLOIDAL CURVE

Exercise 2: A circle of radius 20 mm rolls outside slipping on another circle of radius 60 mm. Draw the path of a point, on the rolling circle, the Epi cycloid (Fig 2)

As the generating circle rolls once on the directing circle, it subtends an angle on the periphery of the directing circle.

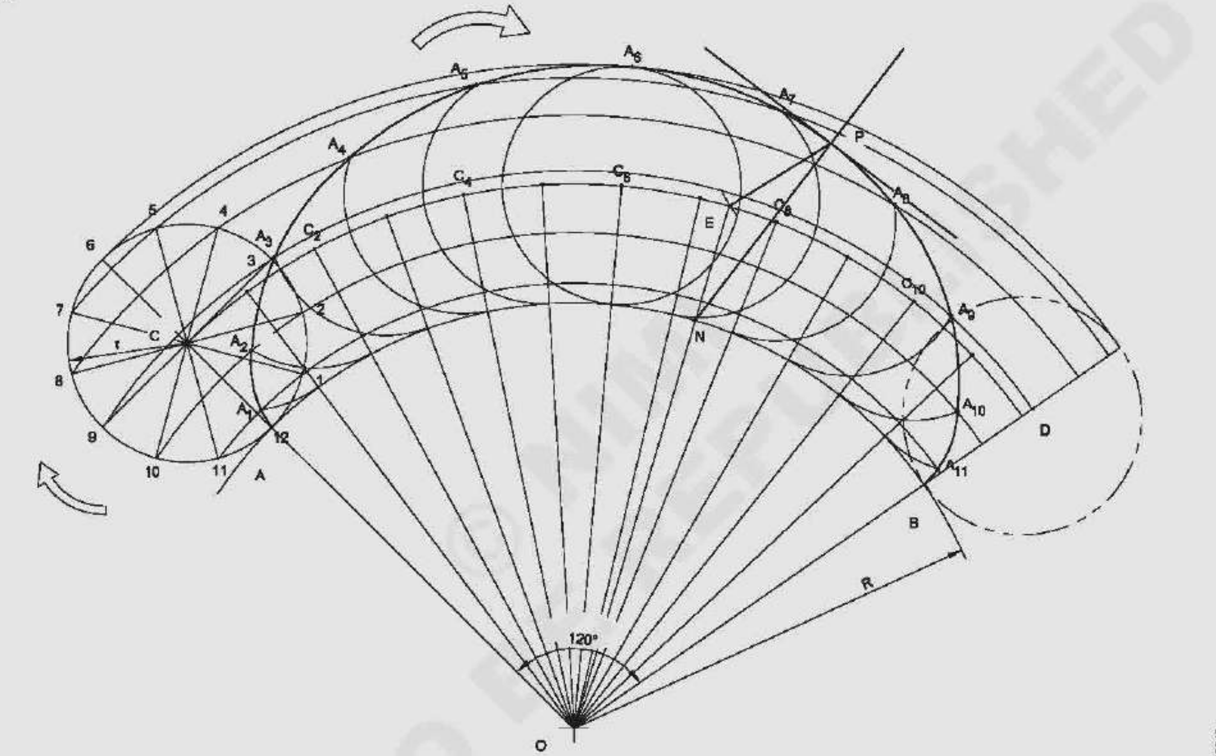
The value of this angle is $= \frac{r}{R} \times 360^\circ$

where r and R are the radii of generating and directing circles (20 mm and 60 mm) respectively.

Angle substandard $= \frac{20}{60} \times 360^\circ = 120^\circ$

- Draw an angle AOB equal to 120° and extend the sides.
- Point '0' as centre, draw the arc AB of radius 60 mm.
- Divide the angle AOB into 12 equal parts and draw lines from 0 as shown.
- On OA extended, with centre A and radius 20, draw an arc to cut the line OA extended at C.
- C as centre, draw a circle of r 20.
- Divide the rolling circle (C) into 12 equal parts 1,2,3....12 as shown.
- Point '0' as centre 0-1 as radius, draw an arc on to the angle AOB as shown.
- Repeat the same and draw arcs through points 2,3,4 & 5 with centre `O'.
- Mark the intersecting points of radial lines drawn from 'O' and drawn through C as C_1, C_2, C_3...etc.
- With radius r and centres C_1, C_2, C_3 etc. draw arcs on the respective radial lines to get points such as A_1, A_2, A_3 etc.
- Join points A_1, A_2, A_3 etc with smooth curve to form the required epicycloid.

Fig 2



EPICYCLOID

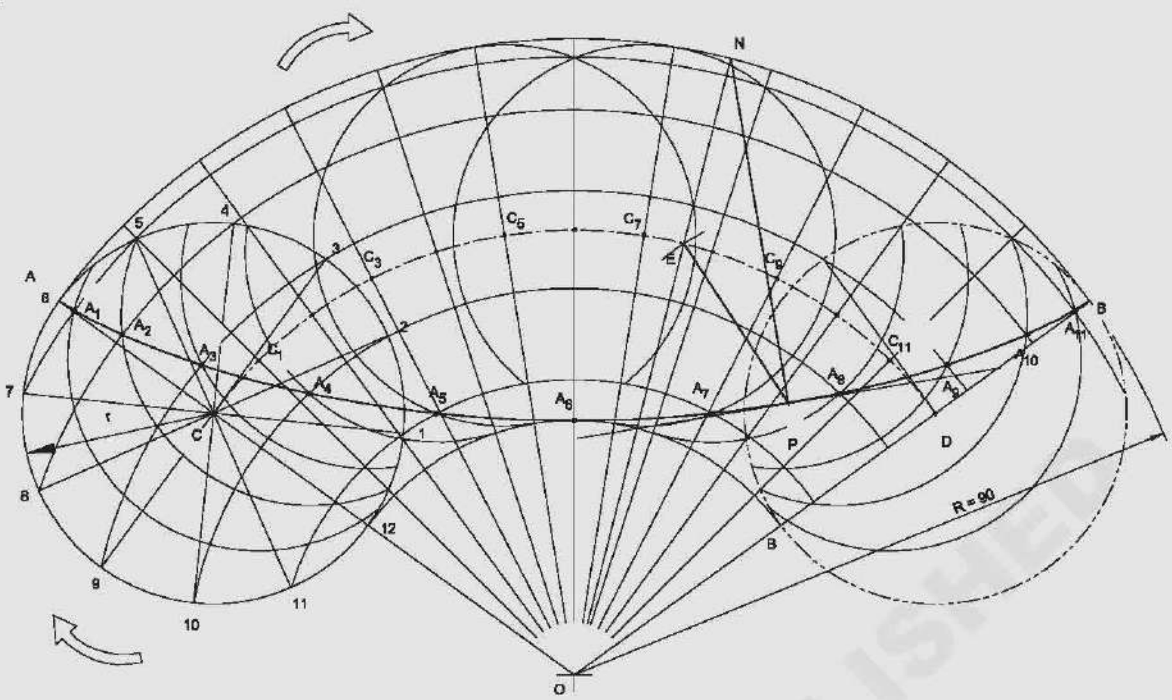
Exercise 3: A circle of diameter 80 mm rolls inside a circle of R90. Draw the resulting hypocycloid (Fig 3)

As in the previous exercise the angle subtended by the rolling circle

$$= \frac{r}{R} \times 360^\circ = \frac{40\,mm}{90\,mm} \times 360^\circ = 160^\circ$$

- Draw an angle AOB equal to 160° and extend the sides.
- Point '0' as centre, draw the arc AB of radius 90 mm.
- Divide the angle AOB into 12 equal parts, and draw lines from 0 as shown.
- On OA with centre A and radius r 40, draw an arc to cut the line CA at C.
- C as centre, draw a circle of r 40.
- Divide the rolling circle into 12 equal parts 1,2,3...12 as shown.
- Point '0' as centre 0-1 as radius draw an arc on to the angle AOB as shown.
- Repeat the same and draw arcs through points 2,3,4 etc with centre `O'.
- Mark the points intersecting points of radial lines drawn from '0' and the arc drawn through 'C' as C_1, C_2, C_3....etc.
- With radius r and centres C_1, C_2, C_3 etc., draw arcs on the respective radial lines to get points such as A_1, A_2, A_3 etc.
- Join A, A_1, A_2....A_{12}, B by a smooth curve and complete the required hypocycloid.

Fig 3

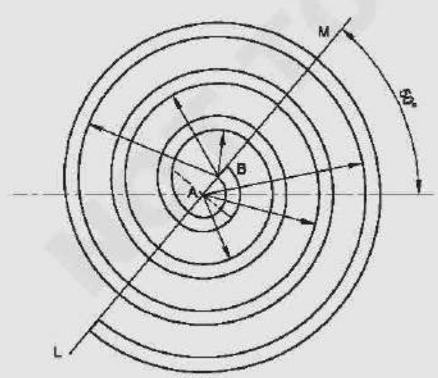


HYPOCYCLOID

Exercise 4: Draw four spirals on a line (Fig 4)

- Draw a line LM at an angle of 50° to the horizontal.
- Mark the points A and B (AB = 10mm)
- 'A' as centre, draw concentric semi-circles of radii 10 mm and 17 mm on the right side of the line LM.
- 'B' as centre, draw two concentric semi-circles/arcs in continuation to the previous arcs on left side of the line LM.
- Repeat the procedure with centres A and B and complete the pattern.

Fig 4



SPIRALS

Exercise 5: Draw the cam as per dimensions given (Fig.5)

- Draw a vertical line and mark the points C_1C_2 such that C_1C_2 = 84 mm.
- C_1 as centre, draw an arc of radius 56 mm (100-44) and C_2 as centre, draw another arc of radius 78 mm (100-22). Both arcs cut at C_3.
- Similarly obtain a point C_4 by drawing two arcs of radii 84 mm (44 + 40) and 62 mm (22 + 40) from points C_1 and C_2.
- Draw a circle of radius 44 mm with C_1 as centre and draw a circle or radius 22 mm with C_2 as centre.
- Produce C_1C_2 and get points A and B.
- C_3 as centre and radius BC_3 (100 mm) draw an arc.
- C_4 as centre and radius 40 mm draw an arc.
- Draw a circle of R10 with centre C_2.
- Rub off the unwanted lines and complete the pattern.
- Draw a polygon (heptagon) and mark the corners 0,1,2,.....6 etc.
- Extend each side of the polygon (heptagon) as shown.
- '1' as centre and 1-O as radius, draw an arc OP_1
- '2' as centre and 2-P_1 as radius, draw an arc P_1P_2
- Proceed in the similar way and draw arcs P_3 P_4 P_5 P_6 & P_7
- Now the curve 'O' $P_1P_2P_3$....P_7 is the required involute.

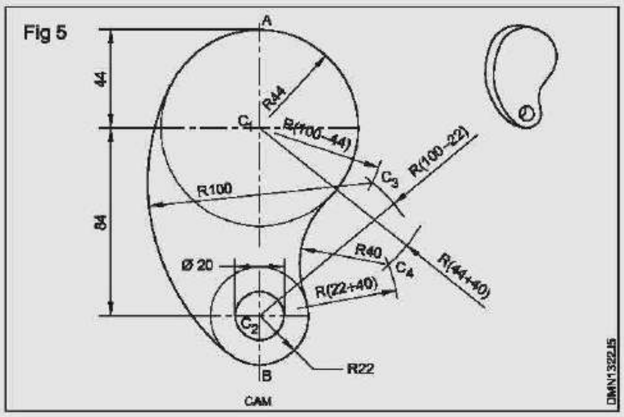


Fig 5

Exercise 6: Archimedean spiral of one convolution of radius 80 mm and starting from pole (Fig 6)

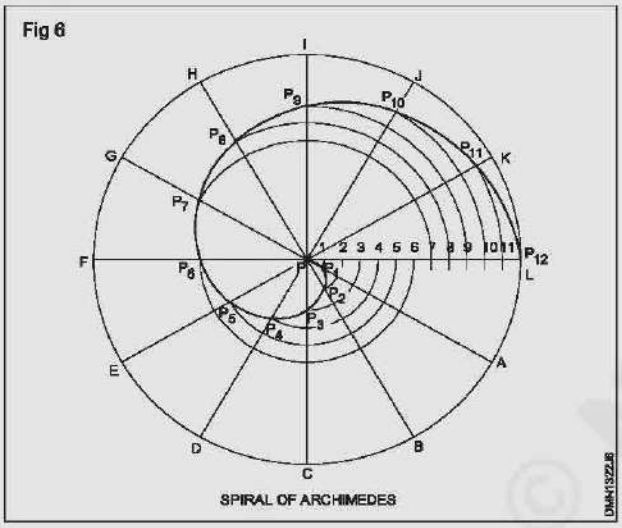


Fig 6

- Any point 'P' as centre, draw a circle of given radius 80 mm.
- Divide the circle into a number of equal (12) parts, number it as ABC, divide the radius into the same number of 12 equal parts and number them as 1,2,3.....12.
- Draw arc of radius P-1 on radial line PA. Mark the interaction as P_1.
- Draw arc of radius P-2 on radius line PB and mark the interaction as P_2.
- Draw arc of radius P-3 on radius line PC and mark the interaction as P_3.
- Similarly draw arcs on all the radial lines and mark points P_4 to P_{12}.
- Join the points P, P1, P2......P12 by a smooth curve and form spiral of the archimedes.

Exercise 7: Construct an Archimedean spiral of $1\frac{1}{2}$ convolution, smallest radius r = 20 mm, largest radius R = 60 mm (Fig 7)

- Any convenient point 'O' as centre, draw a circle of radius 60 mm and draw the radius OH.
- Set of OQ equal to 20 mm.
- Divide the circle into 12 equal parts (i.e 30°) and mark them as 1',2'.....12'.
- Divide the line QH into 18 equal parts and number them as 1,2,3.....18.
- 'O' as centre, draw arcs passing through points 1,2.... 18 to intersect the corresponding radial lines at P_1, P_2..... etc.
- Join the points P1 to P18 by a smooth curve and form the required spiral.

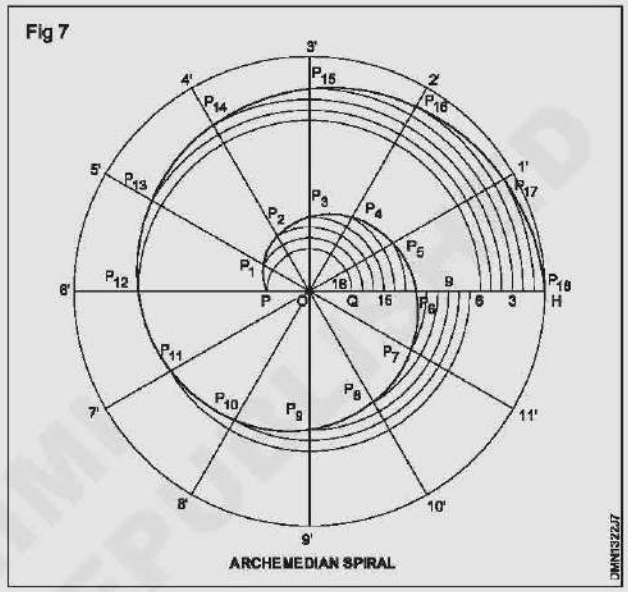
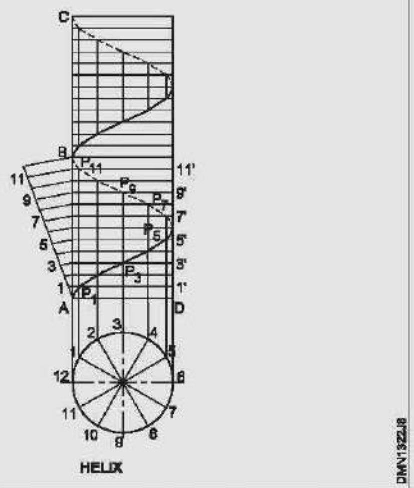


Fig 7

Exercise 8: Construct a helix for two revolutions of a point on a cylinder dia 40 and pitch of helix 30 mm (Fig.8)

- Draw a circle of diameter 40 mm, divide it into 12 equal parts and number them as 1,2.....12.
- Draw vertical projectors of sufficient length (more than 60 mm) from points 1,2.....12, such that the points 1 and 11, 2 and 10, 3 and etc lie on same vertical line.
- Mark the points A,B and C such that AB = BC = lead = 30 mm.
- Divide AB into the same number (12) of equal parts as there are similar divisions in the circle.
- Draw horizontal lines from 0, 1',2'..... etc to intersect with the projections from 1,2,3 etc at points P_1, P_2, P_3......P_{12}.
- Join the points P_1, P_2 etc with a smooth curve and form helix.
- Repeat the above procedure in portion BC and complete the helix for two leads.

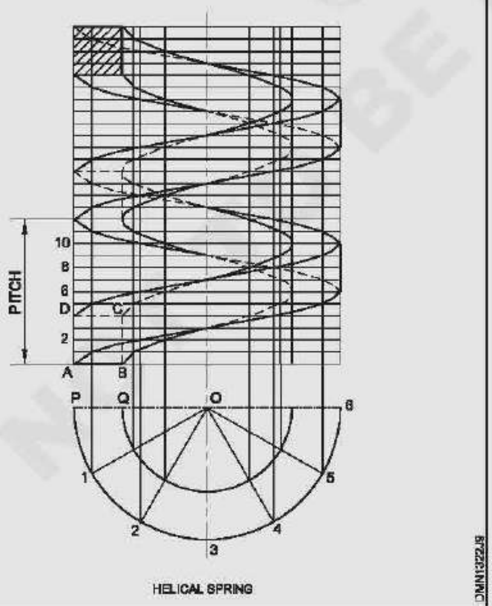
Fig 8



Exercise 9: Construct a helical spring of 10 mm square. Cross-section has an outside dia of 60 mm and pitch 30mm. Draw two pitches of the spring (Fig.9)

- To draw the spring with one pitch we have to draw four helixes.
- Helixes starting from A is to be drawn using procedure of the previous exercise with outside diameter 60 mm. For helix 'D' reproduce same thing at distance 10 mm above helix A (thickness of spring 10 mm).
- Follow the above said procedure complete two helixes B&C with inside diameter 40 mm.
- Repeat four more curves and complete the two pitches of the spring.

Fig 9

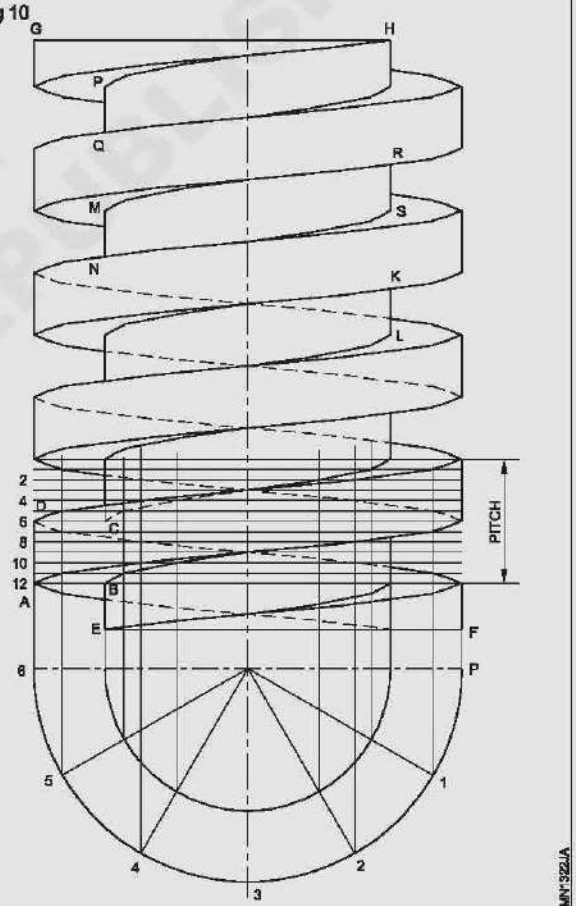


Exercise 10: On a cylindrical rod of dia 80 a square helical groove of depth 12 mm is cut forming square thread. Assuming the pitch equal to 24 mm, draw 2 complete turns of the thread. Find the helix angle. Assume width/thickness of thread = depth of thread (Fig 10)

Outside dia	=	80 mm
Depth of thread	=	12 mm
Pitch of thread	=	24 mm

- Starting from ABCD, draw four helixes using the method followed in the previous exercises and complete 5 pitches by drawing parallel helical curves.
- Draw vertical lines such as PQ, RS, MN etc to represent the core of the all five pitches.
- Draw parallel lines EF and GH at any convenient distance.
- Rub of the invisible portions of the helixes and complete the square thread.

Fig 10



Objects drawing with dimensions of different alignment as per SP-46-2003

Objectives: At the end of this exercise you shall be able to
- **place dimensions in the drawings by aligned system and unidirectional system of dimensioning**
- **give dimensions to the given drawings by following dimensioning principles and as per BIS.**

Procedure

Exercise - 1

To the given drawing of the profiled sheet metal. (Fig 1) Place the dimensions in aligned system.

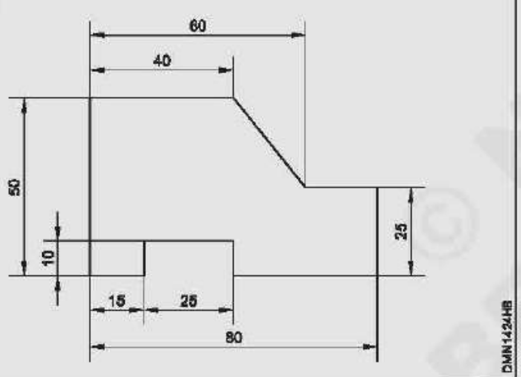
Draw the drawing of the sheet metal to 1.1 scale.

Draw the extension lines in continuation of outlines.

Draw the dimension lines. (Fig 1)

Place the dimension value near the middle and above the dimension line to be read from **"bottom and right hand side"** of the drawing.

Fig 1



Note: Draw the dimension line terminations as per IS:11669-1986.

- Draw the arrow heads with short lines forming borbs at any convenient angle between 15° to 90°.
- Maintain the uniformity of line thickness.
- Dimension lines are not to be broken.

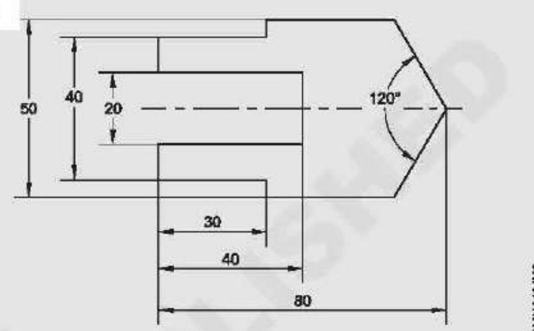
Exercise - 2

To the given drawing of the profiled sheet metal shown in Fig 2. Place the dimensions in unidirectional system.

- Place the horizontal dimensions above and middle of the dimension line without break.
- Break the dimension in the middle of all non-horizontal dimension lines. (Fig 2)

Note: Follow the general principles of dimensioning and show the dimensions as per standard.

Fig 2



Exercise - 3

Draw the four cover plates shown in (Fig 3a,b,c &d) and dimension according to standards.

- Fig 3 & 4 dimensions are placed taking the left and bottom edge as reference lines.
- Fig 5 & 6 centre lines are taken as reference lines for placing the dimensions.

Fig 3

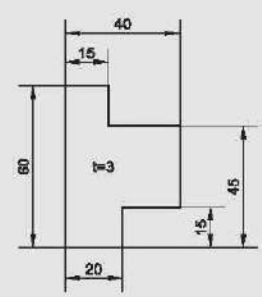


Fig 4

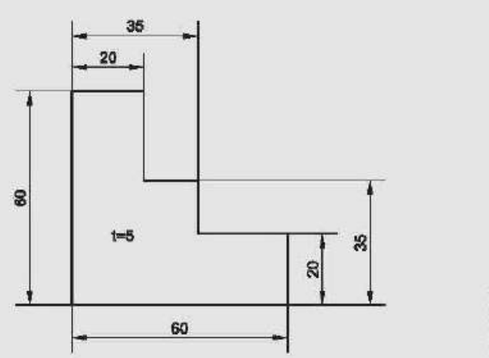


Fig 5

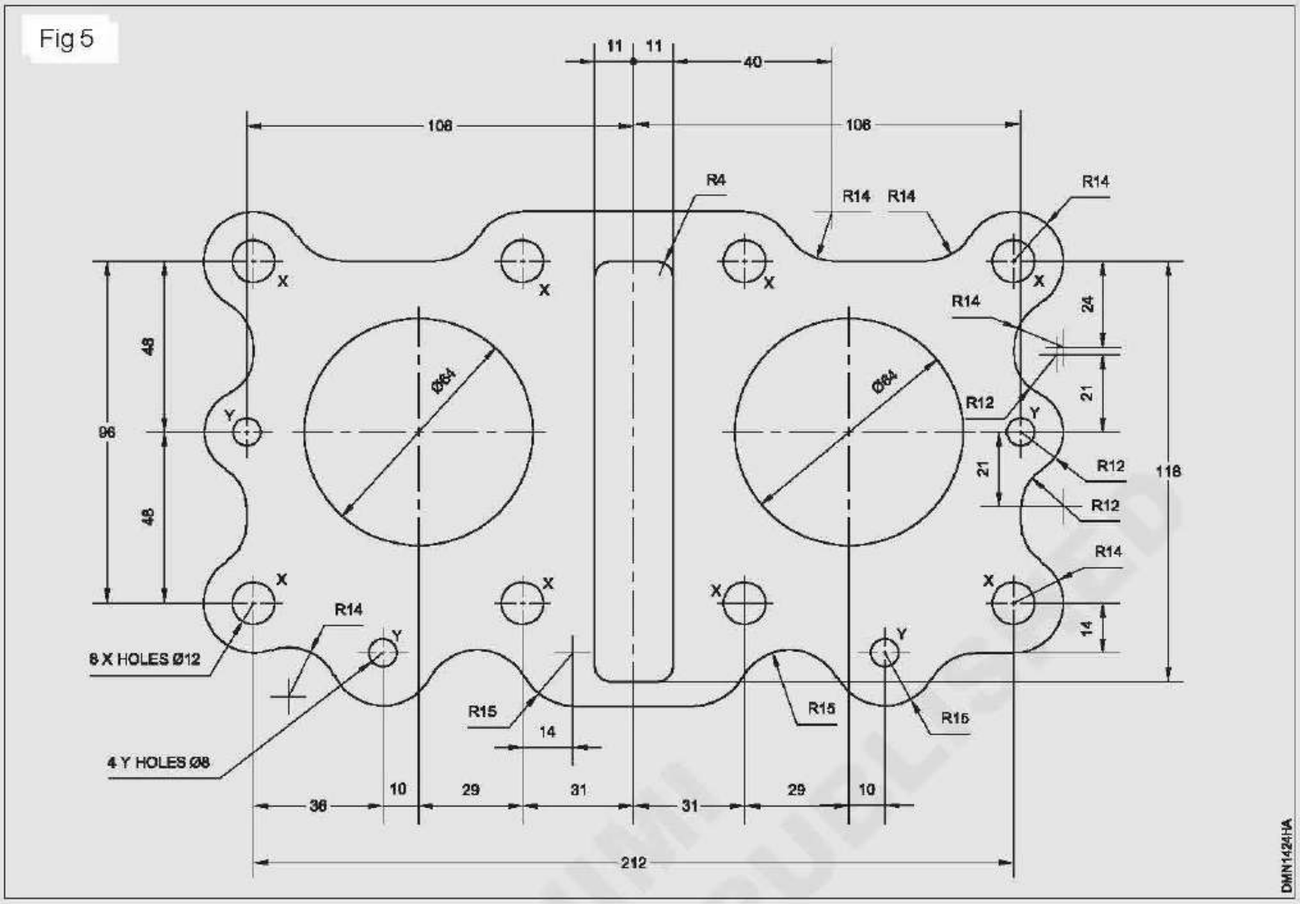


Fig 6

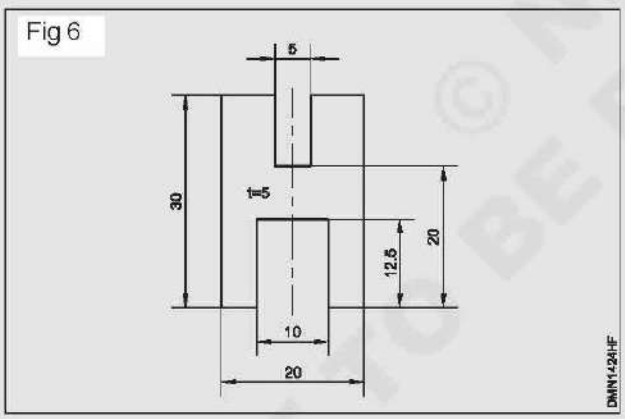
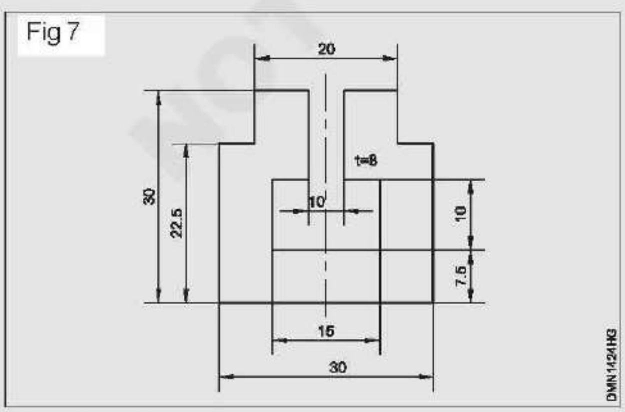


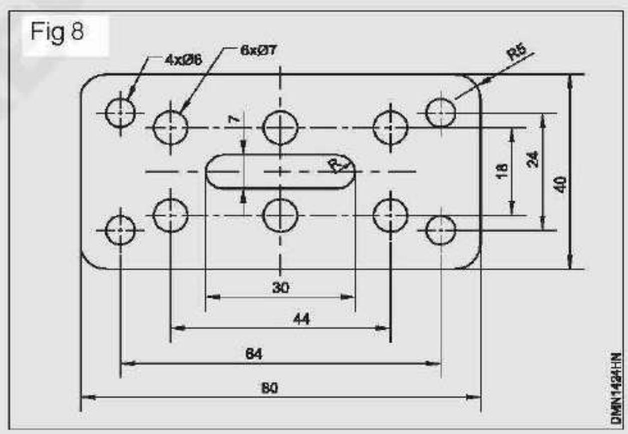
Fig 7



Exercise - 5

Draw the board and seal of a three phase motor shown in Fig 7 .

Fig 8



- Follow the same procedure adopted in previous exercises.
- Centre lines are taken as reference line for the placing the dimensions

Exercise - 6

Draw the special can shown in Fig 8 and place the dimensions.

- Adopt the principles of dimensioning arcs, curves and fillets.
- Centre lines are taken as reference line for giving dimensions. (Fig 9)

Fig 9

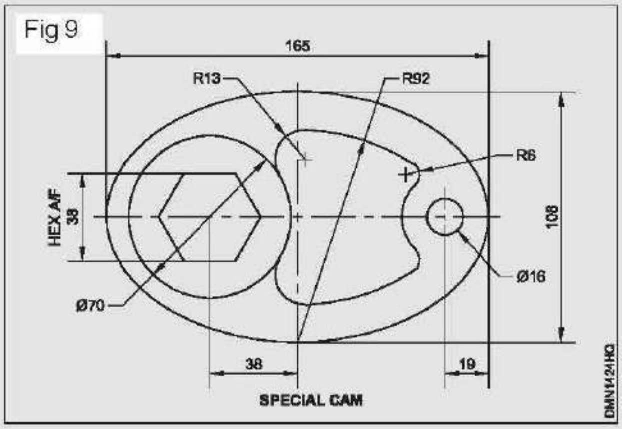
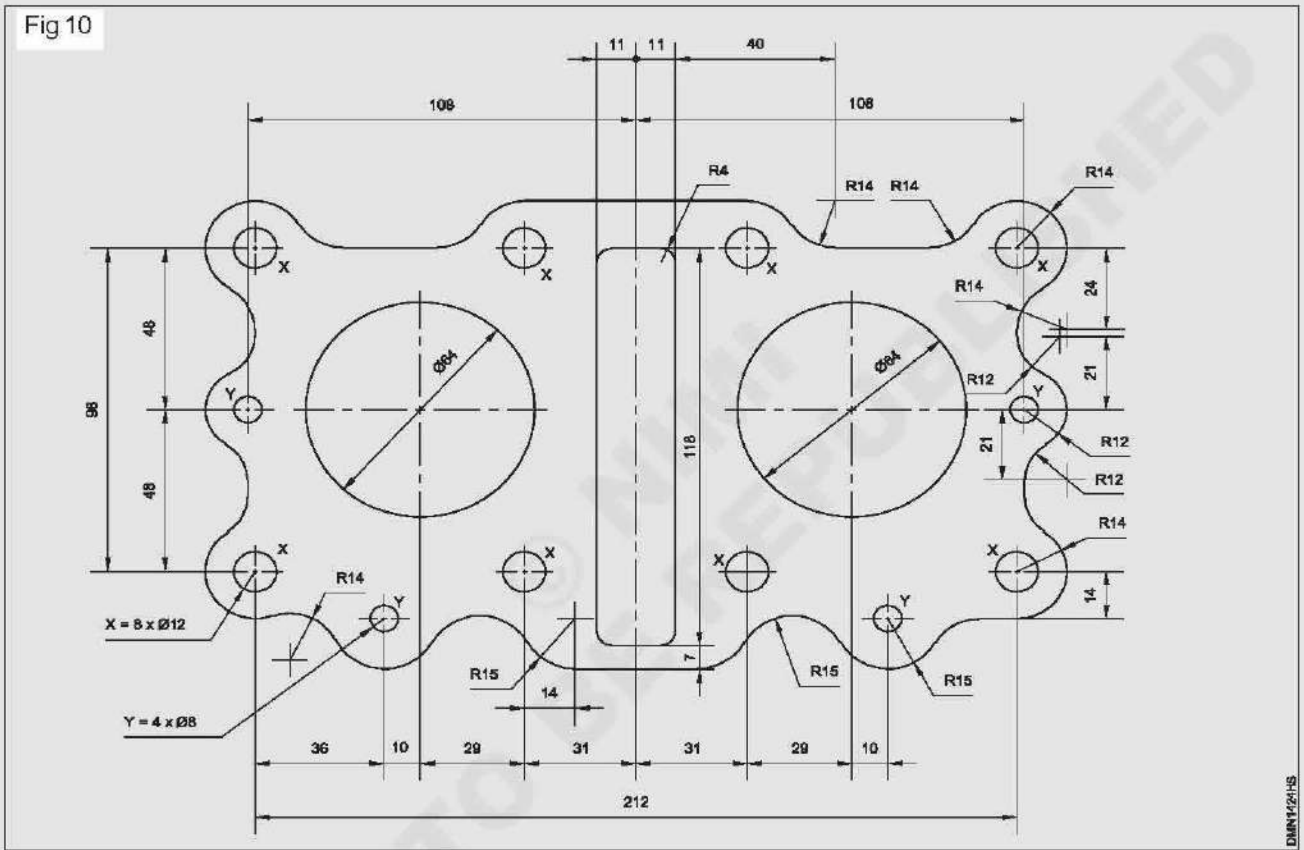

165
R13
R92
R6
HEX A/F 38
108
Ø16
Ø70
38
19
SPECIAL CAM
DMN1424HQ

Fig 10


11
11
40
109
108
R4
R14
R14
R14
R14
R12
R12
R12
R14
R14
24
21
21
14
96
48
48
118
Ø64
Ø64
X = 8 x Ø12
Y = 4 x Ø8
R15
R15
R15
7
14
36
10
29
31
31
29
10
212
DMN1424HS

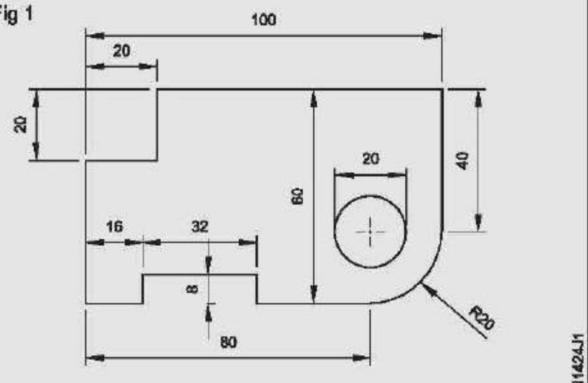
Application of Dimensioning Technique

Objective: At the end of this exercise you shall be able to
- **draw the drawings by correcting the mixtures and drawing in both systems.**

Exercise: Reproduce the given drawings (Figures 1 to 10) & dimensioning them both its aligned system and unidirectional system.

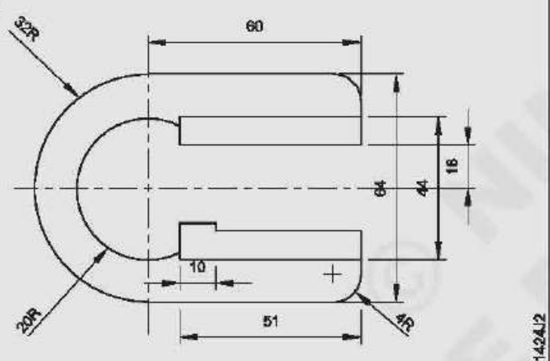
Note: Read the related theory related to dimensioning technic and draw the given figures 1 to 10 as per instructor.

Fig 1



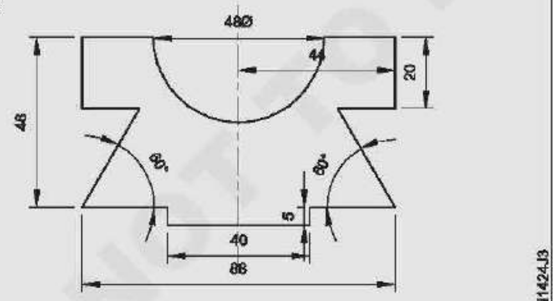
AN ANGLE PLATE (WRONG DIMENSIONING)

Fig 2



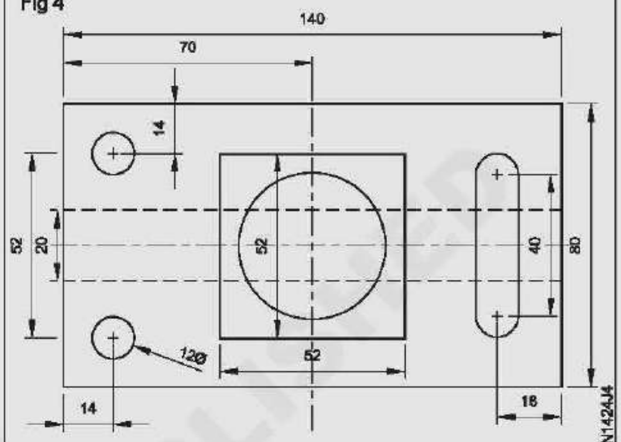
GO-NO-GO GAUGE (WRONG DIMENSIONING)

Fig 3



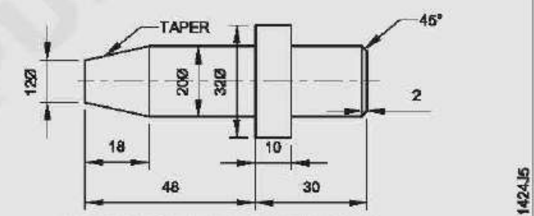
A TEMPLATE (WRONG DIMENSIONING)

Fig 4



A MACHINE PART (WRONG DIMENSIONING)

Fig 5



A MACHINE PART (WRONG DIMENSIONING)

Fig 6

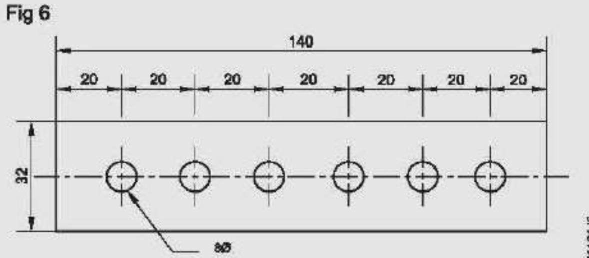
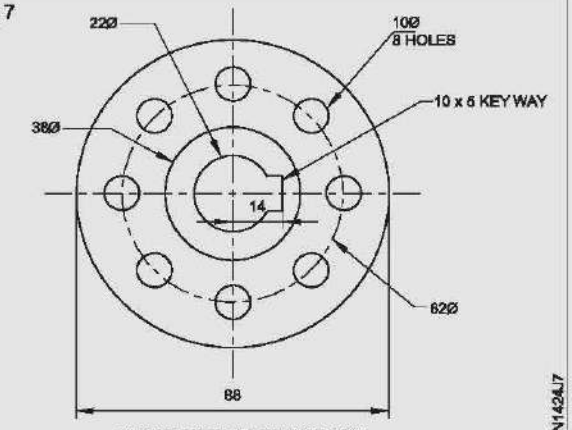


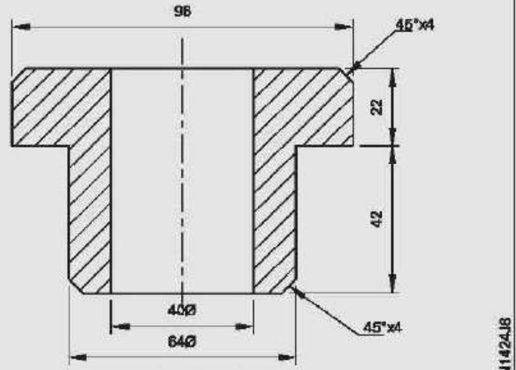
PLATE WITH HOLES (WRONG DIMENSIONING)

Fig 7



FLANGE (WRONG DIMENSIONING)

Fig 8



JIG BUSH (WRONG DIMENSIONING)

Fig 9

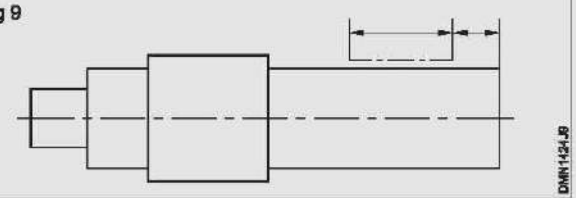
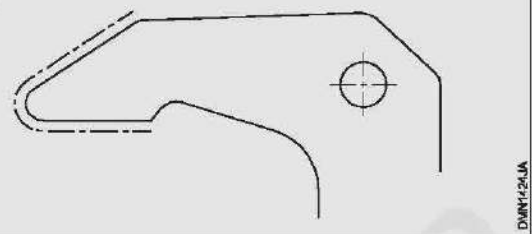


Fig 10



Draw plain scale - Diagonal scale - comparative scale - vernier scale

Objectives: At the end of this exercise you shall be able to
- **construct the plain scale, diagonal scale and comparative scale**
- **vernier scale and scale of chords.**

Scales: It is difficult to draw the components to their actual sizes, because they may be too large to be accommodated on the drawing sheet or too small to draw and cannot be effectively used in the shop floor. For example, think of making the drawing of a motor car. It is too long and wide to be drawn on the drawing sheet to its original size. Similarly small component like wheel of a wrist watch or its needle (hands) if drawn to its original size will not be legible enough for use in the shop floor.

So depending on the situation drawings are drawn smaller or larger than the actual sizes. When we say that the drawings are smaller or larger, we mean that a given length in the drawing will be smaller or larger than the corresponding length in the object.

The ratio of the length in the drawing to its corresponding length of an object, when both the lengths are in the same unit, it is called the **Representative Fraction** (RF).

$$RF = \frac{\text{Size of the component in the Drawing}}{\text{Actual size of the component}}$$

1 Construct a plain scale of RF $\frac{1}{20}$ to measure up to 10 cm (min) and mark a distance of 1.2 metres on the scale.

2 Construct a scale of RF $\frac{1}{6000}$ and using this scale, draw a rectangle of perimeter 2400 m and the sides are in the ratio of 3:4.

3 Construct a plain scale of RF $\frac{1}{40}$ to measure metres and decimetres and mark a distance of 3.7m.

4 Construct a comparative scale to convert °C to °F

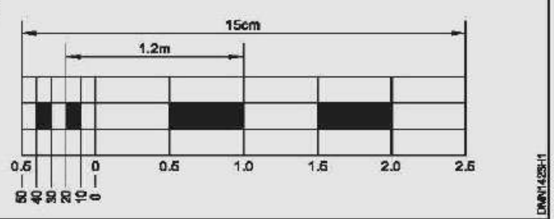
5 Construct a scale of chord and mark angle 35° using the same.

PROCEDURE

Exercise 1: Construct a plain scale of RF 1/20 to read 1.2 m and minimum distance of 10 cm (Fig 1)

- Take a length as 15 cm.
  15 cm x 20 = 300 cm (15 cm will represent 3 m)
- Draw a rectangle of 15 cm x 1.5 cm to represent the length and width of scale.
- Divide the length (15 cm) into 6 equal parts. Each part representing 0.5 m or 50 cm.
- Mark the parts as 0.5; 0; 0.5; 1.0; 1.5 etc.
- Divide the first part into five equal divisions (each of the division is 1 dm or 10 cm) and mark them as shown.
- Divide the width of the scale into 3 parts.
- Fill and number the alternate blocks (middle) as shown.
- Mark 1.2 m on the scale.

Fig 1



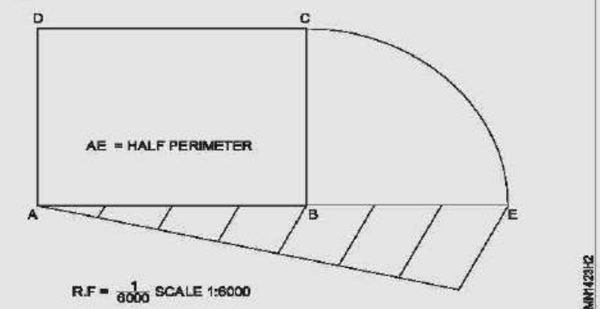
Exercise 2: Construct a rectangle whose perimeter is 2400 m and its sides are in the ratio of 3:4. Using scale of RF 1/6000 (Fig 2)

- Calculate the length of the scale to represent half the perimeter: 1200 m

  Length of the scale = RF x length to be measured

  $\frac{1}{6000} \times 1200 \text{ m} \times 100 = 20 \text{ cm}.$
- Draw a line AE 20 cm long represent half the perimeter.
- Divide AE into 7 equal parts. (3:4); 3+4 = 7)
- Mark AB equals to 4 units.

Fig 2



- Draw perpendiculars at A and B.
- Mark D and C on the perpendiculars to a length of BE.
- Join DC and complete the rectangle ABCD.

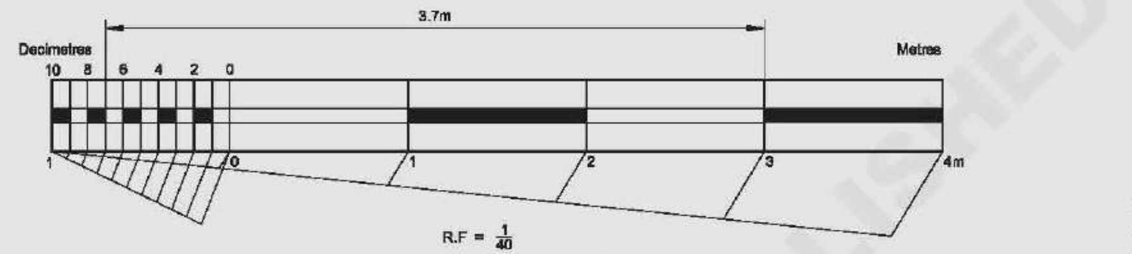
Exercise 3: Construct a plain scale to show metres and decimetres long enough to measure upto 5 m. RF = 1/40. Mark a length of 3.7 m on it (Fig 3)

RF = 1/40
Maximum length to measure = 5 m
Length of the scale required = RF x L

$$= \frac{1}{40} \times 5\,\text{m} \times 100 = 12.5\,\text{cm}$$

- Draw a rectangle of 12.5 cm x 1.5 cm.
- Divide the length into 5 equal parts each representing one metre.
- Mark them as shown.
- Divide the first division into 10 equal parts each representing 1 decimetre and mark them as shown.
- Mark the distance 3.7 as shown in Fig 3.

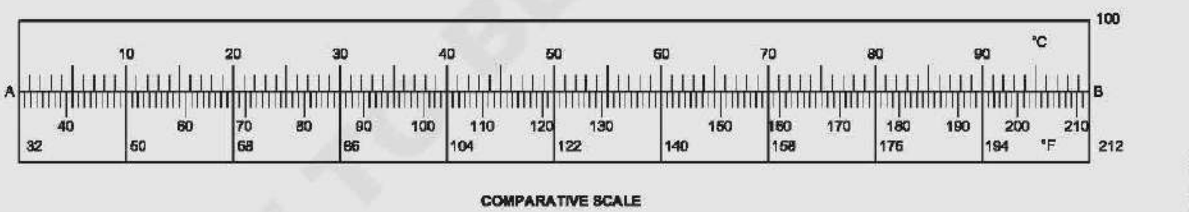
Fig 3



Exercise 4: Construct a comparative scale to convert Fahrenheit (°F) into Celsius °C and vice-versa (Fig 4)

- Draw a line AB of 15 cm long. (Top part will read °C and bottom part will read °F)
- Divide the line into 10 equal divisions.
- Top side mark 0, 10, 20....100 for °C scale (100 divisions) and on bottom side, mark 32, 50, 68.... 212 for °F scale 180 divisions as shown.
- On °C side divide one division into 10 equal parts. (Now each small division represents 1°C)
- On °F side, divide each division into 18 equal parts. (Now each small division represents 1°F)
- Mark other numbers and complete drawing the scale.

Fig 4



Exercise 5: Construct a scale of chord and draw an angle 35° (Fig 5 & 6)

- Draw a quadrant ABC and extend the line AB.
- 'A' as centre AC as radius, draw the arc CD. (Now AD represents the chord of arc CD)
- Divide the arc AC into 18 equal parts, so that each division will represent 5°.
- 'A' as centre, draw arcs with radii A_1', A_2'....A_{18}' to intersect line DA and mark them 5°, 10°.....90° as shown in Fig 5.

To draw the angle of 35°

- Draw a line PQ equal to AB.

Fig 5

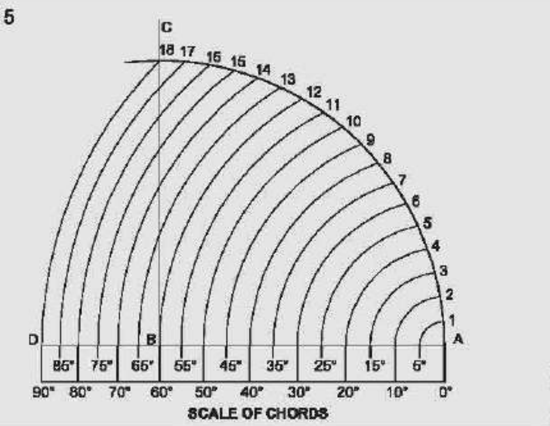
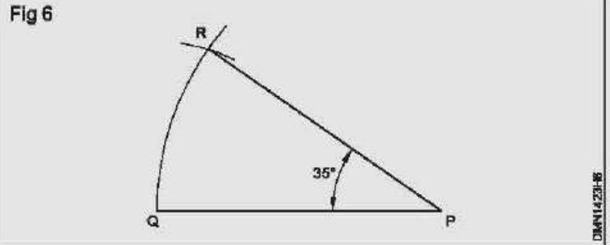


Fig 6



- 'P' as centre PQ as radius, draw arc.
- Set the compass to A-35° and 'Q' as centre, draw an arc to intersect the previous arc at R.
- Join PR by a straight line and now angle RPQ = 35°

6 Construct a diagonal scale of RF = $\frac{1}{25}$, showing metre, decimeter and centimeter to measure upto 4 m. Mark a length of 2.69 m, 1.09 m and 0.08 m..

7 Construct direct vernier scale to read centimeters and to measure upto 4 metres having an RF = $\frac{1}{25}$. Mark distances 0.21 m; 2.74 m & 3.72 m.

8 Construct retrograde vernier scale to read centimeters and to measure upto 4 metres having an RF = $\frac{1}{25}$. Mark distances 0.21 m, 2.74 m & 3.89 m.

Exercise 6: Construct a diagonal scale for 4m length and show the lengths 2.69m, 1.09m and 0.08m. (RF=1/25) (Fig.7)

- Length of scale required = RF x length to be measured

$$= \frac{1}{25} \times 4 \text{ m} \times 100 = 16 \text{ cm.}$$

- Draw a rectangle ABCD of 16 cm x 4 cm.
- Divide the rectangle ABCD into 4 equal parts and mark them EF, GH & IJ and each division represents one metre.
- Divide the line AB into ten equal parts and mark them 1_1, 2_1, 3_1.....10_1.
- Draw vertical lines from points 1, 2...... etc.
- Divide the BF into 10 equal parts and mark them as 1'2'3' etc and each division representing 10 cm (1 dm).
- Draw diagonals on all ten small rectangles in the 1st (lower) block ABFE and complete the diagonal scale.
- Metres are read on EF or line parallel to it i.e GH, IJ & DC. Decimetres are read on the division of line AE and centrimetres are read on points where the diagonals intersect with the vertical parallel lines drawn through the divisions of line AB.
- Mark 2.69 in using the diagonal scale. (Fig 7)

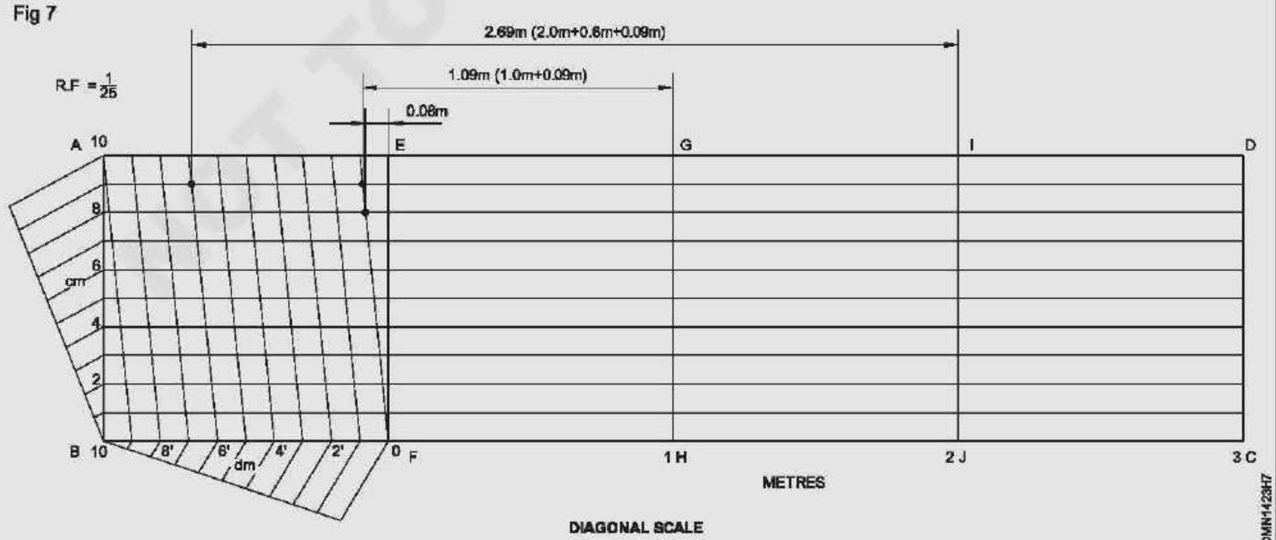
  2.00 m on metre division

  0.60 m on decimetre division

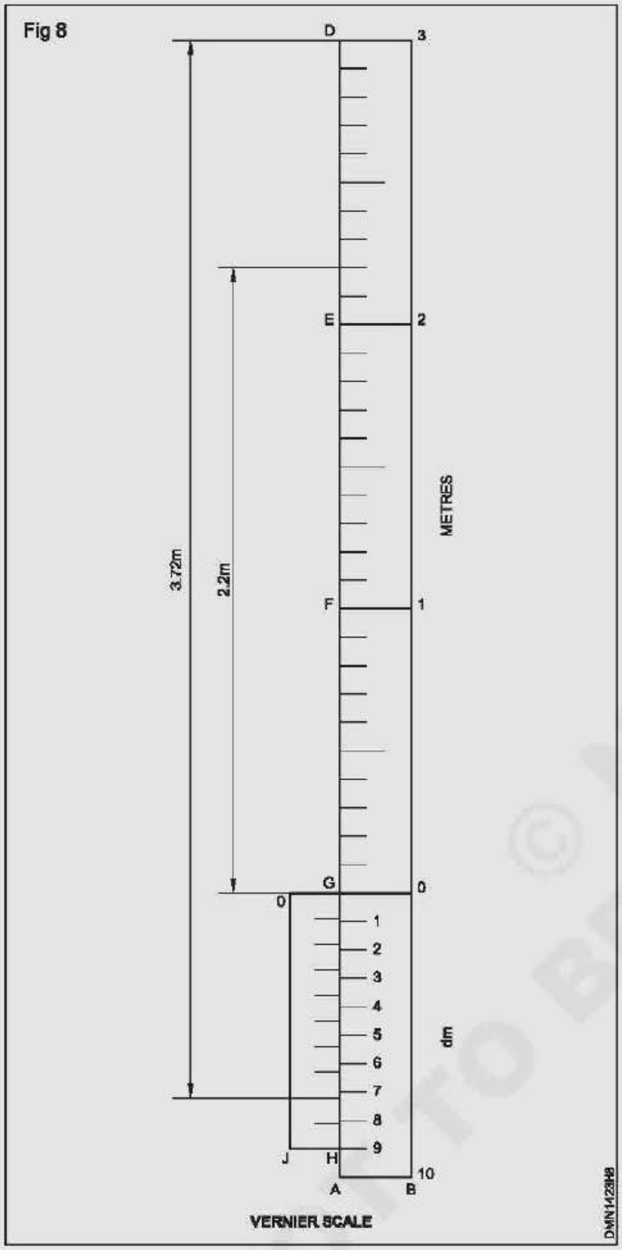
  0.09 in the diagonal cm division

  1.09 m and 0.08 m are also marked in the figure 1 in the same way.

Fig 7

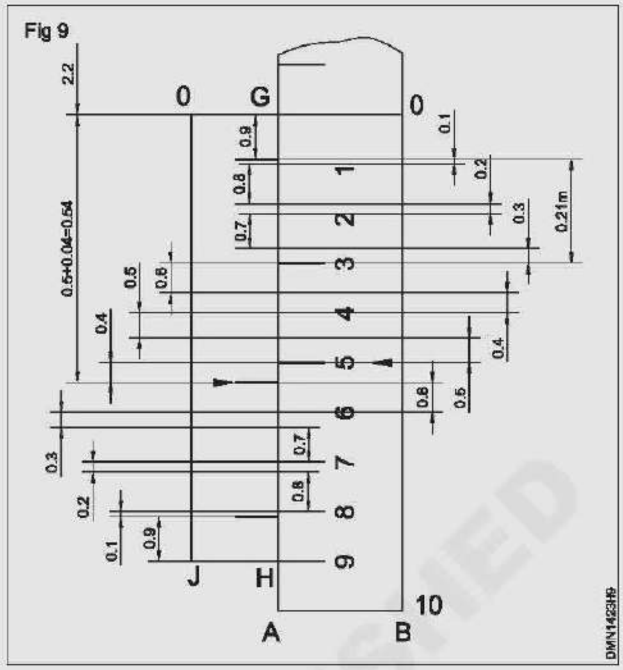


Exercise 7: (Figs 8) Construct a direct vernier scale of RF = $\frac{1}{25}$ to read centimetres for 4 m and 3.72 m, 2.74m and 0.21m



- Draw a rectangle ABCD (16 cm x 1 cm) representing main scale.
- Divide the main scale into 4 length of each part equal and representing 1 metre.

  Divide each line AG, GF, FE & ED into 10 equal divisions and each division is called one main scale division. (1 dm)
- Add another rectangle GO, J, H as secondary scale (vernier) to a length of 9 MSD (9 dm).
- Divide GH into 10 equal division on secondary (vernier scale) side and complete the vernier scale.



To show the reading 0.21 m (Fig 8 & 9)

0.21 m = 0.2 m + 0.01 m

0.2 m = Two full decimetre

0.01 m = difference between 1st MSD and 1st VSD (secondary)

Lowest main scale block and vernier side portion of Fig 8 is shown with more detail in Fig 9.

- To mark the reading 2.74 m refer the figures 8 & 9.

  Follow the marking made on G to D line in figure 8 for 2.2 m

  On the main scale OB 5 MSD marking made on G to D line in figure 3 for 0.5 m (1 MSD = 0.1 m)

  Refer VSD from left to right in Fig 3 for 0.04 m

  Required reading 2.74 m

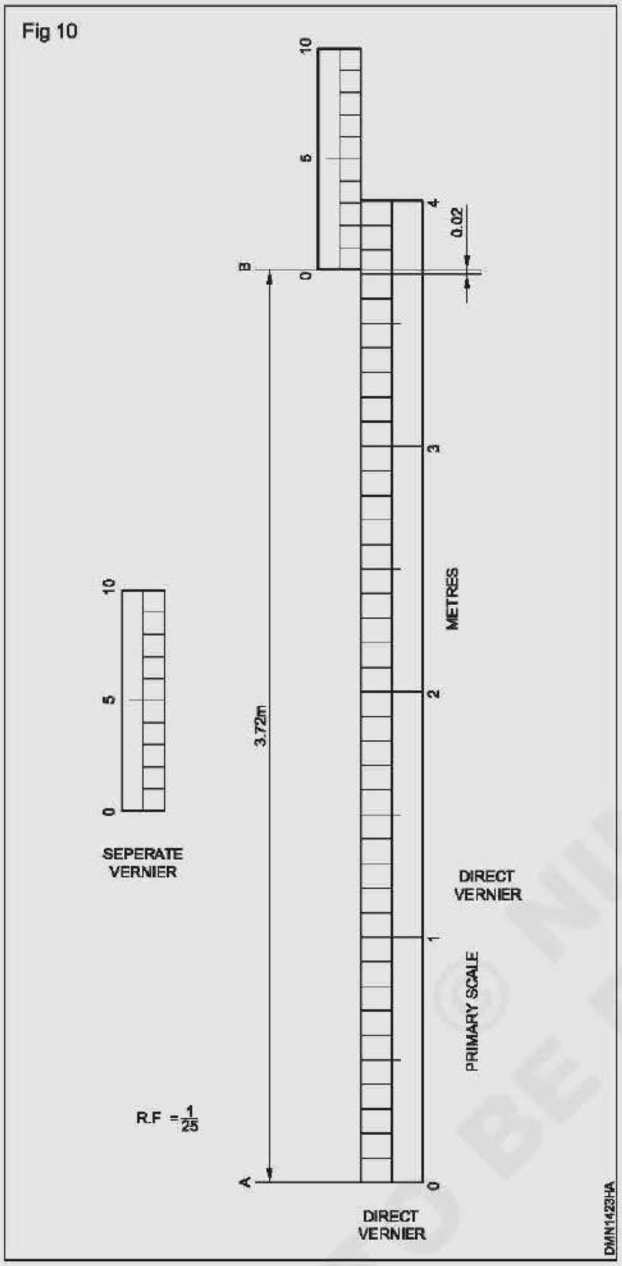
To mark the reading 3.72 m

For reading 3.72 m follow the procedure given for earlier readings. (Fig 8 & 9)

or

Follow the method and procedure given below. (Fig 10)

- Draw the main scale in the form of a rectangle (16 cm x 1 cm) with metre and decimetre divisions.
- Construct the secondary scale (vernier) separately, taking 9 main divisions and dividing it into 10 equal parts.
- To mark 3.72 m, position secondary (vernier) scale along the main scale to the right of 3.7 m and the second division of the secondary (vernier) scale coincides with 3.9 m mark.



Exercise 8: Construct a retrograde vernier scale (Fig 11)

RF = $\frac{1}{25}$; Least count: 1 cm; Maximum length: 4 m Length of scale: $\frac{1}{25}$ x 4 m x 100 = 16 cm

- Construct the main scale (instead 3 metres, draw to 4 metres)
- Extend the left end of the main scale temporarily by one division.
- Draw the secondary (vernier) scale of 11 MSD length as shown.
- Divide the secondary (vernier) scale into 10 equal divisions each representing 1.1 dm or 0.11 mm and complete the retrograde vernier scale.

To mark the reading, refer Figure 11.

(i) 0.21 metres = 1.1 + 0.1 = 0.21

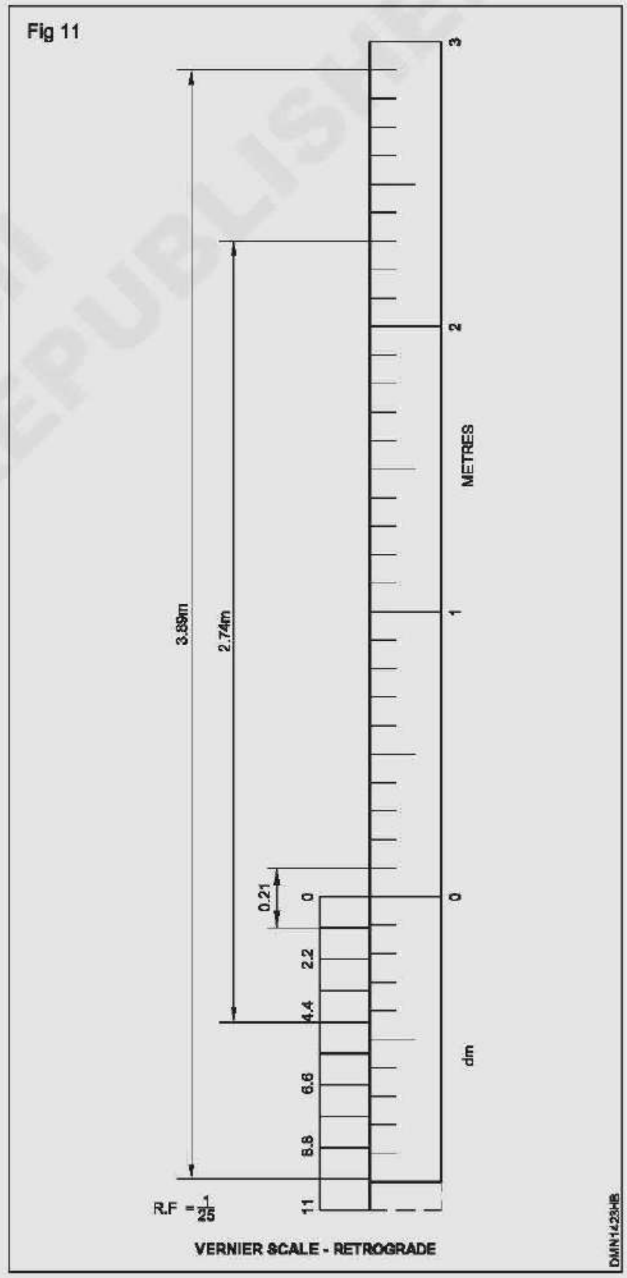
- With reference to zero on the vernier scale, one division on the right hand side and 1 division on the left side.
- Draw extension lines and mark the reading.

(ii) 2.74 metres = 2.3 + 0.44 = 2.74

- Mark 4th division on the vernier scale represents 0.44 from `O' and mark 2.3 m on the main scale.
- Draw extension lines and mark the reading.

(iii) 3.89 metres = 2.9 + 0.99

- Mark the 9th division on the vernier scale represents 0.99 m from `O' and mark 2.9 m on the main scale.
- Draw extension lines and mark the reading.



Orthographic projection of points and lines

Objectives: At the end of this exercise you shall be able to
- **draw the projection of a point for the given positions in the four quadrants**
- **draw the projection of lines in first angle and 3rd angle for the given positions.**

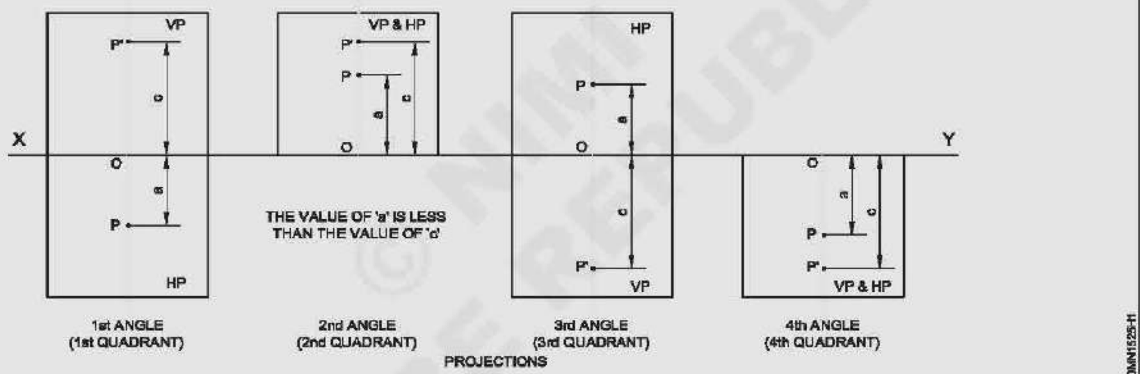
PROCEDURE

Exercise 1: (Fig.1): Draw the plan and elevation of the point 'P' given its distance 'a' and 'c' from the both the plane of projection when the point is lying in first, second, third and fourth quadrants. Refer the table 1 for the values 'a' and 'c' and draw the projections for the points in 1st, 2nd, 3rd and 4th angle.

Table 1

Points/ Distance Angle	1st	2nd	3rd	4th
a in front of VP	20	25	25	20
c above HP	40	30	30	35

Fig 1



Example 1

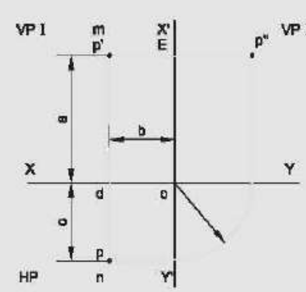
- Draw the plan and elevation of the point 'p' given its distance as a & c from both planes of projection, when the point is lying in first, second, third and fourth quadrants. Refer the table for the values a & c.

Points/ Distance	1st	2nd	3rd	4th
Distance from VPI a	30	35	40	40
Distance from VPII b	20	40	30	20
Distance above HP c	25	30	35	15

- Draw the XY line.
- Represent the HP & VP for each quadrant as shown in Fig 1.
- Draw vertical line perpendicular to intersect/ meet XY at '0'.
- Mark the points P and P' in all four quadrants such that OP = a & OP' = c.
- Now the points P & P' are the plan and elevation as required.
- Adopt the same procedure for points 2nd angle, 3rd angle and 4th angle. (Fig 1)

Draw the plan and elevation of a point given its distance as a, b and c from VP II and HP respectively, when the point is kept in first quadrant. Refer for the values of a, b and c (Fig 2)

Fig 2



- Draw xy and x' y' intersecting each other at 'o' and mark the plane VP I, VP II and HP
- Draw a vertical lines 'xy' at a distance of b as shows in figure
- Mark the point p such that d' is equal to 'a' as shows
- The point p' shows highlighted will be the elevation
- Mark p shows that dp is equal to 'c' as section in figure. now the point p will be the plan.
- Draw a projector through p' parallel xy cutting x' y' at e
- Mark the point p" such that EP" is equal to c
- Now the point p" will be the side elevation on VP II
- Adopt the procedure for the points A, B, C & D and draw the projection.

Exercise 3: Draw the projections of a line on VPI, VPII and HP given its positions as in the table below

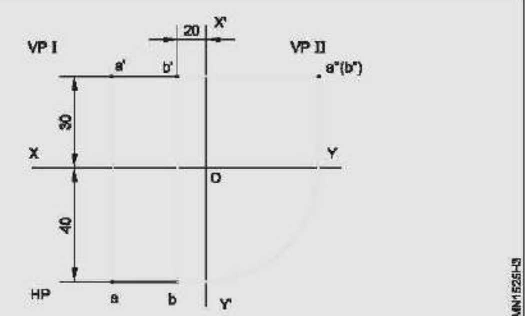
Ex.No.	Line	Length of line	Distance from HP	Distance from VPI	Distance from VPII	Line inclined to HP	Line inclined to VPI
1	ab	40	30	40	20	Parallel	Parallel (Fig 3)
2	cd	45	Nearest end point of line 20 above HP	35	15	90° perpendicular	Parallel (Fig 4)
3	pq	55	25	Farthest end point of line 75 in front of VP	20	Parallel	90° (Fig 5) perpendicular
4	rs*	50	Nearest end point of line 15 above HP	40	60mm to the point nearest to HP	30°	Parallel (Fig 6)
5	mn	60	28	Nearest end point of line 15 from VP	33 mm to the nearest end point of the line	Parallel	55° (Fig 7)
6	kl*x	70	Nearest end point of line 20 above HP	Nearest end point of line 25 in front	35mm to the point nearest to VPII	Either 40°	or 50° (Fig 8)
7	gh***	70	-do-	Nearest end point of line 25 from VP	80 mm to the point nearest to VPII	40°	50° (Fig 9)

* Angle is measured positive in the anti-clockwise direction.
** Line KL is parallel to VPII
*** Line GH is not parallel to VPII

Ex 3-1

- Draw lines XY and X'Y'.
- Draw a line ab (plan) at a distance of 40 mm below XY and point 'b' 20 mm away from X'Y'.
- Draw a line a'b' 30 mm which will be the elevation above XY and point b' is 20 mm away from X'Y'.
- Draw the projectors from ab and a'b'. Projectors meet at a point a" (b") - the side elevation. (Fig 3)

Fig 3

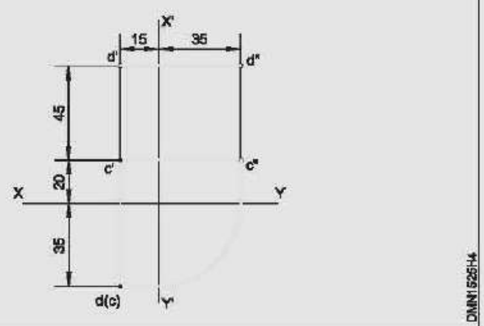


Ex 3.2

- Draw the lines XY and X'Y'.
- Mark the point d (c) - Plan 35 mm below XY and 15 mm to the left of X'Y'.
- Project the point d (c) upwards and mark point c' 20 mm above XY line.
- Mark point d' 45 mm above point c'.
- Join points c'd' (length of the line). Now line c'd' is the elevation.
- Project c'd' to the right and draw the side elevation c''d'' 35 mm away from the X'Y' line. (Fig 4)

For the previous step 35 mm may be taken from a scale or transfered from plan d (c) by construction as shown in Fig 4.

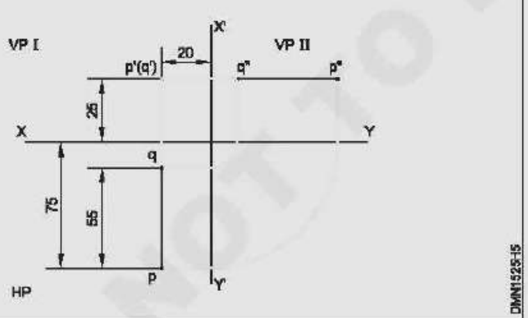
Fig 4



Ex 3.3

- Draw the lines XY and X'Y'.
- Draw pq (plan) of length 55 mm such that it is 20 mm away from X'Y' and point P is 75 mm below the XY line.
- Project pq vertically upwards and mark p' (elevation) ON it and 25 mm above the XY line.
- The side elevation p'' q'' is drawn by projecting p' to the right and transfering the distance pq as shown. (Fig 5)

Fig 5

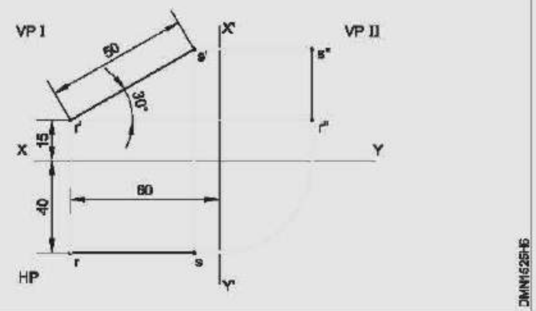


Exercise 3.4 (Fig 6)

In this problem, elevation on r's' will be of actual length (50 mm) at 30° to XY. Plan and side elevation will be shorter than actual length.

- Draw the elevation r's' at an angle 30° and of 50 mm long, with the point r' 15 mm above XY line and 60 mm to the left of X'Y'.
- Project r's' downwards and draw rs the plan at a distance 40 mm below the XY line.
- Draw the projectors from r's' and rs as shown and draw the r''s'' - the side elevation. (Fig 6)

Fig 6

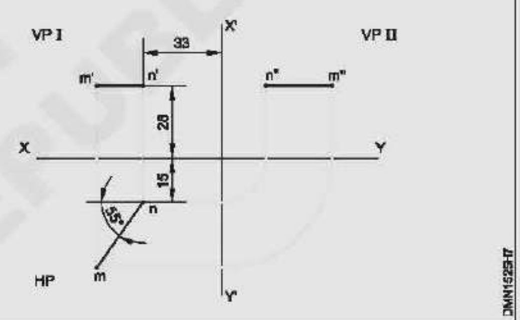


Exercise 3.5 (Fig 7)

Since the line is parallel to HP, projection on HP will be of true length and it will be at an angle of 55° to XY line.

- Draw the line mn (plan) such that it makes an angle of 55° to XY line and is of 60 mm in length. (Fig 7)
- Draw the elevation m' n' and side elevation m'' n'' as given in the earlier exercise.

Fig 7

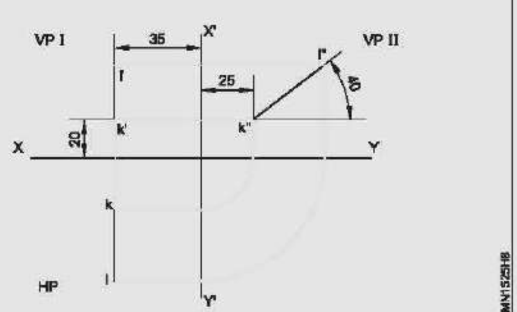


Exercise 3.6 (Fig 8)

Since the line is parallel to VPII, projection of VPII will be of true length and it will be at an angle of 40° as the inclination of the line to HP is 40°.

- Draw k''l'' to length of 70 mm, inclined 40° to XY with point k'' is 20 mm above XY and 25 mm away from X'Y'.
- Draw k'l' (elevation) 35 mm away from X'Y' by projecting points k'' and l''.
- Draw plan kl by drawing projectors from side elevation and elevation. (Fig 8)

Fig 8



In this example projection of the line in all the three plane will be shorter in length than the actual length.

Ex 3.7

- Draw the line gk and g' k', the plan and elevation of the line if it was parallel to VP and inclined 40° to HP.
- Draw a line gh at an angle of 50° to gk and of same length as gk. Now gh will be the plan when the line makes 50° to VP.

The projected length of a line, when projected to one principle plane to which it is inclined will be of same length, no matter what angle it makes with the other principle plane.

- Get the point h' by drawing horizontal projectors through k' and vertical projector through 'h'.
- Join g' h' and this will be the required elevation on VPI.
- Get points g'' h'' by drawing projectors from plan and elevation.
- Join g'' h'', the side elevation. (Fig 9)

Fig 9